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Assessment of patient-reported outcomes after polytrauma: a systematic review

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Assessment of patient-reported outcomes after polytrauma: a systematic review

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Fig. 1 Flow chart

ABSTRACT

Objectives We sought to: (1) collect instruments that assess health-related quality of life (HRQoL), activities of daily living (ADL), and social participation during follow-up after polytrauma, (2) describe their use (e.g. frequency of application during follow-up), and (3) investigate other relevant PROs assessed in the included studies (e.g. pain, depression, anxiety).

Design Systematic Review. The search was conducted in MEDLINE, Embase, CINAHL, PsycINFO, CENTRAL, the trials registers ClinicalTrials.gov, and WHO ICTRP. All original empirical research published between January 2005 and April 2018 was included. Data extraction, narrative content analysis, and a critical appraisal were performed.

Outcome measures Instruments assessing HRQoL, ADL, and social participation after polytrauma. Other PROs assessed in the included studies.

Results The search yielded 3496 hits and 54 publications were included in the review. Predominantly, HRQoL was assessed (in 48 of 54 publications), with Short Form-36 Health Survey (SF-36) applied most frequently. ADL and (social) participation were rarely assessed. The follow-up period lasted between 8 days and 17.5 years; assessments were performed between one and five times. Most used were a single assessment of PROs, a follow-up period of one to one and a half year, and a postal survey. Other relevant PRO areas reported were function, mental disorders, and pain.

Conclusions There is a large variation in the assessment of PROs after polytrauma, impairing comparability of outcomes. First efforts to standardize the collection of PROs have been initiated, but further harmonisation between central players, such as trauma registers, is required. Definition and continuous collection of core outcome sets would improve comparability of study results, creating the basis for more targeted treatment. Additional knowledge on rarely reported PRO areas like (social) participation and social network may lead to their consideration in health services provision.

Registration: PROSPERO (CRD42017060825 on 8 April 2017)

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Keywords: Accident & Emergency Medicine, Adult Intensive & Critical care, Orthopaedic & Trauma Surgery, Public Health

Strengths and limitations of this study

- This review was conducted based on a registered and published protocol.
- First systematic review that provides an overview of patient-oriented instruments used in research after polytrauma.
- This review encourages further improvement in the assessment of PROs after polytrauma since first efforts to standardize the collection of PROs have been achieved to optimize comparability of outcomes, but further consistency between central players (e.g. trauma registers) is needed.
- A limitation might be the restriction of the search period and the language.

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INTRODUCTION

Severe injuries represent a leading cause of death and permanent disability.[1] In the central European region, severely injured people are referred to as having a 'polytrauma' or being 'polytraumatized'.[2] A polytrauma is defined as at least two severe injuries in different body regions or organ systems, at least one of which is potentially life-threatening by itself or in combination with others, with an Injury Severity Score (ISS) ≥16.[1-4] According to the TraumaRegister DGU®, a German registry that covers patients with severe injuries, the number of patients with polytrauma (ISS ≥16) was 15 651 in 2019.[5] Most affected were men (72%) with a mean age of 46.5 years.[6]

The survival rates after polytrauma have increased lately, bringing the quality of survival into the focus of research. One year after polytrauma, survivors often fail to reach pre-traumatic status due to persisting medical problems.[7] Disabilities such as respiratory problems, paraplegia or tetraplegia, prevent 40% from returning to their former workplace.[8, 9] They also have an impact on socio-economic aspects and quality of life.[9-11] Therefore, it is crucial to reduce the short-term and long-term burden after polytrauma.[9, 11] In this context, the patients' perspective on polytrauma outcomes focussing on psychological, physical and social factors is getting more important for the patients themselves, their relatives, and healthcare professionals.[11, 12] Evaluating these patient-reported outcomes (PROs) is essential to complete the holistic assessment of relevant clinical outcomes after injury from a patient's perspective.

To assess the impact of polytrauma on health-related quality of life (HRQoL) and other important PROs, one must consult the person affected directly.[12, 13] HRQoL is already being recorded in registers like the Australian Victoria State Trauma Registry. The German TraumaRegister DGU® will also complement its assessment with HRQoL.[14] Activities of daily living (ADL), defined as basic activities to fulfil independent living, along with physical and psychological needs, are meaningful outcome parameters in case of post-polytrauma impairment. Participation is a health component according to the International Classification of Functioning, Disability and Health (ICF).[15] In recent years, it has gained more attention in political context.[16] Several working groups have controversially discussed which tools should be applied to assess PROs after polytrauma, at which time points, and in which way;[13, 17-25] however, guidelines with clear recommendations are lacking,[13] and no systematic review could be identified.

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Therefore, the aim of this systematic review was to:

1. collect instruments assessing HRQoL, ADL, and social participation during follow-up after polytrauma
2. describe their application in detail (e.g. duration of follow-up period, frequency of application, and time point of measurements within the follow-up period)
3. investigate which other relevant PROs are additionally assessed in included studies (e.g. pain, depression, anxiety, and cognitive function) without detailed analysis.

METHODS

This systematic review was conducted in line with the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guideline (Additional file 1).[26] The review protocol was registered on 8 April 2017 at PROSPERO (CRD42017060825).

Data sources and search strategy

The search was conducted in MEDLINE, Embase, CINAHL (Cumulative Index to Nursing and Allied Health Literature), PsycINFO, CENTRAL (Cochrane Central Register of Controlled Trials), the trials registers ClinicalTrials.gov and WHO ICTRP (WHO International Clinical Trials Registry Platform), covering the period from 1 January 2005 until 5 April 2018. In addition, reference lists of eligible studies were examined. Textwords (tw) and MeSH-terms (mh) like ‘polytrauma’ (tw), ‘multiple trauma’ (mh), ‘quality of life’ (mh), ‘activities of daily living’ (tw), ‘social participation’ (mh) and ‘pain’ (mh) were used. The development of the search strategy was outlined in the study protocol [27] and is provided in Additional file 2.

Inclusion and exclusion criteria

The publications included primarily investigated PROs of patients aged 18-75 years with an Injury Severity Score (ISS) ≥ 16 and/or an Abbreviated Injury Scale (AIS) ≥ 3 during a clearly defined follow-up period. All original empirical research published in English or German was included. Publications examining special types of injury or illness – e.g. low-energy injuries, single or mono injuries, burn injuries, war injuries, or cancer – were excluded. We also excluded some publication types (grey literature, books, abstracts, and editorials). Systematic reviews and meta-analysis were not included, too, but screened to identify further appropriate studies. More details about the inclusion and exclusion criteria were published in the study protocol.[27]

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Selection of publications and studies

Four teams with two reviewers each (MR-IG; KM-SK; MR-SOB; AI-AHF) screened the title and abstracts independently. To foster the process of eligibility decisions, the inclusion or exclusion of the first 50 publications was discussed by two reviewers (MR-IG). In a next step, two reviewers (MR-IG; KM-SK; MR-SOB, KM-SOB) independently screened the full texts of the remaining publications for final inclusion. A third reviewer (MR, SK, IG) solved disagreements regarding the inclusion of publications for full-text screening and their final inclusion in this review. We identified publications relating to the same study and/or population. For this purpose, extensive research was conducted, and authors of the corresponding publications were contacted. The section 'Characteristics of included publications' outlines how we approached publications dealing with different research questions of the same study, comprising subpopulations drawn from a superior study population, reporting subprojects of a study project, or presenting results of the same research question at different follow-up periods.

Data extraction and analysis

Data extraction was performed according to the requirements of Cochrane reviews.[28] For the first and second aim, two researchers (MR-IG) used a piloted data extraction sheet independently comprising the following: first author, publication year and country, study design, aim of the study (objective), study population, treatment, applied instruments (incl. modifications), time of measurements, length of follow-up period, method of assessment (application), study results, and results of the critical appraisal. Regarding the third aim, the other reported PROs were extracted and categorized in a separate extraction sheet. Narrative content analysis was performed regarding the assessment of HRQol, ADL and social participation as well as further reported PROs.

Critical appraisal

Teams with two reviewers each (MR, IG, SK, KM, SOB) independently assessed the methodological quality and the risk of bias in each study type using standardized checklists of the UK National Institute for Health and Care Excellence (NICE),[29, 30] Scottish Intercollegiate Guidelines Network (SIGN),[31] or the Mixed Method Appraisal Tool (MMAT).[32] Discrepancies were resolved by discussion between the reviewers or by a third reviewer.

Patient and public involvement statement

Patients were not involved in the design and implementation of this systematic review.

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RESULTS

Search results

In total, 3485 references were retrieved. After a title and abstract screening as well as a full-text screening, further 11 records were identified by backward citation tracking. Finally, 54 eligible publications were included for data extraction (Fig. 1).

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Fig. 1 Flow chart

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Characteristics of included publications

The publications (n=54) most often came from Germany (n=15, 28%) and the Netherlands (n=11, 20%); rarely from Austria, Denmark, France, Japan, Poland, Spain and Sweden (each n=1, 2%). One study has been carried out in Germany and Austria, another one in Europe, Asia, Australia, Africa, North- and South-America. The publications primarily reported monocentric studies (n=44, 81%) and predominately comprised prospective cohort studies (n=26, 48%) (Table 1).

The number of participants varied between eight[33] and 7612[34], with an average age of 24.6 ± 12.1 [35] to 53.9 ± 18.8 [36] years, and a proportion of men between 37.5% [33] and 84% [37]. The mean ISS ranged between 3.9 ± 1.7 [38] and 56.8 points [39, 40] and the AIS – reported in 18 publications – between zero and five points. Information regarding treatment of injuries was provided in n=31 publications (Table 1).

Six publications were related to the Hannover Rehab study.[35, 41-45] Four reported the results of a polytrauma study project conducted at the University Hospital Basel, which included a subproject of a study project (EK 159/02) and a polytrauma cohort.[46-49] Study projects at the University Medical Centre Utrecht displayed almost identical recruitment periods and locations (Holtslag et al. 2007 a, b: January 1999 to December 2000,[50, 51] Holtslag et al. 2006 and 2008: January 1999 to January 2001,[52, 53] Holtslag et al. (2006) analysed a subgroup out of n=335 participants;[52] Holtslag et al. 2008 additionally included a group 'Fatalities'.[53] Two publications [54, 55] used the same data collection but addressed different research questions (CIHR, grants no. MOP-62873, FRSQ, grant no. 3255). The publication by Sirois et al. (2009) is the original study,[55] the publication by Ouellet et al. (2009) is a secondary analysis.[54] Three publications comprised patients treated at the Murnau Trauma Centre from 2000 to 2005 and recorded in the TraumaRegister DGU®.[39, 40, 56] The publication by von Rueden et al. (2013) [39] and Wurm et al. (2012) [40] reported on the same investigation of a group of severely injured patients with an ISS>50; Simmel et al. (2013) analysed those patients with an ISS>25 [56]. Soberg et al. (2011) [57] investigated the same research question as Soberg et al. (2007) [58] with a follow-up of 5 years (Table 1). Table 1 provides an overview of the included publications.

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Table 1 Overview of included publications

Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results	Critical appraisal
					QoL/HRQoL	(Social) Participation	ADL				
Abraham et al., 2014 USA [59]	Prospective monocentric cohort study	Influence of inpatient delirium on HRQoL after polytrauma; association of depressive symptoms and symptoms of PTSD with domains of HRQoL	N=115 (55.7) Age: Mean 42.4 ± 16.7 ISS: 16 – > 25, Mean 30.4 ± 9.2 AIS: NR Injury: Extremity, head/neck (no brain injury), abdomen/pelvis, thorax	Days intensive care unit (ICU) stay (Mean): 9.4 ± 8.2 Ventilator days (Mean): 3.4 ±6.4	SF-36	-	-	1 year after hospital release[$]	Personal, postal or telephone survey; NR	No association between delirium and HRQoL; higher levels of depressive symptoms and symptoms of PTSD associated with lower HRQoL; stronger association of depressive symptoms and HRQoL compared to symptoms of PTSD	NICE: pp: 3, p: 6, m: /, NA: 4, NR[a]: 4, p[#], m[?]
Ahrberg et al., 2014 Germany [60]	Retrospective monocentric cohort study	Definition and influence of delayed foot fracture diagnosis on the overall outcome of polytraumatized patients	N=47 (68.1) Age: Md 39.0 ISS: Md 27.0 AIS: NR Early fracture diagnosis N=26 Age: Md 38 ISS: Md 29.0 AIS: NR Delayed fracture diagnosis N=21 Age: Md 44 ISS: Md 27.0 AIS: NR Injury: Calcaneus, talus, metatarsalia, navicular, cuboid, cuneiformia, phalanges	Conservative und surgical Days ICU stay (Md): 6 Early fracture diagnosis Conservative und surgical Days ICU stay (Md): 8.0 Delayed fracture diagnosis Conservative und surgical Days ICU stay (Md): 4.0	SF-36	-	-	3.5–7.75 years Mean 5.67 ± 1.58 years[$]	Self-administered questionnaire; NR	No significant differences between the two groups in clinical scores; no significant effects of delayed foot fracture diagnosis on the outcome of polytraumatized patients	NICE: pp: 1, p: 7, m: 2, NA: 4, NR[a]: 3, m[#], m[?]

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results	Critical appraisal
					QoL/HRQoL	(Social) Participation	ADL				
Andruszkow et al., 2013 Germany [41]	Retrospective monocentric cohort study	Effects of additional upper extremity injuries or traumatic brain injuries (TBI) on functional, psychological and vocational outcome in polytraumatized patients	Study population of the Hannover Rehab study Injury upper extremity N=229 (75) Age: Mean 30.2 ± 11.5 ISS: Mean 20.2 ± 8.6 AIS: NR Injury upper extremity and TBI N=32 (72) Age: Mean 32.0 ± 11.0 ISS: Mean 32.1 ± 6.4 AIS: NR Injury: Upper extremity with and without TBI, chest, abdomen, spine, lower extremity	NR	HASPOC, SF-12	-	-	10-28 Years, Md 17.5 years[$]	NR	Limitations of functional and vocational outcome due to additional upper extremity injuries in polytraumatized patients; no deterioration of long-term outcome in combined injury of the upper extremity with TBI compared to isolated TBI; no difference in the PCS and MCS of SF-12 in all three groups	NICE: pp: 2, p: 7, m: 1, NA: 4, NR[a]: 3, m[#], m[?]
Archer et al., 2012 USA [71]	Prospective monocentric cohort study	Connection between fear of movement, catastrophising pain, pain and physical health	N=84 (58) Age: Mean 43.1 ± 17.13 ISS: Mean 30.4 ± 9.5 AIS: NR Injury: Abdomen, chest extremities, external, head/face, neck spine	Surgical Days ICU stay (Mean): 9.3 ± 7.7 Ventilator days (Mean): 3.4 ± 6.4	SF-12 (physical component scale)	-	-	2 years after trauma[$]	NR sent by post	Fear of movement and catastrophising pain contributes to poor long-term outcome of polytraumatized patients	NICE: pp: 3, p: 6, m: 1, NA: 5, NR[a]: 2, p[#], m[?]
Attenberger et al., 2012 Switzerland [46]	Prospective monocentric cohort study	Testing TOP module in the acquisition of long-term results of polytraumatized patients; validation by testing the association with selected variables and test results	N=117 (75,2) Age: 13-92, Mean 39.6 ± 20.5, Md 35.4 ISS: 17-45, 27.5 ± 8.2, Md 27.0 AIS[§]: 0-5, Mean 0.5 ± 0.58-2.7 ±1.48, Md 0.0-3.0 Injury: Inter alia brain injury	NR	EQ-5D, SF-36, ALOS, NHP, TOP* (*dimensions 'pain' and 'function', also pre-injury)	-	-	Mean 2.7 ± 0.9 years post trauma[$]	Self-administered questionnaire sent by post	TOP indicated high rate of relevant problems of polytraumatized people; TOP is a reliable and well-discriminating instrument for recording relevant general HRQoL and trauma-specific aspects, which showed high correlation with other generic outcome measures	NICE: pp: 3, p: 6, m: 1, NA: 5, NR[a]: 2, p[#], m[?]

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results	Critical appraisal
					QoL/HRQoL	(Social) Participation	ADL				
		(criterion validity) and comparison of the results with internationally recognized QoL instruments (convergent validity); testing screening procedure of the TOP									
Baranyi et al., 2010 Austria [61]	Retrospective monocentric cohort study	Occurrence of full or partial symptoms of PTSD including psycho-pathologically relevant dimensions and QoL in polytraumatized patients; relation of PTSD symptoms with QoL, social support, depressive mood, and dissociative symptoms	N=52 (73.1) Age: Mean 37.6 ± 14.2 ISS: Mean 33.1 ± 12.61 AIS: NR Injury: Inter alia head	NR	SF-36	-	-	1 year post trauma[$]	Self-administered questionnaire; outpatient clinic	High risk for PTSD and reduced HRQoL in people with polytrauma; lowest HRQoL in patients with full PTSD, followed by patients with partial symptoms and without PTSD; severe dissociative and depressive symptoms; greater limitations in some QoL dimensions	NICE: pp: 7, p: 3, m:/ , NA: 4, NR[a]: 3, pp[#], p[?]
Bouman et al., 2017 Netherlands [37]	Prospective, multi-centre, non-randomized controlled study	Effectiveness of the Fast Track (FT) rehabilitation programme on QoL and other health-related outcomes in comparison	FT Group N=65 (75) Age: 18-75, Mean 44.7 ± 16.7 ISS: 4-66, Mean 22.1 ± 12.8, Md (IQR) 19.5 (± 12-29) AIS: NR Non-FT group	NR	SF-36* (*Baseline (pre-injury status)	-	-	Baseline, 3, 6, 9, 12 months post trauma[&]	Individual interviews, telephone interviews, postal survey; NR	Both groups improved their functional status and QoL; FT group showed faster functional recovery at six months compared to nine month of Non-FT group; no different effects	NICE: pp: 9, p: 7, m: 3, NA: /, NR[a]: 6, m[#], m[?]

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results	Critical appraisal
					QoL/HRQoL	(Social) Participation	ADL				
		with conventional trauma rehabilitation service	N=67 (84) Age: 18-73, Mean 42.0 ± 16.6 ISS: 4-50, Mean 29.4 ± 11.2, Md (IQR) 29 (± 21-38) AIS: NR Injury: Pelvic fracture, musculoskeletal injuries, neuro-trauma/head injuries							observable in both groups at 12 months	
Burghofer et al., 2005 Germany [62]	Prospective multicentric cohort study	Outcome five years after major blunt trauma	N=273 (70.3) Age: Mean 37.5, Md: 34 ISS: 16-25: 29.3% 26-50: 63.0% 51-75: 7.7% AIS: NR Injury: Severe blunt trauma	NR	SF-36	-	-	5 years after trauma[§]	Self-administered questionnaire or, in case of cognitive impairment, administered by relatives; NR	Significant differences in QoL between the trauma cohort and a reference population; relevant decrease of QoL in patients with initial unconsciousness and in severely injured patients; high proportion of patients with persisting pain (headache and lower limb pain); 61.7% were back in occupation, 43.5% of these in the former profession	NICE: pp: 8, p: 2, m: 1, NA: 6, NR[a]: /, p[#], p[?]
Christensen et al., 2011 26 countries in Europe, Asia, Australia, Africa, North- and South-America [76]	Prospective multicentric randomized placebo-controlled study	Identification of HRQoL dimensions most affected by polytrauma and predictors of poor HRQoL	N=347 (75) Age: Mean 39 ± 14 ISS: Mean 31 ± 11, Md 29 AIS[§]: 4+ Injury: Abdomen, chest, external, extremity, face, head or neck	ICU stay (duration NR)	POLO Chart	-	-	3 Months post trauma[§]	Self-administered questionnaire; NR	Notable reduced HRQoL in polytrauma survivors; physical wellbeing more affected than mental; greater mental impairment detectable by trauma-specific HRQoL instruments than by generic tools; demographic and socio-economic characteristics as well as type of injury and treatment predict HRQoL	NICE: pp: 5, p: 5, m: 2, NA: 10, NR[a]: 3, p[#], p[?]

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results	Critical appraisal
					QoL/HRQoL	(Social) Participation	ADL				
Dienstknecht et al., 2013 Germany [42]	Retrospective monocentric cohort study	Functional and socio-economic long-term results in patients with pelvic ring fractures	Study population of the Hannover Rehab study N=109 (72) Age: 5-55, Mean 28.8 ISS: 16-44, Mean 22.7 AIS: NR Injury: Anterior, posterior and combined pelvic ring injuries, upper and lower limb fracture, thoracic injury, abdominal injury, spine fracture, neurological lesion of lower limb	Conservative und surgical	HASPOC, SF-12	-	-	10-27 years post trauma, Mean 16 years[$]	Self-administered questionnaire as well as assessment via physician; examination in the trauma centre	Long-term functional limitations and socio-economic effects of pelvic ring fractures; reduced QoL in patients with anterior and combined anterior and posterior pelvic ring injuries	NICE: pp: 3, p: 6, m: 1, NA: 4, NR[a]: 3, m[#], m[?]
Frénisy et al., 2006 France [81]	Prospective monocentric cohort study	Comparison of neurobehavioral and psycho-pathological disorders between patients with severe brain injuries (SBI) and patients after polytrauma (MULT)	SBI N=25 (80) Age: Mean 26.97 ± 9.58 ISS: Mean 25.36 ± 10.80 AIS: NR Injury: Brain Polytrauma N=25 (72) Age: Mean 33.78 ± 10.42 ISS: Mean 22.04 ± 10.06 AIS: NR Injury: Bone fractures, thorax, abdomen	NR	-	-	-	6-24 months[$] SBI Mean 1.02 ± 0.39[$] MULT Mean 0.93 ± 0.41[$]	Face to face interview; NR	Similar psychopathological disorders in both groups; SBI-patients show significantly more neurobehavioral disorders and more obsessive symptoms; presence of cognitive disorders in MULT-patients	NICE: pp: 3, p: 8, m: /, NA: 4, NR[a]: 2, p[#], m[?]

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					QoL/HRQoL	(Social) Participation	ADL				
Gribnau et al., 2008 Netherlands [33]	Retrospective monocentric cohort study	Injury characteristics, choice of treatment and QoL of people with U-shaped sacral fractures	N=8 (37.5) Age: 19-55, Md 29 ISS: 17-45, Md 23 AIS: NR Injury: Skull fracture, subdural hematoma, facial fracture, thoracic injury (multiple rib fractures, pneumothorax, haemopneumothorax, sternal fracture, lung contusion), retroperitoneal haematoma, liver laceration	Surgical	EQ-6D	-	-	5-65 months post trauma, Md 36 months[$]	Self-administered questionnaire sent by post	Pain, mood disorders and moderate mobility problems dominate QoL	NICE: pp: 2, p: 1, m: 2, NA: 7, NR[a]: 5, m[#], m[?]
Gross et al., 2010 Switzerland [47]	Prospective monocentric cohort study	Long-term work ability of patients after polytrauma; association of capacity to work with internationally accepted variables of functional outcomes or QoL	N=115 (76) Age: 14-92, Mean 39.5 ± 20.6 ISS: Mean 27.5 ± 8.2 AIS: Mean 2.6 ± 1.5 Reduced capacity to work N=61 (79) Age: Mean 39.4 ± 20.2 ISS: Mean 30.2 ± 7.7 AIS[§]: Mean 0.5 ± 0.6-3.1 ± 1.5 Non-reduced capacity to work N=54 (72) Age: Mean 39.6 ± 21.2 ISS: Mean 24.3 ± 7.7 AIS[§]: Mean 0.5 ± 0.5-2.6 ± 1.6 Injury: NR	Days ICU stay (Mean): 8.4 ± 12.5	EQ-5D*, SF-36*, NHP (*Baseline (pre-injury status) assessed in hospital)	-	-	Inpatient stay, Md 2.5 [2.0; 3.3] years post trauma[#]	Self-administered questionnaire sent by post	50% show reduced capacity to work; reduced QoL and persistent pain compared to the pre-polytrauma condition; patient, injury, and treatment variables have impact on capacity to work; association of capacity to work with long-term results such as various functional outcomes, symptom status, and QoL	NICE: pp: 6, p: 4, m: 2, NA: 3, NR[a]: 2, p[#], p[?]

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					QoL/HRQoL	(Social) Participation	ADL				
Gross and Amsler, 2011 Switzerland [48]	Prospective monocentric cohort study	Prevalence and severity of pain after polytrauma compared to pre-injury status using different pain measures; association of pain and HRQoL	N=102 (74) Age: Mean 39.7 ± 20.5 ISS: Mean 27.1 ± 8.0 AIS: NR Injury: NR	NR	EQ-5D*, SF-36*, NHP, TOP (*separate questionnaire for retrospective assessment of pre-injury status)	-	-	Mean 2.7 ± 0.9 years post trauma, Md 2.4 years[$]	Self-administered questionnaire sent by post	Pain in 85% of patients after polytrauma; higher prevalence and severity of pain compared to pre-injury status; close correlation of posttraumatic pain with HRQoL and reduced capacity to work; differences in type and sensitivity of pain measures	NICE: pp: 4, p: 6, m: 1, NA: 4, NR[a]: 2, p[#], p[?]
Gross et al., 2012 Switzerland [49]	Prospective monocentric cohort study	QoL and functional outcomes after polytrauma compared to pre-injury status; discriminatory potential of different outcome scores between survivors with and without TBI	N=111 (75) Age: Mean 39.5 ± 20.9 ISS: Mean 27.9 ± 8.2 AIS[§]: Mean 0.5 ± 0.6-2.7 ± 1.5 Patients with TBI N=45 Age: NR ISS: Mean 31.1 ± 7.3 AIS[§]: Mean 0.7 ± 0.6-3.9 ± 0.7 Patients without TBI N=66 Age: NR ISS: Mean 25.7 ± 8.1 AIS[§]: Mean 0.4 ± 0.6–2.6 ± 1.6 Injury: Inter alia TBI	Days ICU stay (Mean): 8.9 ± 12.6	EQ-5D*, SF-36*, NHP, TOP (*separate questionnaire for retrospective assessment of pre-injury status)	-	-	Mean 2,7 ± 0,9 years post trauma, Md 2,5 years[$]	Self-administered questionnaire sent by post	Notable reduced QoL and functional outcomes compared to pre-injury status; worse outcomes in patients with TBI; SF-36 and NHP discriminate between patients with and without TBI	NICE: pp: 6, p: 5, m: /, NA: 4, NR[a]: 2, p[#], m[?]
Hldaki et al., 2009 Poland [78]	Prospective monocentric cohort study	Comparison of QoL after polytrauma in three consecutive five-year periods using ICIDH	N=827 (71)* Age: Mean 44.7 ISS: Mean 27.6 AIS: NR Injury: Head/neck, thorax, abdomen, pelvis and its organs, spinal column and spinal cord, upper and lower limbs	Surgical	ICIDH (only impairments)	-	-	3 years after completion of inpatient treatment[$]	NR; outpatient clinic	Reduced QoL in 50% of patients after polytrauma; improvement of QoL in one of four patients over 15 years	NICE: pp: 2, p: 2, m: 4, NA: 6, NR[a]: 3, m[#], m[?]

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					QoL/HRQoL	(Social) Participation	ADL				
Holtslag et al., 2006 Netherlands [52]	Prospective monocentric cohort study	Description of the characteristics of lower extremity injuries in patients after polytrauma and functional long-term outcomes using all domains of ICF and HRQoL and their association	N=186 (75) Age: 16-96, Mean 37 ± 17 ISS: Mean 25 ± 11 AIS: NR Injury: lower extremity injuries, pelvic injuries, TBI	Days ICU stay (Mean): 7 ± 14.1 (range: 1-123)	SIP, SF-36	Assess-ment of criteria such as return to sports and work	GARS	10-18 months post trauma, Mean 14 ± 1.6 months, Md 14 months[$]	Self-administered questionnaire (if necessary, administered by relatives) sent by post	Serious long-term problems of mobility, activity, and participation; mild to moderately limited HRQoL; low association of ICF domain functionality with HRQoL; limited concordance between instruments	NICE: pp: 3, p: 7, m: 1, NA: 4, NR[a]: 2, p[#], p[?]
Holtslag et al., 2007a Netherlands [50]	Prospective monocentric cohort study	Long-term functional health status in patients after polytrauma; relationship between personal and injury-related characteristics vs. long-term HRQoL	N=335 (74) Age: Mean 37.7 ± 17.1 ISS: Mean 24.9 ± 10.6 AIS: NR Injury: Cranial, facial, upper extremity, chest, abdomen, pelvis, femur, lower leg, spine, spinal cord, peripheral nerve	Days ICU stay (Mean): 12.9 ± 18.7	SIP	-	-	12-18 months post trauma, Mean 451 ± 47 days[$]	Self-administered questionnaire sent by post	Long-term mild to moderately limited HRQoL; lower QoL in young male patients and patients with TBI, spinal cord and extremity injuries; greatest limitations in some SIP categories (work, ambulation, home management, recreation and pastimes, alertness behaviour); co-morbidity strong predictor of SIP scores	NICE: pp: 8, p: 2, m: /, NA: 5, NR[a]: 2, p[#], p[?]
Holtslag et al., 2007b Netherlands [51]	Prospective monocentric cohort study	Long-term functional consequences from polytrauma, quantification of the influence of socio-demographic, injury-related and physical	N=335 (74) Age: Mean 37.7 ± 17.1 ISS: Mean 24.9 ± 10.6 AIS: NR Injury: Brain, chest, abdomen, spinal cord, lower extremity, upper extremity	Days ICU stay (Mean): 12.9 ± 18.7	EQ-5D	-	-	12-18 months post trauma, Mean 451 ± 47 days[$]	Self-administered questionnaire (if necessary, administered by relatives) sent by post	Restriction of mobility, self-care and everyday activities, indication of pain and discomfort, anxiety and depression as well as cognitive complaints; localisation of injury, educational level and co-morbidities are important independent predictors of long-term functional	NICE: pp: 8, p: 1, m: /, NA: 5, NRa: 3, p#, p?

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results	Critical appraisal
					QoL/HRQoL	(Social) Participation	ADL				
		factors on outcomes								outcomes after polytrauma; better QoL via EQ-VAS than via utility score	
Holtslag et al., 2008 Netherlands [53]	Prospective monocentric cohort study	Impact of trauma on health in terms of burden of injury, years lived with disability (YLD) in non-fatal diseases, years of life lost (YLL) in fatal diseases and disability-adjusted life year (DALY)	Fatalities: N=567 (62.6) Age: Mean 48.4 ± 23.3 ISS: >15 AIS: NR Injury: NR Severely injured: 335 (74.3) Age: Mean 37.7 ± 17.1 ISS: >15 AIS: NR Injury: NR	NR	EQ-5D	-	-	12-18 months after injury, Mean 451 ± 47 days[$]	NR	Substantial loss of healthy life years; at individual level, major trauma contributed an average of 25 DALYs to the burden of disease; at population level, major trauma caused 10 DALYs per 1000 inhabitants	NICE: pp: 5, p: 8, m: /, NA: 4, NRa: /, pp#, pp?
Jackson et al. 2007 USA [63]	Retrospective monocentric cohort study	Prevalence of and risk factors for persistent cognitive impairment as well as emotional and functional difficulties in people without intracranial haemorrhage	N=58 (67) Age: Mean 45.0 ± 14.3 ISS: 33.0 ± 6.8 AIS: NR Injury: Concussion, skull fracture	NR	SF-36	-	FAQ, KATZ ADL	12-24 months after hospital discharge[$]	Administered by a neuro-psychologist or a psychometrist; evaluation at medical centre or place of residence	Substantial prevalence of persistent cognitive impairment, emotional dysfunction and functional decrements, most of all in people with concussion and skull fracture compared to those without these injuries; cognitive impairment was associated with functional defects, poor QoL, and an inability to return to work	NICE: pp: 5, p: 7, m: /, NA: 4, NR[a]: 1, p[#], p[?]

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					QoL/HRQoL	(Social) Participation	ADL				
Kaske et al., 2014 Germany [77]	Retrospective monocentric cohort study	Assessment of HRQoL using a trauma-specific instrument (POLO Chart) in patients after polytrauma	N=129 (66.9) Age: Mean 43.9 ±17.5 ISS: Mean 21.5 ±13.3 AIS: ≥ 2 Injury: Lower extremity, thorax, upper extremity, head, spine, pelvis, abdomen	Days ICU stay (Mean): 9.2 ± 9.6, Md (IQR_{25-75}): 5 (2-14) Ventilator days (Mean): 8.2 ± 7.6, Md (IQR_{25-75}): 5 (2-14)	POLO Chart	-	-	Mean 24 months ± 6 months post trauma[$]	Self-administered questionnaire sent by post	Physical impairments, mental and socio-economic deficits in patients after polytrauma; reduced HRQoL in 64% of patients two years after polytrauma, in half of these patients severely impaired HRQoL; TOP is able to identify impairments and to guide rehabilitation resources	NICE: pp: 2, p: 5, m: 2, NA: 6, NR[a]: 2, m[#], m[?]
Lefering et al., 2012 Germany and Austria [38]	Diagnostic study	Phase IV validation of the trauma-specific TOP module as part of the POLO Chart	Trauma-Group N=172 (73.8) Age: Mean 38.8 ± 14.4 ISS: 8-75, Mean 26.7 ± 11.2 AIS: NR Control Group N=166 (65.1) Age: Mean 42.9 ± 15.0 ISS: 1-9, Mean 3.9 ± 1.7 AIS: NR Injury: NR	NR	POLO Chart	-	Barthel Index	Trauma Group Mean 24.2 ± 18.7 months post trauma[$] Control Group Mean 26.4 ± 14.2 months post trauma[$]	NR; visit to the hospital for a physical examination and to fill in the questionnaires	Significant differences in all QoL dimensions between trauma and control group; apart from physical domains, also impairment of other dimensions of QoL; results show good discriminatory abilities of the TOP and support its precision in collecting all relevant components of QoL in patients after polytrauma	SIGN: y: 8, n: 3, cs: 2, p*: 1, da: 1
Lippert-Grüner et al., 2007 Germany [64]	Prospective cohort study	HRQoL in patients with severe TBI, with and without concomitant polytrauma	N=49 (78) Age: 15-68, Md 32 ISS: NR AIS: NR Isolated TBI N=28 Age: Mean 36 ISS: Mean 27 AIS: NR TBI and Polytrauma N=21 Age: Mean 26 ISS: Mean 40	ICU (duration NR)	SF-36	-	-	6 and 12 months post trauma[#]	Self-administered questionnaire; during outpatient follow-up examinations	Comparable results for patients with severe TBI with and without concomitant polytrauma in SF-36 with the exception of physical function; indication of TBI as an important factor for HRQoL and outcomes after trauma; improvement of HRQoL over time	NICE: pp: 3, p: 4, m: 1, NA: 4, NR[a]: 5, p[#], m[?]

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					QoL/HRQoL	(Social) Participation	ADL				
			AIS: >3 Injury: Cranial injury								
Livingston et al., 2009 USA [80]	Prospective monocentric cohort study	Long-term outcome of patients with prolonged (>10 days) ICU stay	N=100 (81) Age: 18-94, Mean/Md 42 ISS: 26-30, Mean/Md 28 AIS: 3-5 Injury: TBI, extremity fracture, pelvis fracture, lower extremity fracture	Surgical Days ICU stay: 19-24, Mean 21, Md 19 Ventilator days: Mean 16, Md 14	-	-	-	2.1-5.0 years, Mean/Md 3.3 years[$]	Telephone interview with patient	Overall self-reported QoL is generally good with significant impairments incl. inability to return to work (less than 50%) or regain pre-injury level of activity and reintegration into society	NICE: pp: 1, p: 8, m: 1, NA: 4, NR[a]: 3, m[#], m[?]
Marasco et al., 2015 Australia [36]	Retrospective monocentric cohort study	Long-term morbidity and QoL in patients with rib fractures	N=397 (75) Age: Mean 53.9 ± 18.8 ISS: Mean 22.5 ± 11.8 AIS: NR Thoracic-Group N=216 Age: NR ISS: Mean 16.0 ± 7.3 AIS: max. 3 Polytrauma-Group N=181 Age: NR ISS: Mean 30.1 ± 11.6 AIS: max. 4 Injury: Head, neck, face, spine, thorax, abdomen, upper and lower extremity	NR	SF-12	-	-	6, 12, 24 months post trauma[!]	Telephone interview with patient (if necessary, with proxy)	Significantly reduced QoL 24 months after multiple rib fractures; low return to work rate after 6 months; no differences between the groups in pain and QoL; limited function of the thorax group	
Mc Carthy et al., 2006 USA [34]	Retrospective monocentric cohort study	Subjective psychosocial health of a population-based sample of adults with traumatic brain injury (TBI)	N=7612 (63,9) Age: 15-75+, Mean 43.2 ± 20.0 ISS: 4-18+, Mean 13.7 ± 7.6 AIS: 2-5 Injury: TBI	NR	SF-36 (psychosocial health scales MCS)	-	Assessment of difficulty regarding bathing/ showering, dressing, eating, transferring to a bed/chair,	1 year after injury ± 3 month[$]	Telephone interview	29% reported poor psychosocial health 1 year after injury; less than 50% received mental health services; a high proportion reported unmet needs for mental health services	

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					QoL/HRQoL	(Social) Participation	ADL				
							walking, using the toilet				
Ouellet et al., 2009 Canada [54]	Retrospective multicentric cohort study	Comparison of mental health in patients with polytrauma with and without TBI; factors associated with low mental health; perceived need for and access to mental health-related services	N=405 (64.6) Age: Mean 41.1 ± 15.1 ISS: Mean 23.6 ± 11.7 AIS: NR Patients with TBI N=239 (64.9) Age: Mean 37.4 ± 14.5 ISS: Mean 27.9 ± 9.1 AIS: NR Patients without TBI N=166 (61.0) Age: Mean 46.4 ± 14.5 ISS: Mean 15.7 ± 8.4 AIS: NR Injury: TBI, spine, upper limb, lower limb	Days ICU stay (mean): TBI: 10.3 ± 14.3 Non-TBI: 4.2 ± 8.2	SF-12, Version 2 (MCS as well as all subscales which are related to mental health)	-	-	2-4 years post Trauma[$] TBI: Mean 3.3 ± 0.7 years post Trauma[$] without TBI: Mean 3.3 ± 0.8 years post Trauma[$]	Telephone interview with patients, NR	Not significantly lower MCS-scores in polytrauma patients with TBI compared to patients without TBI; association of 6 factors with low mental health: age, sex, time since trauma, social support, pain and cognitive deficits; high need for mental health-related services but considerable difficulties in maintaining it	
Overgaard et al., 2011 Denmark [65]	Prospective cross-sectional monocentric cohort study	Outcome six to nine years after moderate to severe injury in terms of survival, HRQoL and employment status	N= 322 (71) Age: 15-89, Md 34 ISS: 9-75, Md 17 AIS: NR Injury: Blunt trauma	Surgical Days ICU stay: 1-62, Md 4	SF-36	-	-	6-9 years after admission, Md 7.4[$]	Self-administered questionnaire sent by post or telephone interview	78% of the patients survived the trauma; HRQOL was significantly lower for injured patients than a matched control group; 20% retired early	
Post et al., 2006 Netherlands [11]	Retrospective monocentric cohort study	Return-to-work status and QoL in patients after polytrauma; analysis of injury-related parameters in relation to functional outcome	N=53 (81) Age: 19-64, Mean 37.3 ± 13.2, Md 34.0 ISS: 16-54, Mean 23.5 ± 8.2, Md 21.0 AIS: Md 0-5, 2.0–3.0 Injury: Head, neck, face, thorax, abdomen, extremities, external	NR	SIP	-	-	1.3-2.2 years post trauma, Mean 1.8± 0.3 years, Md 1.8 years[$]	Self-administered questionnaire sent by post	Most patients after polytrauma return to their former work; compared with general population and less severely injured patients good QoL results; age, but not trauma severity or TBI is a predictor for long-term disablement	

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					QoL/HRQoL	(Social) Participation	ADL				
Probst et al., 2010 Germany [43]	Retrospective monocentric cohort study	Association of improved short-term outcomes after polytrauma with gender differences; long-term follow-up using QoL and rehabilitation status	Study population of the Hannover Rehab study N=637 (75.4) Age: M: Mean 26.7 ± 12.2 F: Mean 25.9 ± 13.0 ISS: M: Mean 20.4 ± 9.9 F: NR, 21.7 ± 9.7 AIS[§]: M: Md 1-5 F: Md 1-6 Injury: Head, face/neck, thorax, abdomen, skeletal system	Days ICU stay (Mean): M: 15.4 ± 25.0 F: 12.9 ± 23.2 Ventilator days (Mean): M: 11.2 ± 13.3 F: 9.9 ± 12.5	HASPOC, SF-12	-	-	M: 17.5 ± 4.8 years post Trauma[$], F: 17.4 ± 5.0 after trauma[$]	Interview with patients; assessment via physician and physical examination in the trauma centre	After polytrauma, more severe long-term limitations in women than men; no differences in rehabilitation status; longer sick leave times, significantly lower QoL and higher rates of PTSD and psychological support in women	
Quale et al., 2009 Norway [82]	Prospective monocentric cohort study	Incidence and risk factors of posttraumatic stress symptoms (PTSS)	N= 79 (74.7) Age: Mean 37 ± 14.3 ISS: Mean 25.6 ± 12.3 AIS: NR Injury: Spinal cord, head	Days ICU stay (Mean): 62 ± 61.4 Ventilator days (Mean): 5.8 ± 13.1	-	-	-	1-3 weeks after admittance to rehabilitation, median days since injury 39, quartiles 24-57[$]	Semistructured interviews, questionnaires; NR	Low incidence of PTSD, but higher incidence of PTSS; anxiety, female gender, negative attitudes towards emotional expression are risk factors for PTSS	
Renovell-Ferrer et al., 2017 Spain [66]	Prospective monocentric cohort study	Comparison of aetiology, severity, and functional outcome of people with operated displaced intra-articular calcaneal fractures between polytrauma patients and isolated cases	N=80 (71) Age: 16-74, Mean 48.4, Md 47 ISS: >16 (14%) AIS: NR Injury: Intra-articular calcaneal fractures, pelvic fracture, spine fracture	Surgical	SF-36	-	-	15-62 months after operation, Mean 48 months[$]	NR	No differences in outcome measures and second surgeries; increasing severity of trauma was associated with more second surgeries and worse outcomes; patients with psychiatric comorbidities reported worse HRQoL	NICE: pp: 1, p: 9, m: 1, NA: 2, NR[a]: 5, m[#], m[?]

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					QoL/HRQoL	(Social) Participation	ADL				
Richter et al., 2006 Germany [83]	Prospective monocentric cohort study	Prevalence of PTSD; identification of risk factors for PTSD	N=37 (76) Age: Mean 41.7 ± 17.0 ISS: Mean 30.4 ± 10.7 AIS: NR Injury: NR	Days ICU stay: ≥ 30	-	-	-	7-58 months after discharge from ICU, Mean 35 ± 14 months[$]	Self-administered questionnaire and interview; NR	Compared to non-trauma patients, higher risk for PTSD in trauma patients; ISS and ICU treatment not associated with PTSD	NICE: pp: 5, p: x4 m: 2, NA: 4, NR[a]: 2, m[#], m[?]
Ringburg et al., 2011 Netherlands [75]	Prospective monocentric cohort study	HRQoL in patients after polytrauma; predictive factors for long-term poor functioning in the dimensions of HRQoL instruments	N=246 (66) Age: Md 40 ISS: Md 22 AIS: NR Injury: Head, face, chest, abdomen, extremities	NR	EQ-5D, HUI 2, HUI 3	-	-	12 months post trauma[$]	Self-administered questionnaire sent by post	HRQoL far below general population norms; high prevalence of specific limitations with female gender and comorbidity as predictors of long-term disability	NICE: pp: 6, p: 6, m: /, NA: 4, NR[a]: 1, p[#], p[?]
Sampalis et al. 2006 Canada [67]	Retrospective multicentric cross-sectional cohort study	Functional status and QoL 12 months after injury	N=144 (61.8) Age: 18-85, Mean 44,2 ±17.1 ISS: 1-49, Mean 18.9 ± 9.4 AIS: NR Injury: Head/ neck, extremity, face, thorax, abdominal, burns	Surgical and conservative Days ICU stay: 0.1-11.0, Mean 3.44 ± 0.9	SF-36, SIP	-	-	0.8-1.5 years, Mean 1.3 years post trauma[$]	Self-administered questionnaire sent by post	Residual impairments in functional capacity and QoL 1 year after injury; specific subgroups (injuries resulting of motor vehicle collisions, thoracic and head injuries) report diminished QoL; ICU treatment, prolonged length of hospital stay, and surgical treatment are predictors for poor functional capacity and reduced QoL	NICE: pp: 6, p: 7, m: 1, NA: 3, NR[a]: /, p[#], p[?]
Schmidt-Rohlfing et al., 2011 Germany [44]	Retrospective monocentric cohort study	Correlation between scoring systems focusing on QoL and those focusing on functional status of knee joint	Study population of the Hannover Rehab study Patients with knee injury N=48 (73) Age: Mean 29.7 ± 11.8 ISS: Mean 22 ± 9.8 AIS: NR	Surgical	HASPOC, SF-12	-	-	Mean 16.9 ± 4.9 years post trauma[$]	Assessment via physician, NR	No correlation between scoring systems for functional knee condition and QoL; poorer outcome for patients with knee injuries compared to	NICE: pp: 1, p: 4, m: 3, NA: 5, NR[a]: 4, m[#], m[?]

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					QoL/HRQoL	(Social) Participation	ADL				
			Patients with femur fracture N=107 (72) Age: Mean 26.8 ±13.9 ISS: Mean 23.3 AIS: NR Injury: Knee, femoral fracture, head							patients with femoral fractures	
Simmel et al., 2013 Germany [56]	Retrospective monocentric cohort study	QoL and health status many years after polytrauma and influencing factors	N=127 (76) Age: Mean 36 ISS: 26-75, Mean 35.6 AIS: NR Injury: NR	Days ICU stay (Mean): 21.1	POLO Chart	-	-	38-108 months post trauma, Mean 70 months[$]	Self-administered questionnaire sent by post	Significantly reduced QoL and health status after polytrauma; higher age, female gender, low education and chronic previous illnesses have a negative impact on outcomes as well as difficulties with authorities/institutions, unemployment, long inpatient stays and subjectively inadequate inpatient treatment	NICE: pp: 8, p: 3, m: /, NA: 4, NR[a]: 2, p[#], p[?]
Sirois et al., 2009 Canada [55]	Retrospective multicentric cohort study	Regional differences in perceived need for and barriers to postacute rehabilitation services; long-term functional outcomes and physical health outcomes in patients after polytrauma	N=435 (64.5) Age: 18-65, Mean 41.1 ± 15.1 ISS: 1–75 AIS: NR Injury: Spine, brain, extremity	NR	SF-12, Version 2 (PCS and subscales physical function, role function and general health perception)	-	-	2-4 years, Mean 3.3 ± 0.7 years post trauma[$]	Telephone interview with patient (if necessary, with proxy)	No regional differences in perceived need and barriers to postacute rehabilitation services; no regional differences in long-term functional outcomes and physical health in patients after polytrauma, with the exception of a slightly lower score in SF-12 subscale physical functioning in regions with limited availability of rehabilitation services	NICE: pp: 5, p: 5, m: /, NA: 4, NR[a]: 3, p[#], p[?]

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results	Critical appraisal
					QoL/HRQoL	(Social) Participation	ADL				
Sluys et al., 2005 Sweden [68]	Retrospective monocentric cohort study	Outcome and QoL five years after injury and identification of factors potentially associated with outcome and QoL	N=205 (74) Age: 20-87, Md 39 ISS: 9-57, Md 14 AIS: 3-5 Injury: Extremities, pelvic girdle, head and neck, chest, external/skin, abdominal or pelvic content, face	Surgical, conservative Days ICU stay: 1-86, Mean 4	SF-36	-	-	5 years post trauma[§]	Self-administered questionnaire sent by post, or telephone interview	Relevant decrease in HRQoL and permanent impairments as well as disabilities in most of the affected. Length of hospital stay, days spent on ICU, surgical procedures, in-hospital major complications, age, recurrent injury, and inadequate information are risk factors for poor HRQoL; injury severity did not predict poor HRQoL	NICE: pp: 5, p: 6, m: 3, NA: 3, NR[a]: /, m[#], m[?]
Soberg et al., 2007 Norway [58]	Prospective monocentric cohort study	Complete and partial return to work rate; prognostic factors for complete return to work after polytrauma	N=100 (82) Age: 18-67, 34.5 ± 13.5 ISS: Mean 28.1 ±11.3 AIS: NR Injury: Head/neck, face chest, abdomen/pelvic content, extremity, external, spinal cord	Ventilator days (median): 5 (IR 9)	SF-36	-	-	20.0 ± 9.5 weeks, 1 year and 2 years post trauma[!]	Self-administered questionnaire sent by post, telephone interview of 2 patients	Complete return to work: four out of 10 patients two years after polytrauma, 2/3 actively attempting to return to work. Lower education, low social functioning and > 20 weeks in hospital or rehabilitation are risk factors for not returning to work; tendentially better mental health in returnees than in non-returnees to work, overall lower mental sum scores than for general population; physical health one year after trauma significantly lower than for general population	NICE: pp: 8, p: 2, m: /, NA: 5, NR[a]: 2, p[#], p[?]

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results	Critical appraisal
					QoL/HRQoL	(Social) Participation	ADL				
Soberg et al., 2011 Norway [57]	Prospective monocentric cohort study	Work status and trajectory of return to work after polytrauma; prognostic factors for the return to work of patients after polytrauma over five years	N=75 (83) Age: 18-67, 34.5 ± 13.5 ISS: NRTW-Group Mean 28.9 ± 12.5 RTW-Group Mean 25.8 ± 10.8 AIS: NR Injury: Head (TBI), face chest, abdomen/pelvic content, spinal, upper extremity, lower extremity	NR	SF-36	-	-	20.0 ± 9.5 weeks, 1 year, 2 years and 5 years post trauma[?]	Self-administered questionnaire sent by post	Complete return to work of 50% of patients after polytrauma over 5 years, 23% receive disability benefits. Higher education, better physical, social and cognitive functioning and use of coping strategies are prognostic factors for returning to work; generally better physical and mental functioning after trauma in returnees to work than in non-returnees	NICE: pp: 7, p: 6, m: /, NA: 3, NR[a]: 1, p[#], p[?]
Stulemeijer et al., 2006 Netherlands [79]	Prospective monocentric cohort study	Impact of additional extracranial injuries on functional outcomes and on post-concussion symptoms in patients with mild TBI	Isolated TBI N=210 (63) Age: Mean 35.2 ± 12.3 ISS: 1-17, Mean 4.7 ± 3.4 AIS (head): 1-4. Mean 1.9 ± 0.7 TBI and extracranial injuries N=89 (74) Age: Mean 37.9 ± 12.0 ISS: 5-41, Mean 14.5 ± 7.4 AIS (head): 1-5, Mean 2.3 ± 0.8 Control Group N=261 (45) Age, ISS, AIS: NR Injury: TBI, extremities, face, chest, abdomen, external	NR	-	-	SF-36 (sub-scales of physical and social functioning)	6 Months post trauma[$]	Self-administered questionnaire sent by post	Poorer functional outcomes (i.e. lower levels of physical function, lower rates of return to work and generally worse outcomes) of patients with additional extracranial injuries and mild TBI but equal level of post-concussion symptoms; greater impairment in patients after polytrauma compared to the control group with mild injuries	NICE: pp: 3, p: 7, m: /, NA: 5, NR[a]: 2, p[#], m[?]

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results	Critical appraisal
					QoL/HRQoL	(Social) Participation	ADL				
Suzuki et al., 2007 Japan [69]	Retrospective monocentric cohort study	Evaluation of long-term functional outcome of patients with unstable pelvic ring fractures and correlation with other factors	N=57 (49) Age: 16-72, Mean 42 ISS: 9-50, Mean 24.6 AIS: NR Injury: Pelvic fracture	Surgical and conservative	SF-36	-	-	24.4-107.4 months, Mean 47.2 months[$]	Questionnaire; during clinical reevaluation	Functional outcome was not associated with ISS, fracture location or type of fracture; neurologic injury correlated with poor functional outcome	NICE: pp: 4, p: 9, m: 1, NA: 3, NR[a]: /, p[#], m[?]
Tee et al., 2014 Australia [72]	Prospective monocentric cohort study	Early predictors of suboptimal health status in patients after polytrauma with spine injuries	N: 479 (71.8) Age: Mean 47.5 ± 19.1 ISS: Md14 AIS: NR Injury: TBI, spine	NR	SF-12	-	-	1 year post trauma[$]	NR	Early risk factors of suboptimal physical health or QoL are tachycardia, hyperglycemia, multiple chronic pre-existing medical comorbidities and thoracic spine injuries	NICE: pp: 7, p: 3, m: /, NA: 4, NR[a]: 3, p[#], m[?]
Van Delft-Schreurs et al., 2016 Netherlands [74]	Retrospective monocentric cross-sectional cohort study	Structure and psychometric properties of the Dutch SMFA questionnaire in severely injured patients	N=173 (69) Age: Mean 46 ± 19 ISS: Md 21 AIS: ≥ 2 Injury: Upper and lower extremity, TBI	NR	SMFA, WHOQOL-BREF	-	-	1.3-4.4 years post trauma[$]	Self-administered questionnaire sent by post	Good psychometric properties of the Dutch SMFA in severely injured people and strong correlations between the SMFA and WHOQOL-BREF; Dutch SMFA seems to be valid and useful for generating an overview of physical limitations and emotional problems	SIGN: y: 7, n: 1, cs: 5, p*: 1, si: 1
Victorson et al., 2008 USA [84]	Mixed method monocentric study	Development and validation of the injury distress index (IDI)	N=169 (66.1) Age: Mean 34.5±11.1 ISS: NR AIS: 1-5 Injury: Lower extremity, upper extremity, burn (21%)	Surgical, conservative	-	-	-	Mean 8 days after admission[$]	Self-administered questionnaire; trauma centre	IDI showed acceptable reliability and validity	MMAT: y: 8, n: 2, ct: 2

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results	Critical appraisal
					QoL/HRQoL	(Social) Participation	ADL				
Vles et al., 2005 Netherlands [7]	Prospective monocentric cohort study	Prevalence and determinants of disabilities and return to work after severe injury	N=166 (81) Age: Men: Mean 33 Women: Mean 37 ISS: Men: Mean 23 Women: Mean 23 AIS: ≥ 2 Injury: Cranium, brain, extremities, spine, pelvis, chest, thoracic contents, abdomen, pelvic contents	NR	EQ-5D	-	Change of daily activities and sports	12-69 months, Mean/Md 41 months (3.4 years) after trauma[$]	Self-administered questionnaire sent by post, or telephone interview	Substantial impact of trauma on long-term functioning; follow-up between 12 and 24 months might be suitable for outcome assessment including quality of treatment	NICE: pp: 5, p: 6, m: 3, NA: 3, NR[a]: /, p[#], m[?]
von Rüden et al., 2013 Germany [39]	Retrospective monocentric cohort study	Prognosis, incidence, trauma mechanism, mortality, invalidity, working ability/disability and QoL in patients after polytrauma with ISS ≥ 50	N=88 (77.3) Age: 18-63, Mean 40.3 ± 17.4 ISS: Mean 56.8 AIS[§]: Mean 1.7-4.4 Injury: Head/brain, face, chest, abdomen, spinal cord, pelvis, upper limb, lower limb, soft tissues	Days ICU stay (Mean): ISS <50: 14.4 ISS >50: 25.9 Ventilator days (Mean): ISS <50: 13.4 ISS >50: 20.3	POLO Chart	-	-	18-78 months post trauma, Mean 3.6 years[$]	Self-administered questionnaire sent by post; face to face interview with subgroup	Good prognosis for severely injured persons, but QoL not satisfactory mainly due to psychological problems and chronic pain; 6.5% with ISS ≥;50; main cause: motorbike accident; 36% mortality rate; 57% limited working ability	NICE: pp: 3, p: 5, m: 2, NA: 4, NR[a]: 3, m[#], m[?]
Williamson et al., 2009 Australia [73]	Prospective multicentric cohort study	Predictors of moderate or severe pain six months after injury	N=1290 (61) Age: 14-95 ISS: >15 AIS: NR Injury: Extremities, spinal injuries	NR	SF-12 (only on discharge rating of status in the week before injury)	-	-	6.1-7.1 months after trauma (Md 6.5); before discharge, 6 months (assessment of other PROs than HRQoL)[#]	Face to face interview in acute hospital before discharge and during follow-up by telephone at home	Prevalence of moderate or severe pain at follow-up was 30%; predictors of pain were educational and compensation status, preinjury pain-related disability, pain intensity at discharge	NICE: pp: 7, p: 3, m: /, NA: 4, NR[a]: 3, p[#], p[?]

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results	Critical appraisal
					QoL/HRQoL	(Social) Participation	ADL				
Wurm et al., 2012 Germany [40]	Retrospective monocentric cohort study	Posttraumatic QoL in patients after polytrauma with ISS ≥ 50	N=88 (77) Age: Mean 40.3 ± 17.4 ISS: Mean 56.8 AIS[§]: Mean 3.9-4.4 Injury: Thorax, TBI, abdomen, spine, lower extremity	Days ICU stay (Mean): 23 Ventilator days (Mean): 20	POLO Chart	-	-	18-78 months post trauma, Mean 3.6 years[$]	Self-administered questionnaire sent by post and personal interview with subgroup	Good survival rate but poor posttraumatic QoL mainly due to psychological problems and chronic pain	NICE: pp: 3, p: 5, m: 1, NA: 5, NR[a]: 3, m[#], m[?]
Zelle et al., 2005a Germany [45]	Retrospective monocentric cohort study	Comparison of objective and subjective outcome of patients after polytrauma with injuries below the knee joint and injuries of the lower extremity above the knee joint	Study population of the Hannover Rehab study N=389 (74) Age: Mean 25.4 ± 11.7 ISS: Mean 20.2 ± 9.3 AIS: NR Injury: Lower extremity	NR	HASPOC, SF-12	-	-	Mean 17.3 ± 4.8 years post trauma[$]	Self-administered questionnaire as well as assessment via physician; examination in the trauma centre	Significantly better objective and subjective outcomes of patients after polytrauma with injuries above the knee joint compared to injuries below the knee joint; no significant differences in the MCS of SF-12	NICE: pp: 4, p: 6, m: 1, NA: 4, NR[a]: 2, m[#], m[?]
Zelle et al., 2005b Germany [35]	Retrospective monocentric cohort study	Comparison of long-term functional outcome after polytrauma between workers´ compensation patients (WCP) and non-workers´ compensation patients (N-WCP)	Study population of the Hannover Rehab study N=637 (75) Age: 3-60, Mean 26.4 ISS: 4-54, Mean 20.7 ± 9.7 AIS: NR WCP N=233 (82.0) Age: Mean 29.78 ± 12.09 ISS: Mean 20.60 ± 10.00 AIS: NR N-WCP N=404 (71.5) Age: Mean 24.56 ± 12.12 ISS: Mean 20.80 ± 9.61	NR	HASPOC, SF-12	-	-	10-28 years, Mean 17.5 years[$] WCP: Mean 17.36 ± 4.77[$] N-WCP: Mean 17.0 ± 4.89[$]	Self-administered questionnaire as well as assessment via physician; examination in the trauma centre	Worse subjective and objective outcomes in WCP; no significant differences in the SF-12 MCS	NICE: pp: 8, p: 3, m: /, NA: 4, NR[a]: 2, p[#], p[?]

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results	Critical appraisal
					QoL/HRQoL	(Social) Participation	ADL				
			AIS: NR Injury: Lower extremity, head, face, upper extremity, chest, abdomen, spine, neck, skin								
Zwingmann et al., 2016 Germany [70]	Retrospective monocentric cohort study	Long-term HRQoL and changes in daily life of patients after polytrauma	N=147 (75.5) Age: 6.1-82.1, Mean 40 ± 19 ISS: 17-59, Mean 28 ±10.6 AIS: NR Injury: NR	Days ICU stay (Mean): 7.8 ± 7.5	EQ-5D, SF-36, TOP	-	-	5.0-7.4 years post trauma, Mean 6 ± 0,8 years[$]	NR, verbal interview and/or written questionnaire sent by post	Impaired QoL due to pain and functional disability; association of negative socio-economic effects and emotional repercussion	NICE: pp: 3, p: 3, m: 4, NA: 5, NR[a]: 2, m[#], m[?]

Indices and Abbreviations:

AIS[§]: AIS values shown for various body regions; F: Female; IQR/IR: Interquartile range; M: Male; max.: highest value; MCS: Mental Component Summary score; NR: not reported; Md: Median; N: Number; NRTW: Non-return to work; PCS: Physical Component Summary score; QoL: Quality of Life; RTW: Return to work; SD: Standard deviation; *: Numbers displayed for Overall-Characteristic of study population; [$]: one measurement; [#]: two measurements; [!]: three measurements; [?]: four measurements; [&]: five measurements

Critical appraisal:

n: no; y: yes

MMAT: ct: can´t tell – statement not possible;

NICE: m: there are significant sources of bias.; m[#]: Evaluation of internal validity of entire study – few or no checklist criteria were fulfilled; m[?]: Evaluation of the study's overall external validity – few or no checklist criteria were fulfilled; NA: not applicable – aspects of the study design that are not applicable; NR[a]: not reported – the study does not report how certain aspects have been or should be taken into account; p: either the answer to the checklist question cannot be clearly answered by the way the study is reported, or the study may not have considered all potential sources of bias; p[#]: Evaluation of internal validity of entire study – some checklist criteria were fulfilled; p[?]: Evaluation of the study's overall external validity – some checklist criteria were fulfilled; pp: study was designed or conducted to minimize risk of bias; pp[#]: Evaluation of the study's overall internal validity – all or most checklist criteria were fulfilled; pp[?]: Evaluation of the study's overall external validity – all or most checklist criteria were fulfilled

SIGN: cs: can´t say– statement not possible; da: study directly applicable to target population; p*: Most criteria met. Some flaws in study with an associated risk of bias; si: some indirectness in applicability to target population

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Instruments used to assess HRQoL

In total, 15 different tools were used to assess HRQoL during follow-up after polytrauma. Of those, six were generic instruments: Short Form-36 Health Survey (SF-36, n=21, 39%),[34, 37, 46-49, 52, 57-70] 12-Item Short Form Health Survey (SF-12, n=12, 22%),[35, 36, 41-45, 54, 55, 71-73] Nottingham Health Profile (NHP, n=4, 7%),[46-49] Sickness Impact Profile (SIP, n=4, 7%),[11, 50, 52, 67] Short Musculoskeletal Function Assessment (SMFA, n=1, 2%),[74] and World Health Organization Quality of Life Instrument – Short Form (WHOQOL-BREF, n=1, 2%).[74] In addition, four preference-based tools were applied: European Quality of Life 5-Dimension (EQ-5D, n=9, 17%),[7, 46-49, 51, 53, 70, 75] EQ-6D (n=1, 2%),[33] Health Utility Index Mark 2 (HUI 2, n=1, 2%),[75] and HUI 3 (n=1, 2%).[75] Three trauma-specific instruments were used: Hannover Score for Polytrauma Outcome (HASPOC, n=6, 11%),[35, 41-45] Trauma Outcome Profile (TOP, (n=4, 7%),[46, 48, 49, 70] and Aachen Long-term Outcome Score (ALOS, n=1, 2%).[46] The Polytrauma Outcome Chart (POLO Chart, n=6, 11%) [38-40, 56, 76, 77] was used as a modular instrument consisting of the Glasgow Outcome Profile, SF-36, EQ-5D, and TOP. The International Classification of Impairments, Disabilities and Handicaps (ICIDH) was reported in one publication (n=1, 2%).[78] Fifteen publications combined two to five different instruments to assess HRQoL.[35, 41-49, 52, 67, 70, 74, 75]

In some publications (n=5), instruments to assess HRQoL were only used partially (e.g. solely the mental or physical part, or selected subscales of the SF-36 or SF-12, chapter 'Disability' of the ICIDH).[34, 54, 55, 71, 78] Six publications conducted a retrospective collection of HRQoL via SF-36, SF-12, EQ-5D and TOP capturing the pre-trauma status.[37, 46-49, 73]

Instruments used to assess ADL

ADL was measured by different generic instruments: Groningen Activity Restriction Scale (GARS),[52] Functional Activities Questionnaire (FAQ),[63] Katz Index of Activities of Daily Living (KATZ ADL),[63] and Barthel Index (BI) (each n=1, 2%).[38] The SF-36 subscales of physical and social functioning (n=1, 2%) [79] were also used to record ADL. Two publications employed self-designed questions.[7, 34] Most frequently (n=5), a single instrument was applied to assess ADL.[7, 34, 38, 52, 79]

Instruments used to assess (social) participation

One publication reported the assessment of participation by self-designed questions covering return to sports and work.[52]

Table 2 displays the number of publications and the instruments used to assess HRQoL, ADL, and (social) participation.

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Table 2 Quantitative overview of instruments assessing HRQoL, ADL and (social) participation in n=54 publications

PRO	Category	Instrument	N (%)
HRQoL	Generic instruments	Short Form-36 Health Survey (SF-36)	21 (39)
		12-Item Short Form Health Survey (SF-12)	12 (22)
		Nottingham Health Profile (NHP)	4 (7)
		Sickness Impact Profile (SIP)	4 (7)
		Short Musculoskeletal Function Assessment (SMFA)	1 (2)
		World Health Organization Quality of Life Instrument – Short Form (WHOQOL-BREF)	1 (2)
	Preference-based instruments	European Quality of Life 5-Dimension (EQ-5D)	9 (17)
		EQ-6D*	1 (2)
		Health Utility Index Mark 2 (HUI 2)	1 (2)
		Health Utility Index Mark 3 (HUI 3)	1 (2)
	Trauma-specific instruments	Hannover Score for Polytrauma Outcome (HASPOC)	6 (11)
		Trauma Outcome Profile (TOP)	4 (7)
		Aachen Long-term Outcome Score (ALOS)	1 (2)
	Modular instruments	Polytrauma-Outcome-Chart (POLO Chart)	6 (11)
	Conceptual framework	The International Classification of Impairments, Disabilities and Handicaps (ICIDH)	1 (2)
ADL		Self-designed questions	2 (4)
	Generic instruments	Groningen Activity Restriction Scale (GARS)	each n=1 (2)
		Functional Activities Questionnaire (FAQ)	
		Katz Index of Activities of Daily Living (KATZ ADL)	
		Barthel Index (BI)	
		SF-36 subscales physical and social functioning	
(Social) Participation		Self-designed questions	1 (2)

*EQ-6D=EQ-5D comprising an additional question capturing a cognitive component (memory, concentration, coherence, and intelligence quotient)

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Application of instruments

A postal survey was most often used to assess HRQoL, ADL, and (social) participation (n=17).[11, 33, 46-52, 56, 57, 67, 71, 74, 75, 77, 79] Next in frequency were telephone interviews (n=5),[34, 36, 54, 55, 80] expert assessments (n=2),[44, 63] face to face interview (n=1),[81] and combinations of those (n=16).[7, 35, 37, 39, 40, 42, 43, 45, 58, 59, 65, 68, 70, 73, 82, 83] The research settings were: trauma centre,[35, 42, 43, 45, 84] outpatient clinic,[61, 64, 78] hospital,[38] medical centre or place of patient´s residence,[63] hospital and home.[73] In four publications, no information regarding the application of instruments was provided.[41, 53, 66, 72] Relatives or proxies were reported in five publications.[36, 51, 52, 55, 62] While most instruments are used with several data collection methods and survey settings, the HASPOC, where reported, is only recorded in the trauma centre and is assessed by a physician in addition to the respondent.

Measurements during follow-up

PROs mentioned above were most frequently recorded at a single time point (n=47).[7, 11, 33-35, 38-46, 48-56, 59-63, 65-72, 74-84] Few publications reported two (n=3),[47, 64, 73] three (n=2),[36, 58] four or five (each n=1) [37, 57] time points of assessment.

Duration of follow-up

Most often (n=13), a follow-up of one to one and a half year was reported.[34, 37, 50-53, 59, 61, 64, 67, 72, 75, 81] Six publications described a follow-up period of over 10 and up to 17.5 years.[35, 41-45] Rarely, shorter follow-up periods of eight days [84] to six months [73, 79] were detected.

Further PROs assessed

Beyond HRQoL, ADL, and (social) participation, assessments of further PROs were reported in the included publications. Most frequently these comprised function, pain, and mental disorders (including depression and/or depressive symptoms, anxiety, and PTSD). Only few PROs focused on social support, cognition, subjective health status, and social network.

Table 3 provides an overview of the kind and number of further recorded PROs.

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Table 3 Quantitative overview of kind and number of further reported PROs in n=54 publications

PRO	N (%)
Function	19 (35)
Mental disorder	16 (30)
Pain	14 (26)
Social support	7 (13)
Cognition	6 (11)
Subjective health status	3 (5)
Social network	2 (4)
Neuropsychological disorder	1 (2)

Critical appraisal results

Some methodological weaknesses were identified regarding the selection of the study population, consideration of confounders, data analysis and presentation of results, internal validity and generalisability of study results.[33, 39-42, 44, 60, 64, 66, 70, 76-78, 80, 82, 83] While most criteria of the critical appraisal were fulfilled, a moderate risk of bias was observed.[7, 11, 35-38, 43, 45-52, 55, 56, 58, 59, 61, 62, 65, 68, 71-74, 79, 81, 84] A low risk of bias was observed in eight publications.[34, 53, 54, 57, 63, 67, 69, 75] The results of the critical appraisal are provided in Table 1.

DISCUSSION

Main findings

The present systematic review identified 54 publications contributing to the research questions. A great variety was observed among all investigated aspects. HRQoL was the most frequently recorded PRO, and SF-36, SF-12 and EQ-5D were the most frequently applied instruments to assess it. Data on ADL and (social) participation were rarely reported. Most questionnaires were sent out per mail. Measurements were predominantly performed at a single time point, and the follow-up period lasted mainly between one and one and a half year. Function, mental disorders and pain were the most frequently assessed categories of the other PROs.

Instruments for the assessment of HRQoL, ADL, and (social) participation

In 48 of the included publications, HRQoL was assessed – predominantly with generic and preference-based HRQoL instruments such as SF-36, SF-12, EQ-5D, and NHP. Thus, over a long period of time, results are available for the impact of polytrauma on HRQoL and its possible change over time.

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Unsurprisingly, the SF-36, one of the oldest and most widely used instrument for assessing HRQoL,[85] is the most frequently reported instrument in publications included in this review. Along with the SF-12, SF tools accounted for 61% of all identified instruments. This result is supported by observations of other reviews conducted in the field of polytrauma.[85-88]

In 1999, an international consensus conference on evidence-based guidelines for the systematic evaluation and assessment of HRQoL after polytrauma recommended the use of modular instruments comprising generic and trauma-specific aspects.[89] Generic instruments have been applied continuously over a long time period, have been translated into different languages, and can be applied regardless of the underlying disease. They are suitable for assessing HRQoL after polytrauma since it can affect different body regions, show different injury patterns, and lead to a variety of complications. In contrast, trauma-specific instruments capture targeted relevant problems and measure the longitudinal change of HRQoL.[90] Despite the recommendation, the POLO Chart is the only modular instrument comprising generic and trauma-specific aspects reported in the included publications.[90] HASPOC, TOP and ALOS were the trauma-specific tools used. HASPOC is an instrument designed to evaluate the quality of rehabilitation. Since it comprises instruments assessing inter alia HRQoL, ADL, and function,[91, 92] we decided to subsume it into the category 'HRQoL'.

Apart from assessing HRQoL, we were particularly interested in the measurement of ADL and social participation. Regarding the assessment of ADL, five different instruments and self-designed questions in six publications were observed in our review. (Social) participation was only reported once by Holtslag et al. (2006).[52] In summary, it was notable that ADL and (social) participation after polytrauma were seldom assessed. One reason might be that measures of HRQoL or function already include several questions related to ADL and (social) participation. Predominantly, (social) participation is researched in people with brain or spinal cord injuries. A general understanding and clear definition of the participatory construct is still missing, which might be another reason for the low reporting rates of (social) participation. It has been argued that (social) participation should be one of the main aspects in rehabilitation after polytrauma,[93] besides survival and function, and there are a number of instruments available to assess this PRO.[94-96] This could also be assumed for ADL.

Application of instruments

Most frequently, a postal questionnaire was used. Compared to personal interviews, questionnaires sent by mail are quicker and less expensive. In addition, potential bias introduced by an interviewer is excluded. However, it is not possible to verify the person answering the questionnaire, comprehension questions cannot be clarified, the full completion of the questionnaire cannot be guaranteed, and the proportion of non-responders is higher.[97] The consultation of a proxy answering the questions was rarely reported. Injuries, e.g. traumatic brain injuries (TBI), may require the use of a proxy and could enable the collection of outcome data after polytrauma, despite a risk of bias.[97, 98] The present

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systematic review excluded publications reporting single or mono injuries like single TBIs, which might be the reason only few publications including proxies were detected.

The retrospective assessment of the pre-trauma situation helps enable a comparison of results and obtain a holistic view of the outcome after an injury. Gross et al. (2010, 2012) as well as Gross and Amsler (2011) measured pre-fracture level of HRQoL over two years after trauma.[47-49] However, there are limitations to this approach: Memories of the pre-traumatic situation can be distorted by various circumstances, which might increase the risk of recall bias.[99] Large population screenings or the use of population norms might help to evaluate the pre-traumatic state of HRQoL.[19, 46] Baseline assessments should ideally be made in the first week after the event. However, this is not always possible due to the severity of the injury. Another limitation might be that some instruments are not suitable for use in hospital. For example, changed behaviour patterns measured by the SIP can also result from the very fact of being an inpatient. Apart from HRQoL, functional recovery is also of great interest and an important factor for evaluating outcomes after a polytrauma. In comparison to HRQoL and due to better quantification, the retrospective evaluation of function is more feasible.[19]

Most of the included publications reported a single assessment of the PROs. However, the course of recovery cannot be derived from a single measurement. The aforementioned consensus conference recommended measurements at three, 12 and 24 months after the event.[18] Other groups suggested assessments of PROs after one, two, four and 12 months [100] or six and 12 months after the injury, and beyond 12 months in case of longer rehabilitation needs.[13]

The most frequently reported follow-up period were one to one and a half year. Six publications conducted follow-up periods of more than 10 years.

Further PROs assessed

Further reported PROs assessed mainly function, mental disorders, and pain – outcomes that are highly affected by polytrauma. Social support, subjective health status, cognition, or social network were rarely recorded. This might be explained as follows: Instruments assessing HRQoL mainly capture subjective health status. While cognition is frequently affected by brain injuries, publications reporting study populations with single brain injuries were excluded from this review. Social network and social support seem to influence recovery and return to work after trauma [101, 102] and should therefore receive more attention in future research.

Recommendations for the assessment of PROs

The existing recommendations for the use and application of instruments – as well as time points of assessments and outcomes to be measured after polytrauma – differ substantially,[13, 18] but there are attempts to standardize outcome measurements.[19] Classification systems might be helpful for a standardized assessment of PROs, facilitating the adequate selection of instruments for a specific research objective.[24] The internet platform PROMIS – Patient-reported Outcomes Measurement

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Information System – provides research tools and is continuously updated with the latest developments and newly translated instruments.[103] An important step towards a standardized approach of collecting HRQoL was made in the TraumaRegister DGU® by the implementation of HRQoL assessment with the SF-12.[14]

Furthermore, several groups have focused on the definition of special outcome variables to be recorded as default parameters. Hoffman et al. (2016) recommended the development of a 'minimum data set' based on the ICF.[21] A core outcome set relevant for the respective outcome could be defined. This procedure could guide decision-making for the collection and presentation of outcomes in studies.[23] In addition, ICF linking enabling a check of which ICF components are mapped by an instrument could support the tool selection for particular research questions.[17, 20] Other authors also support the idea of core outcome sets.[22, 25]

Strengths and limitations

This review was conducted thoroughly based on a registered protocol in line with the PRISMA-P quality and the PRISMA statement requirements. We provide a broad overview of instruments used for the collection of HRQoL, ADL and (social) participation, along with information on their application. In addition, this study offers an overview on further PROs reported in the context of polytrauma. First efforts to standardize the collection of PROs have been achieved to optimize comparability of outcomes, but further consistency between central players (e.g. trauma registers) is needed. Therefore, this review encourages further improvement in the assessment of PROs after polytrauma.

The present systematic review has some limitations: Selection bias can be assumed as the search was restricted by the search period and language. However, we performed backward citation tracking and saw that only a few additional articles could be included.

CONCLUSION

The present review reveals variations across study findings, which impairs the comparability of outcomes after polytrauma. To improve this situation, a more standardized approach for recording PROs should be established. First efforts have been initiated, but further alignment between the central players, such as the trauma registers, is required. Continuous collection of core outcomes during the regeneration of persons with polytrauma could create the basis for a targeted use of treatments and their adaptation to the results of the outcome measurement. Moreover, there appears to be little research on (social) participation and on further PROs like social support and social network. PROs need more attention in polytrauma-related outcome research. After all, we need to know the patients' perspectives on relevant outcomes after polytrauma so as to consider their needs when providing health services.

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Supplementary information

Additional file 1: PRISMA Checklist

Additional file 2: Search strategy

Additional file 3: Study protocol of this systematic review

Abbreviations

ADL: Activities of daily living; AH-F: Annegret Herrmann-Frank; AI: Andrea Icks; AIS: Abbreviated Injury Scale; AIS§: AIS values are shown in publication for various body regions; ALOS: Aachen Long-term Outcome Score; AUC: Academy for Trauma Surgery GmbH; BI: Barthel Index; cs: can´t say; ct: can´t tell; CENTRAL: Cochrane Central Register of Controlled Trials; CINAHL: Cumulative Index to Nursing and Allied Health Literature; CW: Charlotte Wittgens; DALY: Disability-adjusted life year; e.g.: For example; EQ-5D: European Quality of Life 5-Dimension; EQ-6D: Six-Dimensional EuroQol instrument; EQ-VAS: EQ-5D visual analogue scale; F: Female; FAQ: Functional Activities Questionnaire; Fig.: Figure; FT: Fast track; GARS: Groningen Activity Restriction Scale; HASPOC: Hannover Score for Polytrauma Outcome; HRQoL: Health-related quality of life, HUI 2: Health Utility Index Mark 2; HUI 3: Health Utility Index Mark 3; ICF: International Classification of Functioning, Disability and Health; ICIDH: International Classification of Impairments, Disabilities and Handicaps; ICU: Intensive Care Unit; IDI: injury distress index; IG: Irmela Gnass; IQR/IR: Interquartile range; ISS: Injury Severity Score; JS: Johannes Sturm; JW: Joachim Windolf; KATZ ADL: Katz Index of Activities of Daily Living; KM: Kai Moschinski; M: Male; max.: highest value; MCS: Mental Component Summary score; Md: Median; MeSH: Medical subject heading; mh: MeSH-term; MIM: Maria-Inti Metzendorf; MMAT: Mixed Method Appraisal Tool; MR: Michaela Ritschel; MULT: Patients after polytrauma; N: Number; n: no; NA: not applicable; NHP: Nottingham Health Profile; NICE: UK National Institute for Health and Care Excellence; NR: not reported; NRTW: Non-return to work; N-WCP: Non-workers´ compensation patients; PCS: Physical Component Summary score; POLO Chart: Polytrauma-Outcome-Chart; PRISMA: Preferred Reporting Items for Systematic Review and Meta-Analysis; PRISMA-P: Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols; PRO: Patient-reported outcome; PROMIS: Patient-reported Outcomes Measurement Information System; PTSD: Posttraumatic stress disorder; PTSS: Posttraumatic stress symptoms; QoL: Quality of Life; SA: Silke Andrich; SBI: Severe brain injuries; SK: Silke Kuske; RTW: Return to work; SD: Standard deviation; SF: Sascha Flohé; SF-12: 12-Item Short Form Health Survey; SF-36: Short Form-36 Health Survey; SIGN: Scottish Intercollegiate Guidelines Network; SIP: Sickness Impact Profile; SMFA: Short Musculoskeletal Function Assessment; SOB: Sandra Olivia Borgmann; TBI: traumatic brain injury; TOP: Trauma Outcome Profile; tw: Textwords; WCP: Workers´ compensation patients; WHO ICTRP: WHO International Clinical Trials Registry Platform WHOQOL-BREF: World Health

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Organization Quality of Life Instrument – Short Form; y: yes; YLD: Years lived with disability; YLL: Years of life lost

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Competing interests

JS was executive secretary of the Academy for Trauma Surgery GmbH (AUC). The authors declare that they have no competing interests.

Patient consent for publication

Not required.

Data availability statement

There are no additional data available.

Differences between protocol and review

We have planned to include publications only when the full text of the study is available. However, since all texts were available, there was nothing to exclude according to this criterion.

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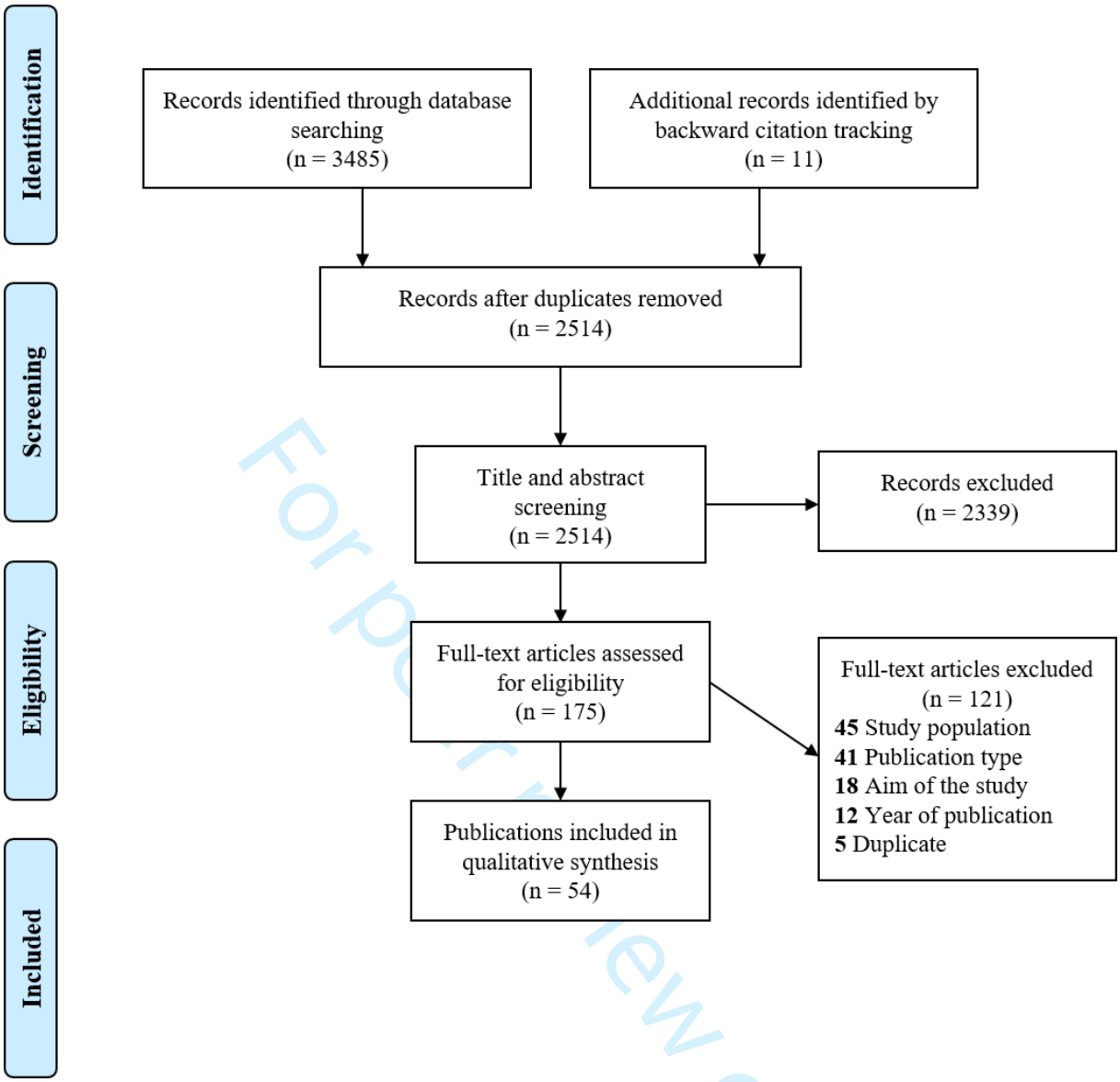
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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	2, 5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5, 6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5, 6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Additional file 2
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5, 6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	NA
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	NA
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6, 7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8-33, Table 1-3
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	33, Table 1 (9-29)
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	NA
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	33-36
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	36
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	36
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	38

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: www.prisma-statement.org.
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Additional file 2: Literature search strategy

PubMed (last searched 5 April 2018)

#1
"Multiple Trauma"[mh] OR multiple trauma*[tw] OR polytrauma*[tw]
#2
"Quality of Life"[mh] or "quality of life"[tw] or "life quality"[tw] OR QoL[tw] OR HRQoL[tw] OR satisf*[tw] OR "self reported health"[tw] OR "Activities of Daily Living"[mh] OR "activities of daily living"[tw] OR "daily living activity"[tw] OR "daily living activities"[tw] OR "activity of daily living"[tw] OR "limitation of activity"[tw] OR "self care"[tw] OR "physical functioning"[tw] OR functional abilit*[tw] OR "functional assessment"[tw] OR "Social Participation"[mh] OR "social participation"[tw] OR social activit*[tw] OR "social functioning"[tw] OR "social interaction"[tw] OR "social isolation"[tw] OR "social integration"[tw] OR "Pain"[mh] OR pain[tw] OR "Depression"[mh] OR "Depressive Disorder"[mh] OR depress*[tw] OR "emotional distress"[tw] OR "Stress, Psychological"[mh] OR stress[tw] OR "Anxiety"[mh] OR anxiet*[tw] OR "Independent Living"[mh] OR "independent living"[tw] OR "community dwelling"[tw] OR "wellbeing"[tw] OR "well being"[tw] OR "Cognition Disorders"[mh] OR cognitive function*[tw] OR "mental health"[tw] OR "cognitive impairment"[tw] OR "Patient Outcome Assessment"[mh] OR "patient reported outcome"[tw] OR "patient reported outcomes"[tw] OR "patient related outcome"[tw] OR "patient related outcomes"[tw] OR "patient centered outcomes"[tw] OR "patient centered outcome"[tw] OR rehabilitation outcome*[tw] OR "Work Capacity Evaluation"[mh] OR "work capacity"[tw] OR "occupational function"[tw] OR "Stress Disorders, Post-Traumatic"[mh] OR PTSD[tw] OR "traumatic stress disorder"[tw] OR "posttraumatic stress disorder"[tw]
#3
#1 AND #2
#4
#3 AND Filters: **Publication date from 2005/01/01 to 2016/12/31; English; German**

UPDATE
#3 AND Filters: **Publication date from 2016/08/01 to 2018/12/31; English; German**

Embase Ovid (<1974 to 2016 August 02>, last searched 3 August 2016)

1. multiple trauma/
2. (multiple trauma* or polytrauma*).tw.
3. or/1-2
4. exp quality of life/
5. (quality of life or life quality or QoL or HRQoL or satisf* or self reported health).tw.
6. daily life activity/
7. ("activities of daily living" or "activity of daily living" or daily living activit* or "limitation of activity" or self care or physical functioning or functional abilit*or functional assessment).tw.
8. social participation/
9. (social participation or social activit* or social functioning or social interaction or social isolation or social integration).tw.
10. pain/
11. pain.tw.
12. exp depression/

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13. (depress* or emotional distress).tw.
14. mental stress/
15. stress.tw.
16. anxiety/
17. anxiet*.tw.
18. independent living/
19. (independent living or community dwelling).tw.
20. wellbeing/
21. (wellbeing or well being).tw.
22. cognitive defect/
23. postoperative cognitive dysfunction/
24. (cognitive adj (disorders or function* or impairment)).tw.
25. (mental health).tw.
26. patient satisfaction/
27. (patient adj (outcome assessment or reported outcome* or related outcome* or centered outcome*)).tw.
28. (rehabiliation outcome*).tw.
29. work capacity/
30. (work capacity or occupational function).tw.
31. posttraumatic stress disorder/
32. (PTSD or posttraumatic stress disorder or traumatic stress disorder).tw.
33. or/4-32
34. 3 and 33
35. limit 34 to ((english or german) and yr="2005 - 2016")

CINAHL EBSCO (last searched 5 April 2018)

1. MH "Multiple Trauma"
2. TI (multiple trauma* or polytrauma*)
3. AB (multiple trauma* or polytrauma*)
4. S1 OR S2 OR S3
5. MH ("Quality of Life+" or "Health and Life Quality (Iowa NOC)+" or "Disability Evaluation")
6. TI ("quality of life" or "life quality" or "QoL" or "HRQoL" or satisf* or "self reported health")
7. AB ("quality of life" or "life quality" or "QoL" or "HRQoL" or satisf* or "self reported health")
8. MH "Activities of Daily Living+"
9. MH "Self-Care Deficit (Saba CCC)+"
10. MH "Activities of Daily Living (Saba CCC)"
11. MH "Self-Care (Iowa NOC)+"
12. TI ("activities of daily living" or "daily living activity" or "daily living activities" or "activity of daily living" or "limitation of activity" or "self care" or "physical functioning" or "functional abilit*" or "functional assessment")
13. AB ("activities of daily living" or "daily living activity" or "daily living activities" or "activity of daily living" or "limitation of activity" or "self care" or "physical functioning" or "functional abilit*" or "functional assessment")
14. MH "Social Participation"
15. TI ("social participation" or "social activit*" or "social functioning" or "social interaction" or "social isolation" or "social integration")
16. AB ("social participation" or "social activit*" or "social functioning" or "social interaction" or "social isolation" or "social integration")
17. MH "Pain+"
18. TI pain

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19. AB pain
20. MH "Depression+"
21. TI (depress* or "emotional distress")
22. AB (depress* or "emotional distress")
23. MH "Stress, Psychological+"
24. TI stress
25. AB stress
26. MH "Anxiety+"
27. TI anxiet*
28. AB anxiet*
29. MH "Community Living+"
30. TI ("independent living" or "community living" or "community dwelling"or "wellbeing"or "well being")
31. AB ("independent living" or "community living" or "community dwelling"or "wellbeing"or "well being")
32. MH "Cognition Disorders+"
33. TI ("cognitive function*" or "mental health" or "cognitive impairment")
34. AB ("cognitive function*" or "mental health" or "cognitive impairment")
35. MH "Outcomes (Health Care)+"
36. TI ("patient reported outcome*" or "patient related outcome*"or "patient centered outcome*" or "rehabilitation outcome*")
37. AB ("patient reported outcome*" or "patient related outcome*"or "patient centered outcome*" or "rehabilitation outcome*")
38. MH "Work Capacity Evaluation"
39. TI ("work capacity" or "occupational function")
40. AB ("work capacity" or "occupational function")
41. MH "Stress Disorders, Post-Traumatic+"
42. TI (PTSD or "posttraumatic stress disorder" or "traumatic stress disorder")
43. AB (PTSD or "posttraumatic stress disorder" or "traumatic stress disorder")
44. S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36 OR S37 OR S38 OR S39 OR S40 OR S41 OR S42 OR S43
45. S4 AND S44
46. EM 20050101-20161231
47. S45 AND S46

UPDATE
EM 20160801-20181231

PsycINFO Ovid (last searched 5 April 2018)

1. (multiple trauma* or polytrauma*).tw.
2. "Quality of Life"/
3. "Life Satisfaction"/
4. (quality of life or life quality or QoL or HRQoL or satisf* or self reported health).tw.
5. "Activities of Daily Living"/
6. "Self-Care Skills"/
7. ("activities of daily living" or "activity of daily living" or daily living activit* or "limitation of activity" or self care or physical functioning or functional abilit*or functional assessment).tw.
8. (social participation or social activit* or social functioning or social interaction or social isolation or social integration).tw.

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9. Pain/
10. pain.tw.
11. exp Major Depression/
12. (depress* or emotional distress).tw.
13. Psychological Stress/
14. "Stress and Coping Measures"/
15. Coping Behavior/
16. Adjustment Disorders/
17. stress.tw.
18. Anxiety/
19. anxiet*.tw.
20. (independent living or community dwelling).tw.
21. Well Being/
22. (wellbeing or well being).tw.
23. Cognitive Impairment/
24. exp Memory Disorders/
25. Cognitive Ability/
26. (cognitive adj (disorders or function* or impairment)).tw.
27. (mental health).tw.
28. Client Satisfaction/
29. (patient adj (outcome assessment or reported outcome* or related outcome* or centered outcome*)).tw.
30. exp Treatment outcomes/
31. (rehabilitation outcome*).tw.
32. (work capacity or occupational function).tw.
33. Post-Traumatic Stress/
34. Posttraumatic Stress Disorder/
35. (PTSD or posttraumatic stress disorder or traumatic stress disorder).tw.
36. or/2-35
37. 1 and 36
38. limit 37 to yr="2005 -Current"

UPDATE

(201608* or 201609* or 20161* or 2017* or 2018*).up.

Cochrane Central Register of Controlled Trials via Cochrane Register of Studies Online (last searched 5 April 2018)

1. (multiple trauma* or polytrauma*):TI,AB,KY
2. ("quality of life" or "life quality" or QoL or HRQoL or satisf* or "self reported health"):TI,AB,KY
3. ("activities of daily living" or "daily living activity" or "daily living activities" or "activity of daily living" or "limitation of activity" or "self care" or "physical functioning" or "functional abilit*" or "functional assessment"):TI,AB,KY
4. ("social participation" or social activit* or "social functioning" or "social interaction" or "social isolation" or "social integration"):TI,AB,KY
5. (pain):TI,AB,KY
6. (depress* or "emotional distress"):TI,AB,KY
7. (stress):TI,AB,KY
8. (anxiet*):TI,AB,KY
9. ("independent living" or "community dwelling" or "wellbeing" or "well being"):TI,AB,KY

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10. ("cognition disorder*" or "cognitive function*" or "mental health" or "cognitive impairment"):TI,AB,KY
11. ("patient reported outcome" or "patient reported outcomes" or "patient related outcome" or "patient related outcomes" or "patient centered outcomes" or "patient centered outcome" or "rehabilitation outcome*"):TI,AB,KY
12. ("work capacity" or "occupational function"):TI,AB,KY
13. (PTSD or "traumatic stress disorder" or "posttraumatic stress disorder"):TI,AB,KY
14. #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13
15. #1 AND #14

ClinicalTrials.gov (Expert search, last searched 5 April 2018)

Condition: "multiple trauma" OR "multiple traumata" OR polytrauma OR polytraumata

Search terms: "quality of life" OR "life quality" OR QoL OR HRQoL OR satisfaction OR "self reported health" OR "activities of daily living" OR "daily living activity" OR "daily living activities" OR "activity of daily living" OR "limitation of activity" OR "self care" OR "physical functioning" OR "functional ability" OR "functional abilities" OR "functional assessment" OR social OR pain OR depression OR "depressive disorder" OR distress OR stress OR anxiety OR "independent living" OR "community dwelling" OR "wellbeing" OR "well being" OR "cognition disorder" OR "memory disorders" OR "cognition disorders" OR "cognitive function" OR "cognitive functions" OR "cognitive abilities" OR "cognitive ability" OR "mental health" OR "cognitive impairment" OR "patient outcome assessment" OR "patient reported outcome" OR "patient reported outcomes" OR "patient related outcome" OR "patient related outcomes" OR "patient centered outcomes" OR "patient centered outcome" OR rehabilitation OR coping OR adjustment OR "work capacity" OR "occupational function" OR PTSD OR "traumatic stress disorder" OR "posttraumatic stress disorder"

UPDATE

("quality of life" OR "life quality" OR QoL OR HRQoL OR satisfaction OR "self reported health" OR "activities of daily living" OR "daily living activity" OR "daily living activities" OR "activity of daily living" OR "limitation of activity" OR "self care" OR "physical functioning" OR "functional ability" OR "functional abilities" OR "functional assessment" OR social OR pain OR depression OR "depressive disorder" OR distress OR stress OR anxiety OR "independent living" OR "community dwelling" OR "wellbeing" OR "well being" OR "cognition disorder" OR "memory disorders" OR "cognition disorders" OR "cognitive function" OR "cognitive functions" OR "cognitive abilities" OR "cognitive ability" OR "mental health" OR "cognitive impairment" OR "patient outcome assessment" OR "patient reported outcome" OR "patient reported outcomes" OR "patient related outcome" OR "patient related outcomes" OR "patient centered outcomes" OR "patient centered outcome" OR rehabilitation OR coping OR adjustment OR "work capacity" OR "occupational function" OR PTSD OR "traumatic stress disorder" OR "posttraumatic stress disorder") | "multiple trauma" OR "multiple traumata" OR polytrauma OR polytraumata | First posted from 08/03/2016 to 12/31/2018

WHO ICTRP search (Standard search, last searched 5 April 2018)

multiple trauma* OR polytrauma*

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Open Access Protocol

BMJ Open

Assessment of patient-reported outcomes after polytrauma: protocol for a systematic review

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ABSTRACT

Introduction Survivors of polytrauma experience long-term and short-term burden that influences their lives. The patients' view of relevant short-term and long-term outcomes should be captured in instruments that measure quality of life and other patient-reported outcomes (PROs) after a polytrauma. The aim of this systematic review is to (1) collect instruments that assess PROs (quality of life, social participation and activities of daily living) during follow-up after polytrauma, (2) describe the instruments' application (eg, duration of period of follow-up) and (3) investigate other relevant PROs that are also assessed in the included studies (pain, depression, anxiety and cognitive function).

Methods and analysis The systematic review protocol is developed in line with the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols statement. MEDLINE, EMBASE, Cumulative Index to Nursing and Allied Health Literature, PsycINFO, Cochrane Central Register of Controlled Trials and the trials registers ClinicalTrials.gov and WHO International Clinical Trials Registry Platform will be searched. Keywords, for example, 'polytrauma', 'multiple trauma', 'quality of life', 'activities of daily living' or 'pain' will be used. Publications published between January 2005 and the most recent date (currently: August 2016) will be included. In order to present the latest possible results, an update of the search is conducted before publication. The data extraction and a content analysis will be carried out systematically. A critical appraisal will be performed.

Ethics and dissemination Formal ethical approval is not required as primary data will not be collected. The results will be published in a peer-reviewed publication.

PROSPERO registration number CRD42017060825.

Strengths and limitations of this study

- ► To the best of our knowledge, there are no published systematic reviews providing an overview of assessed patient-reported outcomes after polytrauma, different instruments used to measure these as well as the application of these instruments.
- ► This systematic review will report the identified instruments used to assess quality of life, social participation and activities of daily living, and describe their application.
- ► We will show additional measures for further relevant patient-reported outcomes following polytrauma, for example, pain, depression, anxiety and cognitive function.
- ► A limitation of the review might be that publications will be included which were published since 2005. It cannot be ruled out in that case that relevant literature published before 2005 is missing.

INTRODUCTION

Severe injuries represent a leading cause of death and permanent disability.[1] Especially in the central European region, such severely injured patients are termed as 'polytrauma patients' or as being 'polytraumatised'. 'Polytrauma' is defined as having at least two severe injuries in different body regions that are potentially life threatening. In the Anglo-American literature, these patients are mostly referred to as 'multiple-injury', 'multiple-trauma' or 'severely injured'. All these descriptions have in common that a certain degree of injury is required. The severity of trauma is usually indicated by the Injury Severity Score (ISS).[2] In general, an ISS ≥16 falls within the definition of polytrauma, multiple-trauma or severely injured patients.[1 3] In the following, we use the term polytrauma. Comparing the incidence of trauma is challenging, considering the different definitions or conditions of trauma, and due to inconsistencies in the available data.[4] In 2015, according to the TraumaRegister DGU®, a German registry that cover patients with severe injuries, the number of severely multiply injured patients was 17 630 (ISS ≥16).[5] In 2012, the number of severely multiply injured patients (ISS ≥16) reached approximately 18 400 per year. This corresponds to a cumulative incidence of 0.02% per year for Germany.[6] Persons affected are mostly men (72%) with a mean age of 46.5 years.[7]

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Due to strategies for early advanced life support, high quality in healthcare services, progress in treatment options and more traffic safety, survival rates after polytrauma are increasing. However, survivors are faced with long-term and short-term burden after polytrauma: 1 year after polytrauma, patients continue to suffer from persisting problems with mobility (34%), self-care (15%) and activities of daily living (ADL) (51%). Furthermore, they experience pain and/or discomfort (58%) as well as anxiety and/or depression (37%).[8] Relevant disabilities, such as respiration limitations, paraplegia or tetraplegia, are major causes for 40% of those affected not returning to their former workplace[6 9] and impact on socioeconomic and quality-of-life aspects.[9–11] For the necessary acute and rehabilitative care, the social economic impact is estimated to be €106 000 per patient. In Germany, additional costs of €935 000 accrue if patients do not return to their former workplace.[9]

Consequently, the aim to reduce short-term and long-term burden after polytrauma is critically important.[9 11] Thus, clinical parameters are necessary to support treatment processes and services in acute care facilities and to display short-term outcomes. Additionally, evaluations capturing a patient's view of short-term and long-term outcomes in terms of, for example, psychological and physical factors, functional status and social interaction, are becoming increasingly important for doctors and nurses as well as for patients and their family members.[11 12] The measurement of these patient-reported outcomes (PROs), which can also be reported, for example, by relatives in case of mental disability, is important for completing the assessment of relevant clinical outcomes after injury. To assess the impact of a polytrauma on quality of life (QoL) and other relevant PROs, it is necessary to question the patients.[12 13] The preferred gold standard is a measurement at multiple times during the follow-up period to track trends and focus on long-term conditions and outcomes associated with polytrauma.[13] Therefore, it is necessary to know which instruments for assessing PROs are applied and how. So far, no systematic review has been identified providing an overview of assessed PROs after polytrauma, different instruments used to measure these as well as the application of these instruments. Thus, this systematic review will provide an overview of instruments used to measure PROs, including QoL and other currently reported outcome measures for patients with polytrauma, and describe their application in detail.

Currently the TraumaRegister DGU® collects data on emergency care, treatment in shock rooms, intensive care unit and discharge.[3 12] Regarding the comprehensive evaluation of the short-term and long-term burden, the register plans to expand their measurement battery by assessing PROs, for example, QoL.

AIM OF THE STUDY

The aim of this systematic review is to:

1. collect instruments that assess PROs (QoL, social participation and ADL) during the follow-up after polytrauma;
2. describe the application of these instruments in detail (eg, duration of period of follow-up, frequency of application and time of measurements during follow-up);
3. investigate other relevant PROs that are additionally assessed in the included studies (eg, pain, depression, anxiety and cognitive function) without reporting the application of instruments in detail.

METHOD

The systematic review protocol is developed in line with the quality requirements of the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) statement,[14] and the PRISMA statement will be considered during the review procedure.[15] The protocol was registered on 8 April 2017 at PROSPERO (registration number CRD42017060825). Prior to publication an update of the search will be performed to be able to present the latest results.

Databases

The literature search will be conducted in the following databases: MEDLINE, EMBASE, Cumulative Index to Nursing and Allied Health Literature, PsycINFO, Cochrane Central Register of Controlled Trials and the trials registers ClinicalTrials.gov and WHO International Clinical Trials Registry Platform. To ensure literature saturation, reference lists of eligible studies will be examined for further relevant publications.

Search strategy

The search strategy will be developed in cooperation with a specialist for systematic reviews using database-specific-controlled vocabulary and additional free-text terms. Online supplementary appendix A provides the MEDLINE search strategy that will be adapted to the other databases accordingly.

Inclusion and exclusion criteria

Studies will be included if they match the following inclusion criteria: (1) the study assesses PROs, such as QoL/health-related QoL, social participation, ADL, pain, depression, anxiety or cognitive function in people aged 18–75 years during a temporally clearly defined follow-up period after polytrauma (injuries involving at least two different areas of the body or organ systems, ISS >15, Abbreviated Injury Scale (AIS) ≥3); (2) all kinds of original empirical research that were published between 1 January 2005 and currently 3 August 2016; (3) English or German language publications will be considered and (4) the full text of the study is available (ie, for contacting the authors). The interest in QoL in research and its implication in practice has been growing since 2005.[16] As one of our main aims is to collect instruments measuring QoL, we decided to include publications as of 2005. Our

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procedure is supported by the burgeoning perspective in clinical research on PROs and its relevance for the future in addition to clinical data.[17] A further reason was to capture the most recent developments in research on PROs. Therefore, we looked specifically for publications from the last 11 years.

Studies will be excluded from this review if they mention one of the following criteria: (1) an ISS ≤15 or an AIS <3 or no reporting of ISS or AIS; (2) low-energy injuries, single or mono injuries or geriatric injuries, burn injuries, war injuries (group of veterans or military staff are excluded), cancer and other chronic diseases as secondary diagnosis; (3) publications in which the primary aim does not focus on PROs and (4) grey literature, books, letters/short reports, abstracts, editorials, comments or discussion papers as well as case studies, systematic reviews and meta-analysis. However, systematic reviews and meta-analysis will be screened to identify further appropriate studies.

Study selection process

On the basis of predefined inclusion and exclusion criteria, suitable publications will be selected by title and abstract, and independently screened by two reviewers. A third reviewer will solve differences concerning the inclusion or exclusion of studies. In order to foster the process of suitability decisions, inclusion and exclusion of the first 50 publications by title and abstract screening will be discussed between the two reviewers. Subsequently, two reviewers assessing eligibility for final inclusion in this review will screen the full texts of the remaining publications again independently. A third reviewer will solve conflicts in the final inclusion or exclusion of studies. Inter-rater reliability will be determined after the title abstract as well as after the full-text screening.

Data extraction and synthesis

The data extraction will be performed according to the requirements of Cochrane reviews.[18] For data extraction, two experienced researchers will use a piloted data extraction sheet independently. Extracted data from the included studies will provide an overview of: first author and publication year, study design, country, study population (number of subjects, proportion of men, mean age with SD and range, kind of injury, ISS, AIS and other characteristics), treatment, aim of the study, findings, and, furthermore, for the reported PROs according to aim 1, applied instruments to assess these PROs, description of the instrument, data collection (method of assessment, time of measurements, length of follow-up period, quality criteria of instruments (eg, validity and reliability) and modifications of the instruments. Furthermore, the result of the critical appraisal of the study quality will be added.

Critical appraisal

In the final selection of eligible studies, two reviewers will independently perform a quality appraisal to assess the methodological quality and the risk of bias in each study type using standardised checklists of the UK National Institute for Health and Care Excellence[19] or Scottish Intercollegiate Guidelines Network[20] or the Mixed Method Appraisal Tool.[21] Reviewers will resolve disagreements through discussion.

DISCUSSION

We will perform the proposed systematic review to generate an overview of the instruments used to assess PROs in the field of polytrauma. The results of this systematic review will serve as a basis to expand the TraumaRegister DGU® with a focus on QoL measures. Likewise, the additional knowledge on further PROs, for example, pain, depression, anxiety and cognitive function, might emphasise the patients' perspectives on relevant outcomes after polytrauma and lead to its consideration in the provision of health services. Subsequently, healthcare providers and policy-makers may draw their attention on this topic and will implement the assessment of PROs in decision-making regarding the treatment process.

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Competing interests JS is executive secretary of the Academy for Trauma Surgery GmbH (AUC).

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Assessment of patient-reported outcomes after polytrauma – instruments and methods: a systematic review

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Tables:

Table 1 Overview of included publications

Table 2 Quantitative overview of instruments assessing HRQoL, ADL and (social) participation in n=54 publications

Table 3 Quantitative overview of kind and number of further reported PROs in n=54 publications

Figures:

Fig. 1 Flow chart

ABSTRACT

Objectives We (1) collected instruments that assess health-related quality of life (HRQoL), activities of daily living (ADL), and social participation during follow-up after polytrauma, (2) described their use, and (3) investigated other relevant patient-reported outcomes (PROs) assessed in the studies.

Design Systematic Review using the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guideline.

Data Sources MEDLINE, Embase, CINAHL, PsycINFO, CENTRAL, as well as the trials registers ClinicalTrials.gov and WHO ICTRP were searched from January 2005 to April 2018.

Eligibility Criteria All original empirical research published in English or German including PROs of patients aged 18-75 years with an Injury Severity Score (ISS) ≥ 16 and/or an Abbreviated Injury Scale (AIS) ≥ 3. Studies with defined injuries or diseases (e.g. low-energy injuries) and some text types (e.g. grey literature and books) were excluded. Systematic reviews and meta-analyses were excluded, but references screened for appropriate studies.

Data Extraction and synthesis Data extraction, narrative content analysis, and a critical appraisal (e.g. UK National Institute for Health and Care Excellence) were performed by two Reviewers independently.

Results The search yielded 3496 hits; 54 publications were included. Predominantly, HRQoL was assessed, with Short Form-36 Health Survey (SF-36) applied most frequently. ADL and (social) participation were rarely assessed. The methods most used were postal surveys and single assessments of PROs, with a follow-up period of one to one and a half years. Other relevant PRO areas reported were function, mental disorders, and pain.

Conclusions There is a large variation in the assessment of PROs after polytrauma, impairing comparability of outcomes. First efforts to standardize the collection of PROs have been initiated, but

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further harmonisation between central players. Additional knowledge on rarely reported PRO areas (e.g. (social) participation, social networks) may lead to their consideration in health services provision.

Registration PROSPERO (CRD42017060825)

Keywords Accident & Emergency Medicine, Adult Intensive & Critical care, Orthopaedic & Trauma Surgery, Public Health

Strengths and limitations of this study

- This systematic review was based on a comprehensive search strategy and conducted according to a registered and published protocol.
- First systematic review that provides an overview of patient-oriented instruments used in research after polytrauma.
- Throughout the review process, we followed the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guideline.
- A limitation might be the restriction of the search period and the language.

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INTRODUCTION

Severe injuries represent a leading cause of death and permanent disability.[1] In the central European region, severely injured people are referred to as having a 'polytrauma' or being 'polytraumatized'.[2] A polytrauma is defined as at least two severe injuries in different body regions or organ systems, at least one of which is potentially life-threatening by itself or in combination with others, with an Injury Severity Score (ISS) ≥16.[1-4] According to the TraumaRegister DGU®, a German registry that covers patients with severe injuries, the number of patients with polytrauma (ISS ≥16) was 15 651 in 2019.[5] Most affected were men (72%) with a mean age of 46.5 years.[6]

The survival rates after polytrauma have increased lately, bringing the quality of survival into the focus of research. One year after polytrauma, survivors often fail to reach pre-traumatic status due to persisting medical problems.[7] Disabilities such as respiratory problems, paraplegia or tetraplegia, prevent 40% from returning to their former workplace.[8, 9] They also have an impact on socio-economic aspects and quality of life.[9-11] Therefore, it is crucial to reduce the short-term and long-term burden after polytrauma.[9, 11] In this context, the patients' perspective on polytrauma outcomes focussing on psychological, physical and social factors is getting more important for the patients themselves, their relatives, and healthcare professionals.[11, 12] Evaluating these patient-reported outcomes (PROs) is essential to complete the holistic assessment of relevant clinical outcomes after injury from a patient's perspective.

To assess the impact of polytrauma on health-related quality of life (HRQoL) and other important PROs, one must consult the person affected directly. [12, 13] HRQoL is already being recorded in registers like the Australian Victoria State Trauma Registry. The German TraumaRegister DGU® will also complement its assessment with HRQoL.[14] Activities of daily living (ADL), defined as basic activities to fulfill independent living, along with physical and psychological needs, are meaningful outcome parameters in case of post-polytrauma impairment. In recent years, the concept of social participation has gained more attention in the political context [15] and was therefore included in the review as an outcome of interest, along with HRQoL and ADL. Several working groups have controversially discussed which tools should be applied to assess PROs after polytrauma, at which time points, and in which way. [13, 16-24] However, guidelines with clear recommendations are lacking,[13] and no systematic review could be identified.

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Therefore, the aim of this systematic review was to:

1. collect instruments assessing HRQoL, ADL, and social participation during follow-up after polytrauma
2. describe their application in detail (e.g. duration of follow-up period, frequency of application, and time point of measurements within the follow-up period)
3. investigate which other relevant PROs are additionally assessed in included studies (e.g. pain, depression, anxiety, and cognitive function) without detailed analysis.

METHODS

This systematic review was conducted in line with the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guideline (Additional file 1).[25] The review protocol was registered on 8 April 2017 at PROSPERO (CRD42017060825).

Data sources and search strategy

The search was conducted in MEDLINE, Embase, CINAHL (Cumulative Index to Nursing and Allied Health Literature), PsycINFO, CENTRAL (Cochrane Central Register of Controlled Trials), the trials registers ClinicalTrials.gov and WHO ICTRP (WHO International Clinical Trials Registry Platform), initially covering the period from 1 January 2005 until 3 August 2016, with a later update of the search until 5 April 2018. The search period was chosen because since 2005 interest in HRQoL in research and its implication in practice has been growing.[26] As one of our main aims was to collect instruments measuring HRQoL, we decided to include publications beginning with 2005. Our approach is supported by the emerging importance of PROs in clinical research and their relevance to the future as a complement to clinical data.[27] Furthermore, the search period was selected to capture the most recent developments in research on PROs.

In addition, reference lists of eligible studies were examined. Textwords (tw) and MeSH-terms (mh) like ‘polytrauma’ (tw), ‘multiple trauma’ (mh), ‘quality of life’ (mh), ‘activities of daily living’ (tw), ‘social participation’ (mh) and ‘pain’ (mh) were used. The development of the search strategy was outlined in the study protocol [28] (Additional file 2) and is provided in more detail in Additional file 3.

Inclusion and exclusion criteria

The publications included primarily investigated PROs of patients aged 18-75 years with an Injury Severity Score (ISS) ≥ 16 and/or an Abbreviated Injury Scale (AIS) ≥ 3 during a clearly defined follow-up period. All original empirical research published in English or German was included. Publications examining special types of injury or illness – e.g. low-energy injuries, single or mono injuries, burn injuries, war injuries, or cancer – were excluded. We also excluded some publication types (grey

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literature, books, abstracts, and editorials). Systematic reviews and meta-analyses were not included, too, but screened to identify further appropriate studies. More details about the inclusion and exclusion criteria were published in the study protocol.[28]

Selection of publications and studies

Four teams with two reviewers each (MR-IG; KM-SK; MR-SOB; AI-AHF) screened the title and abstracts independently. To foster the process of eligibility decisions, the inclusion or exclusion of the first 50 publications was discussed by two reviewers (MR-IG). In a next step, two reviewers (MR-IG; KM-SK; MR-SOB, KM-SOB) independently screened the full texts of the remaining publications for final inclusion. A third reviewer (MR, SK, IG) solved disagreements regarding the inclusion of publications for full-text screening and their final inclusion in this review. We identified publications relating to the same study and/or population. For this purpose, extensive research was conducted, and authors of the corresponding publications were contacted. The section 'Characteristics of included publications' outlines how we approached publications dealing with different research questions of the same study, comprising subpopulations drawn from a superior study population, reporting subprojects of a study project, or presenting results of the same research question at different follow-up periods.

Data extraction and analysis

Data extraction was performed according to the requirements of Cochrane reviews.[29] For the first and second aim, two researchers (MR-IG) used a piloted data extraction sheet independently comprising the following: first author, publication year and country, study design, aim of the study (objective), study population, treatment, applied instruments (incl. modifications), time of measurements, length of follow-up period, method of assessment (application), study results, and results of the critical appraisal. Regarding the third aim, the other reported PROs were extracted and categorized in a separate extraction sheet. Narrative content analysis was performed regarding the assessment of HRQol, ADL and social participation as well as further reported PROs.

Critical appraisal

Teams with two reviewers each (MR, IG, SK, KM, SOB) independently assessed the methodological quality and the risk of bias in each study type using standardized checklists of the UK National Institute for Health and Care Excellence (NICE),[30, 31] Scottish Intercollegiate Guidelines Network (SIGN),[32] or the Mixed Method Appraisal Tool (MMAT).[33] Discrepancies were resolved by discussion between the reviewers or by a third reviewer.

Patient and public involvement statement

Patients were not involved in the design and implementation of this systematic review.

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RESULTS

Search results

In total, 3485 references were retrieved. After a title and abstract screening as well as a full-text screening, further 11 records were identified by backward citation tracking. Finally, 54 eligible publications were included for data extraction (Fig. 1).

Fig. 1 Flow chart

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Characteristics of included publications

The publications (n=54) most often came from Germany (n=15, 28%) and the Netherlands (n=11, 20%); rarely from Austria, Denmark, France, Japan, Poland, Spain and Sweden (each n=1, 2%). One study has been carried out in Germany and Austria, another one in Europe, Asia, Australia, Africa, North- and South-America. The publications primarily reported monocentric studies (n=44, 81%) and predominately comprised prospective cohort studies (n=26, 48%) (Table 1).

The number of participants varied between eight [(34)] and 7612 [35], with an average age of 24.6 ± 12.1 [36] to 53.9 ± 18.8 [37] years, and a proportion of men between 37.5% [34] and 84% [38]. The mean ISS ranged between 3.9 ± 1.7 [39] and 56.8 points [40, 41] and the AIS – reported in 18 publications – between zero and five points. Information regarding treatment of injuries was provided in n=31 publications (Table 1).

Six publications were related to the Hannover Rehab study.[36, 42-46] Four reported the results of a polytrauma study project conducted at the University Hospital Basel, which included a subproject of a study project (EK 159/02) and a polytrauma cohort.[47-50] Study projects at the University Medical Centre Utrecht displayed almost identical recruitment periods and locations (Holtslag et al. 2007 a, b: January 1999 to December 2000,[51, 52] Holtslag et al. 2006 and 2008: January 1999 to January 2001,[53, 54] Holtslag et al. (2006) analysed a subgroup out of n=335 participants;[53] Holtslag et al. 2008 additionally included a group 'Fatalities'.[54] Two publications [55, 56] used the same data collection but addressed different research questions (CIHR, grants no. MOP-62873, FRSQ, grant no. 3255). The publication by Sirois et al. (2009) is the original study,[56] the publication by Ouellet et al. (2009) is a secondary analysis.[55] Three publications comprised patients treated at the Murnau Trauma Centre from 2000 to 2005 and recorded in the TraumaRegister DGU®.[40, 41, 57] The publication by von Rueden et al. (2013) [40] and Wurm et al. (2012) [41] reported on the same investigation of a group of severely injured patients with an ISS>50; Simmel et al. (2013) analysed those patients with an ISS>25.[57] Soberg et al. (2011) [58] investigated the same research question as Soberg et al. (2007) [59] with a follow-up of 5 years (Table 1).

More than half of the publications appeared between 2005 and 2009; the number of annual publications seems to be decreasing since 2015. Table 1 provides an overview of the included publications.

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Table 1 Overview of included publications

Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results
					QoL/HRQoL	(Social) Participation	ADL			
Abraham et al., 2014 USA [60]	Prospective monocentric cohort study	Influence of inpatient delirium on HRQoL after polytrauma; association of depressive symptoms and symptoms of PTSD with domains of HRQoL	N=115 (55.7) Age: Mean 42.4 ± 16.7 ISS: 16 – > 25, Mean 30.4 ± 9.2 AIS: NR Injury: Extremity, head/neck (no brain injury), abdomen/pelvis, thorax	Days intensive care unit (ICU) stay (Mean): 9.4 ± 8.2 Ventilator days (Mean): 3.4 ±6.4	SF-36	-	-	1 year after hospital release[$]	Personal, postal or telephone survey; NR	No association between delirium and HRQoL; higher levels of depressive symptoms and symptoms of PTSD associated with lower HRQoL; stronger association of depressive symptoms and HRQoL compared to symptoms of PTSD
Ahrberg et al., 2014 Germany [61]	Retrospective monocentric cohort study	Definition and influence of delayed foot fracture diagnosis on the overall outcome of polytraumatized patients	N=47 (68.1) Age: Md 39.0 ISS: Md 27.0 AIS: NR Early fracture diagnosis N=26 Age: Md 38 ISS: Md 29.0 AIS: NR Delayed fracture diagnosis N=21 Age: Md 44 ISS: Md 27.0 AIS: NR Injury: Calcaneus, talus, metatarsalia, navicular, cuboid, cuneiformia, phalanges	Conservative und surgical Days ICU stay (Md): 6 Early fracture diagnosis Conservative und surgical Days ICU stay (Md): 8.0 Delayed fracture diagnosis Conservative und surgical Days ICU stay (Md): 4.0	SF-36	-	-	3.5–7.75 years Mean 5.67 ± 1.58 years[$]	Self-administered questionnaire; NR	No significant differences between the two groups in clinical scores; no significant effects of delayed foot fracture diagnosis on the outcome of polytraumatized patients

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results
					QoL/HRQoL	(Social) Participation	ADL			
Andruszkow et al., 2013 Germany [42]	Retrospective monocentric cohort study	Effects of additional upper extremity injuries or traumatic brain injuries (TBI) on functional, psychological and vocational outcome in polytraumatized patients	Study population of the Hannover Rehab study Injury upper extremity N=229 (75) Age: Mean 30.2 ± 11.5 ISS: Mean 20.2 ± 8.6 AIS: NR Injury upper extremity and TBI N=32 (72) Age: Mean 32.0 ± 11.0 ISS: Mean 32.1 ± 6.4 AIS: NR Injury: Upper extremity with and without TBI, chest, abdomen, spine, lower extremity	NR	HASPOC, SF-12	-	-	10-28 Years, Md 17.5 years[$]	NR	Limitations of functional and vocational outcome due to additional upper extremity injuries in polytraumatized patients; no deterioration of long-term outcome in combined injury of the upper extremity with TBI compared to isolated TBI; no difference in the PCS and MCS of SF-12 in all three groups
Archer et al., 2012 USA [62]	Prospective monocentric cohort study	Connection between fear of movement, catastrophising pain, pain and physical health	N=84 (58) Age: Mean 43.1 ± 17.13 ISS: Mean 30.4 ± 9.5 AIS: NR Injury: Abdomen, chest extremities, external, head/face, neck spine	Surgical Days ICU stay (Mean): 9.3 ± 7.7 Ventilator days (Mean): 3.4 ± 6.4	SF-12 (physical component scale)	-	-	2 years after trauma[$]	NR sent by post	Fear of movement and catastrophising pain contributes to poor long-term outcome of polytraumatized patients
Attenberger et al., 2012 Switzerland [47]	Prospective monocentric cohort study	Testing TOP module in the acquisition of long-term results of polytraumatized patients; validation by testing the association with selected variables and test results	N=117 (75,2) Age: 13-92, Mean 39.6 ± 20.5, Md 35.4 ISS: 17-45, 27.5 ± 8.2, Md 27.0 AIS[§]: 0-5, Mean 0.5 ± 0.58-2.7 ±1.48, Md 0.0-3.0 Injury: Inter alia brain injury	NR	EQ-5D, SF-36, ALOS, NHP, TOP* (*dimensions 'pain' and 'function', also pre-injury)	-	-	Mean 2.7 ± 0.9 years post trauma[$]	Self-administered questionnaire sent by post	TOP indicated high rate of relevant problems of polytraumatized people; TOP is a reliable and well-discriminating instrument for recording relevant general HRQoL and trauma-specific aspects, which showed high correlation with other generic outcome measures

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					QoL/HRQoL	(Social) Participation	ADL			
		(criterion validity) and comparison of the results with internationally recognized QoL instruments (convergent validity); testing screening procedure of the TOP								
Baranyi et al., 2010 Austria [63]	Retrospective monocentric cohort study	Occurrence of full or partial symptoms of PTSD including psycho-pathologically relevant dimensions and QoL in polytraumatized patients; relation of PTSD symptoms with QoL, social support, depressive mood, and dissociative symptoms	N=52 (73.1) Age: Mean 37.6 ± 14.2 ISS: Mean 33.1 ± 12.61 AIS: NR Injury: Inter alia head	NR	SF-36	-	-	1 year post trauma[$]	Self-administered questionnaire; outpatient clinic	High risk for PTSD and reduced HRQoL in people with polytrauma; lowest HRQoL in patients with full PTSD, followed by patients with partial symptoms and without PTSD; severe dissociative and depressive symptoms; greater limitations in some QoL dimensions
Bouman et al., 2017 Netherlands [38]	Prospective, multi-centre, non-randomized controlled study	Effectiveness of the Fast Track (FT) rehabilitation programme on QoL and other health-related outcomes in comparison	FT Group N=65 (75) Age: 18-75, Mean 44.7 ± 16.7 ISS: 4-66, Mean 22.1 ± 12.8, Md (IQR) 19.5 (± 12-29) AIS: NR Non-FT group	NR	SF-36* (*Baseline (pre-injury status)	-	-	Baseline, 3, 6, 9, 12 months post trauma[&]	Individual interviews, telephone interviews, postal survey; NR	Both groups improved their functional status and QoL; FT group showed faster functional recovery at six months compared to nine month of Non-FT group; no different effects

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					QoL/HRQoL	(Social) Participation	ADL			
		with conventional trauma rehabilitation service	N=67 (84) Age: 18-73, Mean 42.0 ± 16.6 ISS: 4-50, Mean 29.4 ± 11.2, Md (IQR) 29 (± 21-38) AIS: NR Injury: Pelvic fracture, musculoskeletal injuries, neuro-trauma/head injuries							observable in both groups at 12 months
Burghofer et al., 2005 Germany [64]	Prospective multicentric cohort study	Outcome five years after major blunt trauma	N=273 (70.3) Age: Mean 37.5, Md: 34 ISS: 16-25: 29.3% 26-50: 63.0% 51-75: 7.7% AIS: NR Injury: Severe blunt trauma	NR	SF-36	-	-	5 years after trauma[§]	Self-administered questionnaire or, in case of cognitive impairment, administered by relatives; NR	Significant differences in QoL between the trauma cohort and a reference population; relevant decrease of QoL in patients with initial unconsciousness and in severely injured patients; high proportion of patients with persisting pain (headache and lower limb pain); 61.7% were back in occupation, 43.5% of these in the former profession
Christensen et al., 2011 26 countries in Europe, Asia, Australia, Africa, North- and South-America [65]	Prospective multicentric randomized placebo-controlled study	Identification of HRQoL dimensions most affected by polytrauma and predictors of poor HRQoL	N=347 (75) Age: Mean 39 ± 14 ISS: Mean 31 ± 11, Md 29 AIS[§]: 4+ Injury: Abdomen, chest, external, extremity, face, head or neck	ICU stay (duration NR)	POLO Chart	-	-	3 Months post trauma[§]	Self-administered questionnaire; NR	Notable reduced HRQoL in polytrauma survivors; physical wellbeing more affected than mental; greater mental impairment detectable by trauma-specific HRQoL instruments than by generic tools; demographic and socio-economic characteristics as well as type of injury and treatment predict HRQoL

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					QoL/HRQoL	(Social) Participation	ADL			
Dienstknecht et al., 2013 Germany [43]	Retrospective monocentric cohort study	Functional and socio-economic long-term results in patients with pelvic ring fractures	Study population of the Hannover Rehab study N=109 (72) Age: 5-55, Mean 28.8 ISS: 16-44, Mean 22.7 AIS: NR Injury: Anterior, posterior and combined pelvic ring injuries, upper and lower limb fracture, thoracic injury, abdominal injury, spine fracture, neurological lesion of lower limb	Conservative und surgical	HASPOC, SF-12	-	-	10-27 years post trauma, Mean 16 years[$]	Self-administered questionnaire as well as assessment via physician; examination in the trauma centre	Long-term functional limitations and socio-economic effects of pelvic ring fractures; reduced QoL in patients with anterior and combined anterior and posterior pelvic ring injuries
Frénisy et al., 2006 France [66]	Prospective monocentric cohort study	Comparison of neurobehavioral and psycho-pathological disorders between patients with severe brain injuries (SBI) and patients after polytrauma (MULT)	SBI N=25 (80) Age: Mean 26.97 ± 9.58 ISS: Mean 25.36 ± 10.80 AIS: NR Injury: Brain Polytrauma N=25 (72) Age: Mean 33.78 ± 10.42 ISS: Mean 22.04 ± 10.06 AIS: NR Injury: Bone fractures, thorax, abdomen	NR	-	-	-	6-24 months[$] SBI Mean 1.02 ± 0.39[$] MULT Mean 0.93 ± 0.41[$]	Face to face interview; NR	Similar psychopathological disorders in both groups; SBI-patients show significantly more neurobehavioral disorders and more obsessive symptoms; presence of cognitive disorders in MULT-patients

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					QoL/HRQoL	(Social) Participation	ADL			
Gribnau et al., 2008 Netherlands [34]	Retrospective monocentric cohort study	Injury characteristics, choice of treatment and QoL of people with U-shaped sacral fractures	N=8 (37.5) Age: 19-55, Md 29 ISS: 17-45, Md 23 AIS: NR Injury: Skull fracture, subdural hematoma, facial fracture, thoracic injury (multiple rib fractures, pneumothorax, haemopneumothorax, sternal fracture, lung contusion), retroperitoneal haematoma, liver laceration	Surgical	EQ-6D	-	-	5-65 months post trauma, Md 36 months[$]	Self-administered questionnaire sent by post	Pain, mood disorders and moderate mobility problems dominate QoL
Gross et al., 2010 Switzerland [48]	Prospective monocentric cohort study	Long-term work ability of patients after polytrauma; association of capacity to work with internationally accepted variables of functional outcomes or QoL	N=115 (76) Age: 14-92, Mean 39.5 ± 20.6 ISS: Mean 27.5 ± 8.2 AIS: Mean 2.6 ± 1.5 Reduced capacity to work N=61 (79) Age: Mean 39.4 ± 20.2 ISS: Mean 30.2 ± 7.7 AIS[§]: Mean 0.5 ± 0.6-3.1 ± 1.5 Non-reduced capacity to work N=54 (72) Age: Mean 39.6 ± 21.2 ISS: Mean 24.3 ± 7.7 AIS[§]: Mean 0.5 ± 0.5-2.6 ± 1.6 Injury: NR	Days ICU stay (Mean): 8.4 ± 12.5	EQ-5D*, SF-36*, NHP (*Baseline (pre-injury status) assessed in hospital)	-	-	Inpatient stay, Md 2.5 [2.0; 3.3] years post trauma[#]	Self-administered questionnaire sent by post	50% show reduced capacity to work; reduced QoL and persistent pain compared to the pre-polytrauma condition; patient, injury, and treatment variables have impact on capacity to work; association of capacity to work with long-term results such as various functional outcomes, symptom status, and QoL

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					QoL/HRQoL	(Social) Participation	ADL			
Gross and Amsler, 2011 Switzerland [49]	Prospective monocentric cohort study	Prevalence and severity of pain after polytrauma compared to pre-injury status using different pain measures; association of pain and HRQoL	N=102 (74) Age: Mean 39.7 ± 20.5 ISS: Mean 27.1 ± 8.0 AIS: NR Injury: NR	NR	EQ-5D*, SF-36*, NHP, TOP (*separate questionnaire for retrospective assessment of pre-injury status)	-	-	Mean 2.7 ± 0.9 years post trauma, Md 2.4 years[$]	Self-administered questionnaire sent by post	Pain in 85% of patients after polytrauma; higher prevalence and severity of pain compared to pre-injury status; close correlation of posttraumatic pain with HRQoL and reduced capacity to work; differences in type and sensitivity of pain measures
Gross et al., 2012 Switzerland [50]	Prospective monocentric cohort study	QoL and functional outcomes after polytrauma compared to pre-injury status; discriminatory potential of different outcome scores between survivors with and without TBI	N=111 (75) Age: Mean 39.5 ± 20.9 ISS: Mean 27.9 ± 8.2 AIS[§]: Mean 0.5 ± 0.6-2.7 ± 1.5 Patients with TBI N=45 Age: NR ISS: Mean 31.1 ± 7.3 AIS[§]: Mean 0.7 ± 0.6-3.9 ± 0.7 Patients without TBI N=66 Age: NR ISS: Mean 25.7 ± 8.1 AIS[§]: Mean 0.4 ± 0.6–2.6 ± 1.6 Injury: Inter alia TBI	Days ICU stay (Mean): 8.9 ± 12.6	EQ-5D*, SF-36*, NHP, TOP (*separate questionnaire for retrospective assessment of pre-injury status)	-	-	Mean 2,7 ± 0,9 years post trauma, Md 2,5 years[$]	Self-administered questionnaire sent by post	Notable reduced QoL and functional outcomes compared to pre-injury status; worse outcomes in patients with TBI; SF-36 and NHP discriminate between patients with and without TBI
Hldaki et al., 2009 Poland [67]	Prospective monocentric cohort study	Comparison of QoL after polytrauma in three consecutive five-year periods using ICIDH	N=827 (71)* Age: Mean 44.7 ISS: Mean 27.6 AIS: NR Injury: Head/neck, thorax, abdomen, pelvis and its organs, spinal column and spinal cord, upper and lower limbs	Surgical	ICIDH (only impairments)	-	-	3 years after completion of inpatient treatment[$]	NR; outpatient clinic	Reduced QoL in 50% of patients after polytrauma; improvement of QoL in one of four patients over 15 years

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					QoL/HRQoL	(Social) Participation	ADL			
Holtslag et al., 2006 Netherlands [53]	Prospective monocentric cohort study	Description of the characteristics of lower extremity injuries in patients after polytrauma and functional long-term outcomes using all domains of ICF and HRQoL and their association	N=186 (75) Age: 16-96, Mean 37 ± 17 ISS: Mean 25 ± 11 AIS: NR Injury: lower extremity injuries, pelvic injuries, TBI	Days ICU stay (Mean): 7 ± 14.1 (range: 1-123)	SIP, SF-36	Assess-ment of criteria such as return to sports and work	GARS	10-18 months post trauma, Mean 14 ± 1.6 months, Md 14 months[$]	Self-administered questionnaire (if necessary, administered by relatives) sent by post	Serious long-term problems of mobility, activity, and participation; mild to moderately limited HRQoL; low association of ICF domain functionality with HRQoL; limited concordance between instruments
Holtslag et al., 2007a Netherlands [51]	Prospective monocentric cohort study	Long-term functional health status in patients after polytrauma; relationship between personal and injury-related characteristics vs. long-term HRQoL	N=335 (74) Age: Mean 37.7 ± 17.1 ISS: Mean 24.9 ± 10.6 AIS: NR Injury: Cranial, facial, upper extremity, chest, abdomen, pelvis, femur, lower leg, spine, spinal cord, peripheral nerve	Days ICU stay (Mean): 12.9 ± 18.7	SIP	-	-	12-18 months post trauma, Mean 451 ± 47 days[$]	Self-administered questionnaire sent by post	Long-term mild to moderately limited HRQoL; lower QoL in young male patients and patients with TBI, spinal cord and extremity injuries; greatest limitations in some SIP categories (work, ambulation, home management, recreation and pastimes, alertness behaviour); co-morbidity strong predictor of SIP scores
Holtslag et al., 2007b Netherlands [52]	Prospective monocentric cohort study	Long-term functional consequences from polytrauma, quantification of the influence of socio-demographic, injury-related and physical	N=335 (74) Age: Mean 37.7 ± 17.1 ISS: Mean 24.9 ± 10.6 AIS: NR Injury: Brain, chest, abdomen, spinal cord, lower extremity, upper extremity	Days ICU stay (Mean): 12.9 ± 18.7	EQ-5D	-	-	12-18 months post trauma, Mean 451 ± 47 days[$]	Self-administered questionnaire (if necessary, administered by relatives) sent by post	Restriction of mobility, self-care and everyday activities, indication of pain and discomfort, anxiety and depression as well as cognitive complaints; localisation of injury, educational level and co-morbidities are important independent predictors of long-term functional

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					QoL/HRQoL	(Social) Participation	ADL			
		factors on outcomes								outcomes after polytrauma; better QoL via EQ-VAS than via utility score
Holtslag et al., 2008 Netherlands [54]	Prospective monocentric cohort study	Impact of trauma on health in terms of burden of injury, years lived with disability (YLD) in non-fatal diseases, years of life lost (YLL) in fatal diseases and disability-adjusted life year (DALY)	Fatalities: N=567 (62.6) Age: Mean 48.4 ± 23.3 ISS: >15 AIS: NR Injury: NR Severely injured: 335 (74.3) Age: Mean 37.7 ± 17.1 ISS: >15 AIS: NR Injury: NR	NR	EQ-5D	-	-	12-18 months after injury, Mean 451 ± 47 days[$]	NR	Substantial loss of healthy life years; at individual level, major trauma contributed an average of 25 DALYs to the burden of disease; at population level, major trauma caused 10 DALYs per 1000 inhabitants
Jackson et al. 2007 USA [68]	Retrospective monocentric cohort study	Prevalence of and risk factors for persistent cognitive impairment as well as emotional and functional difficulties in people without intracranial haemorrhage	N=58 (67) Age: Mean 45.0 ± 14.3 ISS: 33.0 ± 6.8 AIS: NR Injury: Concussion, skull fracture	NR	SF-36	-	FAQ, KATZ ADL	12-24 months after hospital discharge[$]	Administered by a neuro-psychologist or a psychometrist; evaluation at medical centre or place of residence	Substantial prevalence of persistent cognitive impairment, emotional dysfunction and functional decrements, most of all in people with concussion and skull fracture compared to those without these injuries; cognitive impairment was associated with functional defects, poor QoL, and an inability to return to work

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					QoL/HRQoL	(Social) Participation	ADL			
Kaske et al., 2014 Germany [69]	Retrospective monocentric cohort study	Assessment of HRQoL using a trauma-specific instrument (POLO Chart) in patients after polytrauma	N=129 (66.9) Age: Mean 43.9 ±17.5 ISS: Mean 21.5 ±13.3 AIS: ≥ 2 Injury: Lower extremity, thorax, upper extremity, head, spine, pelvis, abdomen	Days ICU stay (Mean): 9.2 ± 9.6, Md (IQR_{25-75}): 5 (2-14) Ventilator days (Mean): 8.2 ± 7.6, Md (IQR_{25-75}): 5 (2-14)	POLO Chart	-	-	Mean 24 months ± 6 months post trauma[$]	Self-administered questionnaire sent by post	Physical impairments, mental and socio-economic deficits in patients after polytrauma; reduced HRQoL in 64% of patients two years after polytrauma, in half of these patients severely impaired HRQoL; TOP is able to identify impairments and to guide rehabilitation resources
Lefering et al., 2012 Germany and Austria [39]	Diagnostic study	Phase IV validation of the trauma-specific TOP module as part of the POLO Chart	Trauma-Group N=172 (73.8) Age: Mean 38.8 ± 14.4 ISS: 8-75, Mean 26.7 ± 11.2 AIS: NR Control Group N=166 (65.1) Age: Mean 42.9 ± 15.0 ISS: 1-9, Mean 3.9 ± 1.7 AIS: NR Injury: NR	NR	POLO Chart	-	Barthel Index	Trauma Group Mean 24.2 ± 18.7 months post trauma[$] Control Group Mean 26.4 ± 14.2 months post trauma[$]	NR; visit to the hospital for a physical examination and to fill in the questionnaires	Significant differences in all QoL dimensions between trauma and control group; apart from physical domains, also impairment of other dimensions of QoL; results show good discriminatory abilities of the TOP and support its precision in collecting all relevant components of QoL in patients after polytrauma
Lippert-Grüner et al., 2007 Germany [70]	Prospective cohort study	HRQoL in patients with severe TBI, with and without concomitant polytrauma	N=49 (78) Age: 15-68, Md 32 ISS: NR AIS: NR Isolated TBI N=28 Age: Mean 36 ISS: Mean 27 AIS: NR TBI and Polytrauma N=21 Age: Mean 26 ISS: Mean 40	ICU (duration NR)	SF-36	-	-	6 and 12 months post trauma[#]	Self-administered questionnaire; during outpatient follow-up examinations	Comparable results for patients with severe TBI with and without concomitant polytrauma in SF-36 with the exception of physical function; indication of TBI as an important factor for HRQoL and outcomes after trauma; improvement of HRQoL over time

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					QoL/HRQoL	(Social) Participation	ADL			
			AIS: >3 Injury: Cranial injury							
Livingston et al., 2009 USA [71]	Prospective monocentric cohort study	Long-term outcome of patients with prolonged (>10 days) ICU stay	N=100 (81) Age: 18-94, Mean/Md 42 ISS: 26-30, Mean/Md 28 AIS: 3-5 Injury: TBI, extremity fracture, pelvis fracture, lower extremity fracture	Surgical Days ICU stay: 19-24, Mean 21, Md 19 Ventilator days: Mean 16, Md 14	-	-	-	2.1-5.0 years, Mean/Md 3.3 years[$]	Telephone interview with patient	Overall self-reported QoL is generally good with significant impairments incl. inability to return to work (less than 50%) or regain pre-injury level of activity and reintegration into society
Marasco et al., 2015 Australia [37]	Retrospective monocentric cohort study	Long-term morbidity and QoL in patients with rib fractures	N=397 (75) Age: Mean 53.9 ± 18.8 ISS: Mean 22.5 ± 11.8 AIS: NR Thoracic-Group N=216 Age: NR ISS: Mean 16.0 ± 7.3 AIS: max. 3 Polytrauma-Group N=181 Age: NR ISS: Mean 30.1 ± 11.6 AIS: max. 4 Injury: Head, neck, face, spine, thorax, abdomen, upper and lower extremity	NR	SF-12	-	-	6, 12, 24 months post trauma[!]	Telephone interview with patient (if necessary, with proxy)	Significantly reduced QoL 24 months after multiple rib fractures; low return to work rate after 6 months; no differences between the groups in pain and QoL; limited function of the thorax group
Mc Carthy et al., 2006 USA [35]	Retrospective monocentric cohort study	Subjective psychosocial health of a population-based sample of adults with traumatic brain injury (TBI)	N=7612 (63,9) Age: 15-75+, Mean 43.2 ± 20.0 ISS: 4-18+, Mean 13.7 ± 7.6 AIS: 2-5 Injury: TBI	NR	SF-36 (psychosocial health scales MCS)	-	Assessment of difficulty regarding bathing/ showering, dressing, eating, transferring to a bed/chair,	1 year after injury ± 3 month[$]	Telephone interview	29% reported poor psychosocial health 1 year after injury; less than 50% received mental health services; a high proportion reported unmet needs for mental health services

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					QoL/HRQoL	(Social) Participation	ADL			
							walking, using the toilet			
Ouellet et al., 2009 Canada [55]	Retrospective multicentric cohort study	Comparison of mental health in patients with polytrauma with and without TBI; factors associated with low mental health; perceived need for and access to mental health-related services	N=405 (64.6) Age: Mean 41.1 ± 15.1 ISS: Mean 23.6 ± 11.7 AIS: NR Patients with TBI N=239 (64.9) Age: Mean 37.4 ± 14.5 ISS: Mean 27.9 ± 9.1 AIS: NR Patients without TBI N=166 (61.0) Age: Mean 46.4 ± 14.5 ISS: Mean 15.7 ± 8.4 AIS: NR Injury: TBI, spine, upper limb, lower limb	Days ICU stay (mean): TBI: 10.3 ± 14.3 Non-TBI: 4.2 ± 8.2	SF-12, Version 2 (MCS as well as all subscales which are related to mental health)	-	-	2-4 years post Trauma[$] TBI: Mean 3.3 ± 0.7 years post Trauma[$] without TBI: Mean 3.3 ± 0.8 years post Trauma[$]	Telephone interview with patients, NR	Not significantly lower MCS-scores in polytrauma patients with TBI compared to patients without TBI; association of 6 factors with low mental health: age, sex, time since trauma, social support, pain and cognitive deficits; high need for mental health-related services but considerable difficulties in maintaining it
Overgaard et al., 2011 Denmark [72]	Prospective cross-sectional monocentric cohort study	Outcome six to nine years after moderate to severe injury in terms of survival, HRQoL and employment status	N= 322 (71) Age: 15-89, Md 34 ISS: 9-75, Md 17 AIS: NR Injury: Blunt trauma	Surgical Days ICU stay: 1-62, Md 4	SF-36	-	-	6-9 years after admission, Md 7.4[$]	Self-administered questionnaire sent by post or telephone interview	78% of the patients survived the trauma; HRQOL was significantly lower for injured patients than a matched control group; 20% retired early
Post et al., 2006 Netherlands [11]	Retrospective monocentric cohort study	Return-to-work status and QoL in patients after polytrauma; analysis of injury-related parameters in relation to functional outcome	N=53 (81) Age: 19-64, Mean 37.3 ± 13.2, Md 34.0 ISS: 16-54, Mean 23.5 ± 8.2, Md 21.0 AIS: Md 0-5, 2.0–3.0 Injury: Head, neck, face, thorax, abdomen, extremities, external	NR	SIP	-	-	1.3-2.2 years post trauma, Mean 1.8± 0.3 years, Md 1.8 years[$]	Self-administered questionnaire sent by post	Most patients after polytrauma return to their former work; compared with general population and less severely injured patients good QoL results; age, but not trauma severity or TBI is a predictor for long-term disablement

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results
					QoL/HRQoL	(Social) Participation	ADL			
Probst et al., 2010 Germany [44]	Retrospective monocentric cohort study	Association of improved short-term outcomes after polytrauma with gender differences; long-term follow-up using QoL and rehabilitation status	Study population of the Hannover Rehab study N=637 (75.4) Age: M: Mean 26.7 ± 12.2 F: Mean 25.9 ± 13.0 ISS: M: Mean 20.4 ± 9.9 F: NR, 21.7 ± 9.7 AIS[§]: M: Md 1-5 F: Md 1-6 Injury: Head, face/neck, thorax, abdomen, skeletal system	Days ICU stay (Mean): M: 15.4 ± 25.0 F: 12.9 ± 23.2 Ventilator days (Mean): M: 11.2 ± 13.3 F: 9.9 ± 12.5	HASPOC, SF-12	-	-	M: 17.5 ± 4.8 years post Trauma[$], F: 17.4 ± 5.0 after trauma[$]	Interview with patients; assessment via physician and physical examination in the trauma centre	After polytrauma, more severe long-term limitations in women than men; no differences in rehabilitation status; longer sick leave times, significantly lower QoL and higher rates of PTSD and psychological support in women
Quale et al., 2009 Norway [73]	Prospective monocentric cohort study	Incidence and risk factors of posttraumatic stress symptoms (PTSS)	N= 79 (74.7) Age: Mean 37 ± 14.3 ISS: Mean 25.6 ± 12.3 AIS: NR Injury: Spinal cord, head	Days ICU stay (Mean): 62 ± 61.4 Ventilator days (Mean): 5.8 ± 13.1	-	-	-	1-3 weeks after admittance to rehabilitation, median days since injury 39, quartiles 24-57[$]	Semistructured interviews, questionnaires; NR	Low incidence of PTSD, but higher incidence of PTSS; anxiety, female gender, negative attitudes towards emotional expression are risk factors for PTSS
Renovell-Ferrer et al., 2017 Spain [74]	Prospective monocentric cohort study	Comparison of aetiology, severity, and functional outcome of people with operated displaced intra-articular calcaneal fractures between polytrauma patients and isolated cases	N=80 (71) Age: 16-74, Mean 48.4, Md 47 ISS: >16 (14%) AIS: NR Injury: Intra-articular calcaneal fractures, pelvic fracture, spine fracture	Surgical	SF-36	-	-	15-62 months after operation, Mean 48 months[$]	NR	No differences in outcome measures and second surgeries; increasing severity of trauma was associated with more second surgeries and worse outcomes; patients with psychiatric comorbidities reported worse HRQoL

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results
					QoL/HRQoL	(Social) Participation	ADL			
Richter et al., 2006 Germany [75]	Prospective monocentric cohort study	Prevalence of PTSD; identification of risk factors for PTSD	N=37 (76) Age: Mean 41.7 ± 17.0 ISS: Mean 30.4 ± 10.7 AIS: NR Injury: NR	Days ICU stay: ≥ 30	-	-	-	7-58 months after discharge from ICU, Mean 35 ± 14 months[$]	Self-administered questionnaire and interview; NR	Compared to non-trauma patients, higher risk for PTSD in trauma patients; ISS and ICU treatment not associated with PTSD
Ringburg et al., 2011 Netherlands [1]	Prospective monocentric cohort study	HRQoL in patients after polytrauma; predictive factors for long-term poor functioning in the dimensions of HRQoL instruments	N=246 (66) Age: Md 40 ISS: Md 22 AIS: NR Injury: Head, face, chest, abdomen, extremities	NR	EQ-5D, HUI 2, HUI 3	-	-	12 months post trauma[$]	Self-administered questionnaire sent by post	HRQoL far below general population norms; high prevalence of specific limitations with female gender and comorbidity as predictors of long-term disability
Sampalis et al. 2006 Canada [76]	Retrospective multicentric cross-sectional cohort study	Functional status and QoL 12 months after injury	N=144 (61.8) Age: 18-85, Mean 44,2 ±17.1 ISS: 1-49, Mean 18.9 ± 9.4 AIS: NR Injury: Head/ neck, extremity, face, thorax, abdominal, burns	Surgical and conservative Days ICU stay: 0.1-11.0, Mean 3.44 ± 0.9	SF-36, SIP	-	-	0.8-1.5 years, Mean 1.3 years post trauma[$]	Self-administered questionnaire sent by post	Residual impairments in functional capacity and QoL 1 year after injury; specific subgroups (injuries resulting of motor vehicle collisions, thoracic and head injuries) report diminished QoL; ICU treatment, prolonged length of hospital stay, and surgical treatment are predictors for poor functional capacity and reduced QoL
Schmidt-Rohlfing et al., 2011 Germany [45]	Retrospective monocentric cohort study	Correlation between scoring systems focusing on QoL and those focusing on functional status of knee joint	Study population of the Hannover Rehab study Patients with knee injury N=48 (73) Age: Mean 29.7 ± 11.8 ISS: Mean 22 ± 9.8 AIS: NR	Surgical	HASPOC, SF-12	-	-	Mean 16.9 ± 4.9 years post trauma[$]	Assessment via physician, NR	No correlation between scoring systems for functional knee condition and QoL; poorer outcome for patients with knee injuries compared to

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results
					QoL/HRQoL	(Social) Participation	ADL			
			Patients with femur fracture N=107 (72) Age: Mean 26.8 ±13.9 ISS: Mean 23.3 AIS: NR Injury: Knee, femoral fracture, head							patients with femoral fractures
Simmel et al., 2013 Germany [57]	Retrospective monocentric cohort study	QoL and health status many years after polytrauma and influencing factors	N=127 (76) Age: Mean 36 ISS: 26-75, Mean 35.6 AIS: NR Injury: NR	Days ICU stay (Mean): 21.1	POLO Chart	-	-	38-108 months post trauma, Mean 70 months[$]	Self-administered questionnaire sent by post	Significantly reduced QoL and health status after polytrauma; higher age, female gender, low education and chronic previous illnesses have a negative impact on outcomes as well as difficulties with authorities/institutions, unemployment, long inpatient stays and subjectively inadequate inpatient treatment
Sirois et al., 2009 Canada [56]	Retrospective multicentric cohort study	Regional differences in perceived need for and barriers to post-acute rehabilitation services; long-term functional outcomes and physical health outcomes in patients after polytrauma	N=435 (64.5) Age: 18-65, Mean 41.1 ± 15.1 ISS: 1–75 AIS: NR Injury: Spine, brain, extremity	NR	SF-12, Version 2 (PCS and subscales physical function, role function and general health perception)	-	-	2-4 years, Mean 3.3 ± 0.7 years post trauma[$]	Telephone interview with patient (if necessary, with proxy)	No regional differences in perceived need and barriers to post-acute rehabilitation services; no regional differences in long-term functional outcomes and physical health in patients after polytrauma, with the exception of a slightly lower score in SF-12 subscale physical functioning in regions with limited availability of rehabilitation services

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results
					QoL/HRQoL	(Social) Participation	ADL			
Sluys et al., 2005 Sweden [77]	Retrospective monocentric cohort study	Outcome and QoL five years after injury and identification of factors potentially associated with outcome and QoL	N=205 (74) Age: 20-87, Md 39 ISS: 9-57, Md 14 AIS: 3-5 Injury: Extremities, pelvic girdle, head and neck, chest, external/skin, abdominal or pelvic content, face	Surgical, conservative Days ICU stay: 1-86, Mean 4	SF-36	-	-	5 years post trauma[§]	Self-administered questionnaire sent by post, or telephone interview	Relevant decrease in HRQoL and permanent impairments as well as disabilities in most of the affected. Length of hospital stay, days spent on ICU, surgical procedures, in-hospital major complications, age, recurrent injury, and inadequate information are risk factors for poor HRQoL; injury severity did not predict poor HRQoL
Soberg et al., 2007 Norway [59]	Prospective monocentric cohort study	Complete and partial return to work rate; prognostic factors for complete return to work after polytrauma	N=100 (82) Age: 18-67, 34.5 ± 13.5 ISS: Mean 28.1 ±11.3 AIS: NR Injury: Head/neck, face chest, abdomen/pelvic content, extremity, external, spinal cord	Ventilator days (median): 5 (IR 9)	SF-36	-	-	20.0 ± 9.5 weeks, 1 year and 2 years post trauma[!]	Self-administered questionnaire sent by post, telephone interview of 2 patients	Complete return to work: four out of 10 patients two years after polytrauma, 2/3 actively attempting to return to work. Lower education, low social functioning and > 20 weeks in hospital or rehabilitation are risk factors for not returning to work; tendentially better mental health in returnees than in non-returnees to work, overall lower mental sum scores than for general population; physical health one year after trauma significantly lower than for general population

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results
					QoL/HRQoL	(Social) Participation	ADL			
Soberg et al., 2011 Norway [58]	Prospective monocentric cohort study	Work status and trajectory of return to work after polytrauma; prognostic factors for the return to work of patients after polytrauma over five years	N=75 (83) Age: 18-67, 34.5 ± 13.5 ISS: NRTW-Group Mean 28.9 ± 12.5 RTW-Group Mean 25.8 ± 10.8 AIS: NR Injury: Head (TBI), face chest, abdomen/pelvic content, spinal, upper extremity, lower extremity	NR	SF-36	-	-	20.0 ± 9.5 weeks, 1 year, 2 years and 5 years post trauma[?]	Self-administered questionnaire sent by post	Complete return to work of 50% of patients after polytrauma over 5 years, 23% receive disability benefits. Higher education, better physical, social and cognitive functioning and use of coping strategies are prognostic factors for returning to work; generally better physical and mental functioning after trauma in returnees to work than in non-returnees
Stulemeijer et al., 2006 Netherlands [78]	Prospective monocentric cohort study	Impact of additional extracranial injuries on functional outcomes and on post-concussion symptoms in patients with mild TBI	Isolated TBI N=210 (63) Age: Mean 35.2 ± 12.3 ISS: 1-17, Mean 4.7 ± 3.4 AIS (head): 1-4. Mean 1.9 ± 0.7 TBI and extracranial injuries N=89 (74) Age: Mean 37.9 ± 12.0 ISS: 5-41, Mean 14.5 ± 7.4 AIS (head): 1-5, Mean 2.3 ± 0.8 Control Group N=261 (45) Age, ISS, AIS: NR Injury: TBI, extremities, face, chest, abdomen, external	NR	-	-	SF-36 (sub-scales of physical and social functioning)	6 Months post trauma[$]	Self-administered questionnaire sent by post	Poorer functional outcomes (i.e. lower levels of physical function, lower rates of return to work and generally worse outcomes) of patients with additional extracranial injuries and mild TBI but equal level of post-concussion symptoms; greater impairment in patients after polytrauma compared to the control group with mild injuries

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results
					QoL/HRQoL	(Social) Participation	ADL			
Suzuki et al., 2007 Japan [79]	Retrospective monocentric cohort study	Evaluation of long-term functional outcome of patients with unstable pelvic ring fractures and correlation with other factors	N=57 (49) Age: 16-72, Mean 42 ISS: 9-50, Mean 24.6 AIS: NR Injury: Pelvic fracture	Surgical and conservative	SF-36	-	-	24.4-107.4 months, Mean 47.2 months[$]	Questionnaire; during clinical reevaluation	Functional outcome was not associated with ISS, fracture location or type of fracture; neurologic injury correlated with poor functional outcome
Tee et al., 2014 Australia [80]	Prospective monocentric cohort study	Early predictors of suboptimal health status in patients after polytrauma with spine injuries	N: 479 (71.8) Age: Mean 47.5 ± 19.1 ISS: Md14 AIS: NR Injury: TBI, spine	NR	SF-12	-	-	1 year post trauma[$]	NR	Early risk factors of suboptimal physical health or QoL are tachycardia, hyperglycemia, multiple chronic pre-existing medical comorbidities and thoracic spine injuries
Van Delft-Schreurs et al., 2016 Netherlands [81]	Retrospective monocentric cross-sectional cohort study	Structure and psychometric properties of the Dutch SMFA questionnaire in severely injured patients	N=173 (69) Age: Mean 46 ± 19 ISS: Md 21 AIS: ≥ 2 Injury: Upper and lower extremity, TBI	NR	SMFA, WHOQOL-BREF	-	-	1.3-4.4 years post trauma[$]	Self-administered questionnaire sent by post	Good psychometric properties of the Dutch SMFA in severely injured people and strong correlations between the SMFA and WHOQOL-BREF; Dutch SMFA seems to be valid and useful for generating an overview of physical limitations and emotional problems
Victorson et al., 2008 USA [82]	Mixed method monocentric study	Development and validation of the injury distress index (IDI)	N=169 (66.1) Age: Mean 34.5±11.1 ISS: NR AIS: 1-5 Injury: Lower extremity, upper extremity, burn (21%)	Surgical, conservative	-	-	-	Mean 8 days after admission[$]	Self-administered questionnaire; trauma centre	IDI showed acceptable reliability and validity

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results
					QoL/HRQoL	(Social) Participation	ADL			
Vles et al., 2005 Netherlands [7]	Prospective monocentric cohort study	Prevalence and determinants of disabilities and return to work after severe injury	N=166 (81) Age: Men: Mean 33 Women: Mean 37 ISS: Men: Mean 23 Women: Mean 23 AIS: ≥ 2 Injury: Cranium, brain, extremities, spine, pelvis, chest, thoracic contents, abdomen, pelvic contents	NR	EQ-5D	-	Change of daily activities and sports	12-69 months, Mean/Md 41 months (3.4 years) after trauma[$]	Self-administered questionnaire sent by post, or telephone interview	Substantial impact of trauma on long-term functioning; follow-up between 12 and 24 months might be suitable for outcome assessment including quality of treatment
von Rüden et al., 2013 Germany [40]	Retrospective monocentric cohort study	Prognosis, incidence, trauma mechanism, mortality, invalidity, working ability/disability and QoL in patients after polytrauma with ISS ≥ 50	N=88 (77.3) Age: 18-63, Mean 40.3 ± 17.4 ISS: Mean 56.8 AIS[§]: Mean 1.7-4.4 Injury: Head/brain, face, chest, abdomen, spinal cord, pelvis, upper limb, lower limb, soft tissues	Days ICU stay (Mean): ISS <50: 14.4 ISS >50: 25.9 Ventilator days (Mean): ISS <50: 13.4 ISS >50: 20.3	POLO Chart	-	-	18-78 months post trauma, Mean 3.6 years[$]	Self-administered questionnaire sent by post; face to face interview with subgroup	Good prognosis for severely injured persons, but QoL not satisfactory mainly due to psychological problems and chronic pain; 6.5% with ISS ≥;50; main cause: motorbike accident; 36% mortality rate; 57% limited working ability
Williamson et al., 2009 Australia [83]	Prospective multicentric cohort study	Predictors of moderate or severe pain six months after injury	N=1290 (61) Age: 14-95 ISS: >15 AIS: NR Injury: Extremities, spinal injuries	NR	SF-12 (only on discharge rating of status in the week before injury)	-	-	6.1-7.1 months after trauma (Md 6.5); before discharge, 6 months (assessment of other PROs than HRQoL)[#]	Face to face interview in acute hospital before discharge and during follow-up by telephone at home	Prevalence of moderate or severe pain at follow-up was 30%; predictors of pain were educational and compensation status, preinjury pain-related disability, pain intensity at discharge

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results
					QoL/HRQoL	(Social) Participation	ADL			
Wurm et al., 2012 Germany [41]	Retrospective monocentric cohort study	Posttraumatic QoL in patients after polytrauma with ISS ≥ 50	N=88 (77) Age: Mean 40.3 ± 17.4 ISS: Mean 56.8 AIS[§]: Mean 3.9-4.4 Injury: Thorax, TBI, abdomen, spine, lower extremity	Days ICU stay (Mean): 23 Ventilator days (Mean): 20	POLO Chart	-	-	18-78 months post trauma, Mean 3.6 years[$]	Self-administered questionnaire sent by post and personal interview with subgroup	Good survival rate but poor posttraumatic QoL mainly due to psychological problems and chronic pain
Zelle et al., 2005a Germany [46]	Retrospective monocentric cohort study	Comparison of objective and subjective outcome of patients after polytrauma with injuries below the knee joint and injuries of the lower extremity above the knee joint	Study population of the Hannover Rehab study N=389 (74) Age: Mean 25.4 ± 11.7 ISS: Mean 20.2 ± 9.3 AIS: NR Injury: Lower extremity	NR	HASPOC, SF-12	-	-	Mean 17.3 ± 4.8 years post trauma[$]	Self-administered questionnaire as well as assessment via physician; examination in the trauma centre	Significantly better objective and subjective outcomes of patients after polytrauma with injuries above the knee joint compared to injuries below the knee joint; no significant differences in the MCS of SF-12
Zelle et al., 2005b Germany [36]	Retrospective monocentric cohort study	Comparison of long-term functional outcome after polytrauma between workers´ compensation patients (WCP) and non-workers´ compensation patients (N-WCP)	Study population of the Hannover Rehab study N=637 (75) Age: 3-60, Mean 26.4 ISS: 4-54, Mean 20.7 ± 9.7 AIS: NR WCP N=233 (82.0) Age: Mean 29.78 ± 12.09 ISS: Mean 20.60 ± 10.00 AIS: NR N-WCP N=404 (71.5) Age: Mean 24.56 ± 12.12 ISS: Mean 20.80 ± 9.61	NR	HASPOC, SF-12	-	-	10-28 years, Mean 17.5 years[$] WCP: Mean 17.36 ± 4.77[$] N-WCP: Mean 17.0 ± 4.89[$]	Self-administered questionnaire as well as assessment via physician; examination in the trauma centre	Worse subjective and objective outcomes in WCP; no significant differences in the SF-12 MCS

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Author, year, country	Study design	Objective	Study population (Number: N=(% Male), age, ISS/AIS (Range, Mean ± SD/Median), Kind of injury))	Treatment	Instrument(s) for assessment of			Time of measurement/ length of follow-up period	Application/ setting	Results
					QoL/HRQoL	(Social) Participation	ADL			
			AIS: NR Injury: Lower extremity, head, face, upper extremity, chest, abdomen, spine, neck, skin							
Zwingmann et al., 2016 Germany [84]	Retrospective monocentric cohort study	Long-term HRQoL and changes in daily life of patients after polytrauma	N=147 (75.5) Age: 6.1-82.1, Mean 40 ± 19 ISS: 17-59, Mean 28 ±10.6 AIS: NR Injury: NR	Days ICU stay (Mean): 7.8 ± 7.5	EQ-5D, SF-36, TOP	-	-	5.0-7.4 years post trauma, Mean 6 ± 0,8 years[$]	NR, verbal interview and/or written questionnaire sent by post	Impaired QoL due to pain and functional disability; association of negative socio-economic effects and emotional repercussion

Indices and Abbreviations:

AIS[§]: AIS values shown for various body regions; F: Female; IQR/IR: Interquartile range; M: Male; max.: highest value; MCS: Mental Component Summary score; NR: not reported; Md: Median; N: Number; NRTW: Non-return to work; PCS: Physical Component Summary score; QoL: Quality of Life; RTW: Return to work; SD: Standard deviation; *: Numbers displayed for Overall-Characteristic of study population; [$]: one measurement; [#]: two measurements; [!]: three measurements; [?]: four measurements; [&]: five measurements

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Instruments used to assess HRQoL

In total, 15 different tools were used to assess HRQoL during follow-up after polytrauma. Of those, six were generic instruments: Short Form-36 Health Survey (SF-36, n=21, 39%),[35, 38, 47-50, 53, 58-61, 63, 64, 68, 70, 72, 74, 76, 77, 79, 84] 12-Item Short Form Health Survey (SF-12, n=12, 22%),[36, 37, 42-46, 55, 56, 62, 80, 83] Nottingham Health Profile (NHP, n=4, 7%),[47-50] Sickness Impact Profile (SIP, n=4, 7%),[11, 51, 53, 76] Short Musculoskeletal Function Assessment (SMFA, n=1, 2%),[81] and World Health Organization Quality of Life Instrument – Short Form (WHOQOL-BREF, n=1, 2%).[81] In addition, four preference-based tools were applied: European Quality of Life 5-Dimension (EQ-5D, n=9, 17%),[1, 7, 47-50, 52, 54, 84] EQ-6D (n=1, 2%),[34] Health Utility Index Mark 2 (HUI 2, n=1, 2%),[1] and HUI 3 (n=1, 2%).[1] Three trauma-specific instruments were used: Hannover Score for Polytrauma Outcome (HASPOC, n=6, 11%),[36, 42-46] Trauma Outcome Profile (TOP, (n=4, 7%),[47, 49, 50, 84] and Aachen Long-term Outcome Score (ALOS, n=1, 2%).[47] The Polytrauma Outcome Chart (POLO Chart, n=6, 11%) [39-41, 57, 65, 69] was used as a modular instrument consisting of the Glasgow Outcome Profile, SF-36, EQ-5D, and TOP. The International Classification of Impairments, Disabilities and Handicaps (ICIDH) was reported in one publication (n=1, 2%).[67] Fifteen publications combined two to five different instruments to assess HRQoL.[1, 36, 42-50, 53, 76, 81, 84]

In some publications (n=5), instruments to assess HRQoL were only used partially (e.g. solely the mental or physical part, or selected subscales of the SF-36 or SF-12, chapter 'Disability' of the ICIDH). [35, 55, 56, 62, 67] Six publications conducted a retrospective collection of HRQoL via SF-36, SF-12, EQ-5D and TOP capturing the pre-trauma status. [38, 47-50, 83]

Instruments used to assess ADL

ADL was measured by different generic instruments: Groningen Activity Restriction Scale (GARS), [53] Functional Activities Questionnaire (FAQ), [68] Katz Index of Activities of Daily Living (KATZ ADL),[68] and Barthel Index (BI) (each n=1, 2%).[39] The SF-36 subscales of physical and social functioning (n=1, 2%) [78] were also used to record ADL. Two publications employed self-designed questions.[7, 35] Most frequently (n=5), a single instrument was applied to assess ADL.[7, 35, 39, 53, 78]

Instruments used to assess (social) participation

One publication reported the assessment of participation by self-designed questions covering return to sports and work.[53]

Table 2 displays the number of publications and the instruments used to assess HRQoL, ADL, and (social) participation.

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Table 2 Quantitative overview of instruments assessing HRQoL, ADL and (social) participation in n=54 publications

PRO	Category	Instrument	N (%)
HRQoL	Generic instruments	Short Form-36 Health Survey (SF-36)	21 (39)
		12-Item Short Form Health Survey (SF-12)	12 (22)
		Nottingham Health Profile (NHP)	4 (7)
		Sickness Impact Profile (SIP)	4 (7)
		Short Musculoskeletal Function Assessment (SMFA)	1 (2)
		World Health Organization Quality of Life Instrument – Short Form (WHOQOL-BREF)	1 (2)
	Preference-based instruments	European Quality of Life 5-Dimension (EQ-5D)	9 (17)
		EQ-6D*	1 (2)
		Health Utility Index Mark 2 (HUI 2)	1 (2)
		Health Utility Index Mark 3 (HUI 3)	1 (2)
	Trauma-specific instruments	Hannover Score for Polytrauma Outcome (HASPOC)	6 (11)
		Trauma Outcome Profile (TOP)	4 (7)
		Aachen Long-term Outcome Score (ALOS)	1 (2)
	Modular instruments	Polytrauma-Outcome-Chart (POLO Chart)	6 (11)
	Conceptual framework	The International Classification of Impairments, Disabilities and Handicaps (ICIDH)	1 (2)
ADL		Self-designed questions	2 (4)
	Generic instruments	Groningen Activity Restriction Scale (GARS)	each n=1 (2)
		Functional Activities Questionnaire (FAQ)	
		Katz Index of Activities of Daily Living (KATZ ADL)	
		Barthel Index (BI)	
		SF-36 subscales physical and social functioning	
(Social) Participation		Self-designed questions	1 (2)

*EQ-6D=EQ-5D comprising an additional question capturing a cognitive component (memory, concentration, coherence, and intelligence quotient)

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Application of instruments

A postal survey was most often used to assess HRQoL, ADL, and (social) participation (n=17).[1, 11, 34, 47-53, 57, 58, 62, 69, 76, 78, 81] Next in frequency were telephone interviews (n=5),[35, 37, 55, 56, 71] expert assessments (n=2),[45, 68] face to face interview (n=1),[66] and combinations of those (n=16).[7, 36, 38, 40, 41, 43, 44, 46, 59, 60, 72, 73, 75, 77, 83, 84] The research settings were: trauma centre,[36, 43, 44, 46, 82] outpatient clinic,[63, 67, 70] hospital,[39] medical centre or place of patient´s residence,[68] hospital and home.[83] In four publications, no information regarding the application of instruments was provided.[42, 54, 74, 80] Relatives or proxies were reported in five publications.[37, 52, 53, 56, 64] While most instruments are used with several data collection methods and survey settings, the HASPOC, where reported, is only recorded in the trauma centre and is assessed by a physician in addition to the respondent.

Measurements during follow-up

PROs mentioned above were most frequently recorded at a single time point (n=47).[1, 7, 11, 34-36, 39-47, 49-57, 60-69, 71-82, 84] Few publications reported two (n=3),[48, 70, 83] three (n=2),[37, 59] four or five (each n=1) [38, 58] time points of assessment.

Duration of follow-up

Most often (n=13), a follow-up of one to one and a half year was reported.[1, 35, 38, 51-54, 60, 63, 66, 70, 76, 80] Six publications described a follow-up period of over 10 and up to 17.5 years.[36, 42-46] Rarely, shorter follow-up periods of eight days [82] to six months [78, 83] were detected.

Further PROs assessed

Beyond HRQoL, ADL, and (social) participation, assessments of further PROs were reported in the included publications. Most frequently these comprised function, pain, and mental disorders (including depression and/or depressive symptoms, anxiety, and PTSD). Only few PROs focused on social support, cognition, subjective health status, and social network.

Table 3 provides an overview of the kind and number of further recorded PROs.

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Table 3 Quantitative overview of kind and number of further reported PROs in n=54 publications

PRO	N (%)
Function	19 (35)
Mental disorder	16 (30)
Pain	14 (26)
Social support	7 (13)
Cognition	6 (11)
Subjective health status	3 (5)
Social network	2 (4)
Neuropsychological disorder	1 (2)

Critical appraisal results

The methodological quality and the risk of bias was appraised using the NICE-checklist for quantitative intervention studies,[30] [38, 65] the NICE-checklist for quantitative studies reporting correlations and associations,[31] [1, 7, 11, 34-37, 40-64, 66-80, 83, 84] the SIGN-checklist for diagnostic studies,[32] [39, 81] and the Mixed Method Appraisal Tool.[33] [(82)]

Some methodological weaknesses were identified regarding the selection of the study population, consideration of confounders (e.g. age, sex), data analysis and presentation of results, internal validity and generalisability of study results.[34, 40-43, 45, 50, 60, 61, 65, 67, 69-71, 73-75, 84] While most criteria of the critical appraisal were fulfilled, a moderate risk of bias was observed.[7, 11, 36-39, 44, 46-53, 56, 57, 59, 60, 62-64, 66, 72, 77, 78, 80-83] A low risk of bias was observed in eight publications.[1, 35, 54, 55, 58, 68, 76, 79] The results of the critical appraisal are provided in (Additional file 4).

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DISCUSSION

Main findings

The present systematic review identified 54 publications contributing to the research questions. A great variety was observed among all investigated aspects. HRQoL was the most frequently recorded PRO, and SF-36, SF-12 and EQ-5D were the most frequently applied instruments to assess it. Data on ADL and (social) participation were rarely reported. Most questionnaires were sent out per mail. Measurements were predominantly performed at a single time point, and the follow-up period lasted mainly between one and one and a half year. Function, mental disorders and pain were the most frequently assessed categories of the other PROs.

Instruments for the assessment of HRQoL, ADL, and (social) participation

In 48 of the included publications, HRQoL was assessed – predominantly with generic and preference-based HRQoL instruments such as SF-36, SF-12, EQ-5D, and NHP. Thus, over a long period of time, results are available for the impact of polytrauma on HRQoL and its possible change over time.

Unsurprisingly, the SF-36, one of the oldest and most widely used instrument for assessing HRQoL,[85] is the most frequently reported instrument in publications included in this review. Along with the SF-12, SF tools accounted for 61% of all identified instruments. This result is supported by observations of other reviews conducted in the field of polytrauma.[85-88]

Considering the use of the different instruments assessing HRQoL by the different trauma groups, it becomes apparent that instruments are used without a clearly recognizable pattern or reason. Similarly, no pattern can be identified to explain the use of the SF-12 instead of the SF-36. No information was provided in this regard in the publications. However, it can be assumed that the reason for using the SF-12 is that it takes less time to administer than the SF-36. In 1999, an international consensus conference on evidence-based guidelines for the systematic evaluation and assessment of HRQoL after polytrauma recommended the use of modular instruments comprising generic and trauma-specific aspects.[89] Generic instruments have been applied continuously over a long time period, have been translated into different languages, and can be applied regardless of the underlying disease. They are suitable for assessing HRQoL after polytrauma since it can affect different body regions, show different injury patterns, and lead to a variety of complications. In contrast, trauma-specific instruments capture targeted relevant problems and measure the longitudinal change of HRQoL.[90] Despite the recommendation, the POLO Chart is the only modular instrument comprising generic and trauma-specific aspects reported in the included publications.[90] HASPOC, TOP and ALOS were the trauma-specific tools used. HASPOC is an instrument designed to evaluate the quality of rehabilitation. Since it comprises instruments assessing inter alia HRQoL, ADL, and function,[91, 92] we decided to subsume it into the category ‘HRQoL’.

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Apart from assessing HRQoL, we were particularly interested in the measurement of ADL and social participation. Regarding the assessment of ADL, five different instruments and self-designed questions in six publications were observed in our review. (Social) participation was only reported once by Holtslag et al. (2006).[53] In summary, it was notable that ADL and (social) participation after polytrauma were seldom assessed. One reason might be that measures of HRQoL or function already include several questions related to ADL and (social) participation. Predominantly, (social) participation is researched in people with brain or spinal cord injuries. A general understanding and clear definition of the participatory construct is still missing, which might be another reason for the low reporting rates of (social) participation. It has been argued that (social) participation should be one of the main aspects in rehabilitation after polytrauma,[93] besides survival and function, and there are a number of instruments available to assess this PRO.[94-96] This could also be assumed for ADL.

Application of instruments

Most frequently, a postal questionnaire was used. Compared to personal interviews, questionnaires sent by mail are quicker and less expensive.[97] In addition, potential bias introduced by an interviewer is excluded. However, it is not possible to verify the person answering the questionnaire, comprehension questions cannot be clarified, the full completion of the questionnaire cannot be guaranteed, and the proportion of non-responders is higher.[97] However, other methods of data collection like personal or telephone interviews also have limitations (e.g. influence of the interviewer on the person being interviewed),[97] thus all data collection methods may incorporate (different) biases. The consultation of a proxy answering the questions was rarely reported. Injuries, e.g. traumatic brain injuries (TBI), may require the use of a proxy and could enable the collection of outcome data after polytrauma, despite a risk of bias.[97, 98] The present systematic review excluded publications reporting single or mono injuries like single TBIs, which might be the reason only few publications including proxies were detected.

The retrospective assessment of the pre-trauma situation helps enable a comparison of results and obtain a holistic view of the outcome after an injury. Gross et al. (2010, 2012) as well as Gross and Amsler (2011) measured pre-fracture level of HRQoL over two years after trauma.[48-50] However, there are limitations to this approach: Memories of the pre-traumatic situation can be distorted by various circumstances, which might increase the risk of recall bias.[99] Large population screenings or the use of population norms might help to evaluate the pre-traumatic state of HRQoL.[18, 47] Baseline assessments should ideally be made in the first week after the event. However, this is not always possible due to the severity of the injury. Another limitation might be that some instruments are not suitable for use in hospital. For example, changed behaviour patterns measured by the SIP can also result from the very fact of being an inpatient. Apart from HRQoL, functional recovery is also of great interest and an

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important factor for evaluating outcomes after a polytrauma. In comparison to HRQoL and due to better quantification, the retrospective evaluation of function is more feasible.[18]

Most of the included publications reported a single assessment of the PROs. However, the course of recovery cannot be derived from a single measurement. The aforementioned consensus conference recommended measurements at three, 12 and 24 months after the event.[17] Other groups suggested assessments of PROs after one, two, four and 12 months [100] or six and 12 months after the injury, and beyond 12 months in case of longer rehabilitation needs.[13]

The most frequently reported follow-up period were one to one and a half year. Six publications conducted follow-up periods of more than 10 years.

Further PROs assessed

Further reported PROs assessed mainly function, mental disorders, and pain – outcomes that are highly affected by polytrauma. Social support, subjective health status, cognition, or social network were rarely recorded. This might be explained as follows: Instruments assessing HRQoL mainly capture subjective health status. While cognition is frequently affected by brain injuries, publications reporting study populations with single brain injuries were excluded from this review. Social network and social support seem to influence recovery and return to work after trauma [101, 102] and should therefore receive more attention in future research.

Recommendations for the assessment of PROs

The existing recommendations for the use and application of instruments – as well as time points of assessments and outcomes to be measured after polytrauma – differ substantially,[13, 17] but there are attempts to standardize outcome measurements.[18] Classification systems might be helpful for a standardized assessment of PROs, facilitating the adequate selection of instruments for a specific research objective.[23] The internet platform PROMIS – Patient-reported Outcomes Measurement Information System – provides research tools and is continuously updated with the latest developments and newly translated instruments.[103] An important step towards a standardized approach of collecting HRQoL was made in the TraumaRegister DGU® by the implementation of HRQoL assessment with the SF-12.[14]

Furthermore, several groups have focused on the definition of special outcome variables to be recorded as default parameters. Hoffman et al. (2016) recommended the development of a ‘minimum data set’ based on the ICF.[20] A core outcome set relevant for the respective outcome could be defined. This procedure could guide decision-making for the collection and presentation of outcomes in studies.[22] In addition, ICF linking enabling a check of which ICF components are mapped by an instrument could support the tool selection for particular research questions.[16, 19] Other authors also support the idea of core outcome sets.[21, 24]

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Strengths and limitations

This review was conducted thoroughly based on a registered protocol in line with the PRISMA-P quality and the PRISMA statement requirements. We provide a broad overview of instruments used for the collection of HRQoL, ADL and (social) participation, along with information on their application. In addition, this study offers an overview on further PROs reported in the context of polytrauma. First efforts to standardize the collection of PROs have been achieved to optimize comparability of outcomes, but further consistency between central players (e.g. trauma registers) is needed. Therefore, this review encourages further improvement in the assessment of PROs after polytrauma.

The present systematic review has some limitations: Selection bias can be assumed as the search was restricted by the search period and language. However, we performed backward citation tracking and saw that only a few additional articles could be included. Our study only included English and German language literature and thus does not provide insight into published results in other languages. Therefore, it cannot be ruled out that relevant publications from low and middle income countries containing local results in local languages were not included. However, we assume that scientific articles are predominantly published in English; thus, the results presented here should give a comprehensive impression related to the research questions of this systematic review.

We did not use the term 'major trauma'. Therefore, we cannot rule out that relevant publications reporting major trauma were not included. However, major trauma seemed to be increasingly mentioned in connection with single and psychological trauma – neither of which met the inclusion criteria of our systematic review. We screened systematic reviews and meta-analyses to identify further appropriate studies and would have included publications reporting major trauma if they fulfilled the inclusion criteria of this systematic review. Still, we believe that the non-use of the term 'major trauma' has no impact on the results of this systematic review and the answers to the research questions.

CONCLUSION

The present review reveals variations across study findings, which impairs the comparability of outcomes after polytrauma. To improve this situation, a more standardized approach for recording PROs should be established. First efforts have been initiated, but further alignment between the central players, such as the trauma registers, is required. Continuous collection of core outcomes during the regeneration of persons with polytrauma could create the basis for a targeted use of treatments and their adaptation to the results of the outcome measurement. Moreover, there appears to be little research on (social) participation and on further PROs like social support and social network. PROs need more attention in polytrauma-related outcome research. After all, we need to know the patients' perspectives on relevant outcomes after polytrauma so as to consider their needs when providing health services.

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Supplementary information

Additional file 1: PRISMA Checklist

Additional file 2: Study protocol of this systematic review

Additional file 3: Search strategy

Additional file 4: Results of the critical appraisal

Abbreviations

ADL: Activities of daily living; AH-F: Annegret Herrmann-Frank; AI: Andrea Icks; AIS: Abbreviated Injury Scale; AIS§: AIS values are shown in publication for various body regions; ALOS: Aachen Long-term Outcome Score; AUC: Academy for Trauma Surgery GmbH; BI: Barthel Index; cs: can´t say; ct: can´t tell; CENTRAL: Cochrane Central Register of Controlled Trials; CINAHL: Cumulative Index to Nursing and Allied Health Literature; CW: Charlotte Wittgens; DALY: Disability-adjusted life year; e.g.: For example; EQ-5D: European Quality of Life 5-Dimension; EQ-6D: Six-Dimensional EuroQol instrument; EQ-VAS: EQ-5D visual analogue scale; F: Female; FAQ: Functional Activities Questionnaire; Fig.: Figure; FT: Fast track; GARS: Groningen Activity Restriction Scale; HASPOC: Hannover Score for Polytrauma Outcome; HRQoL: Health-related quality of life, HUI 2: Health Utility Index Mark 2; HUI 3: Health Utility Index Mark 3; ICF: International Classification of Functioning, Disability and Health; ICIDH: International Classification of Impairments, Disabilities and Handicaps; ICU: Intensive Care Unit; IDI: injury distress index; IG: Irmela Gnass; IQR/IR: Interquartile range; ISS: Injury Severity Score; JS: Johannes Sturm; JW: Joachim Windolf; KATZ ADL: Katz Index of Activities of Daily Living; KM: Kai Moschinski; M: Male; max.: highest value; MCS: Mental Component Summary score; Md: Median; MeSH: Medical Subject Heading; mh: MeSH-term; MIM: Maria-Inti Metzendorf; MMAT: Mixed Method Appraisal Tool; MR: Michaela Ritschel; MULT: Patients after polytrauma; N: Number; n: no; NA: not applicable; NHP: Nottingham Health Profile; NICE: UK National Institute for Health and Care Excellence; NR: not reported; NRTW: Non-return to work; N-WCP: Non-workers´ compensation patients; PCS: Physical Component Summary score; POLO Chart: Polytrauma-Outcome-Chart; PRISMA: Preferred Reporting Items for Systematic Review and Meta-Analysis; PRISMA-P: Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols; PRO: Patient-reported outcome; PROMIS: Patient-reported Outcomes Measurement Information System; PTSD: Posttraumatic stress disorder; PTSS: Posttraumatic stress symptoms; QoL: Quality of Life; SA: Silke Andrich; SBI: Severe brain injuries; SK: Silke Kuske; RTW: Return to work; SD: Standard deviation; SF: Sascha Flohé; SF-12: 12-Item Short Form Health Survey; SF-36: Short Form-36 Health Survey; SIGN: Scottish Intercollegiate Guidelines Network; SIP: Sickness Impact Profile; SMFA: Short Musculoskeletal Function Assessment; SOB: Sandra Olivia Borgmann; TBI: traumatic brain injury; TOP: Trauma Outcome Profile; tw: Textwords; WCP: Workers´ compensation patients;

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WHO ICTRP: WHO International Clinical Trials Registry Platform WHOQOL-BREF: World Health Organization Quality of Life Instrument – Short Form; y: yes; YLD: Years lived with disability; YLL: Years of life lost

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Competing interests

JS was executive secretary of the Academy for Trauma Surgery GmbH (AUC). The authors declare that they have no competing interests.

Patient consent for publication

Not required.

Data availability statement

There are no additional data available.

Differences between protocol and review

We have planned to include publications only when the full text of the study is available. However, since all texts were available, there was nothing to exclude according to this criterion.

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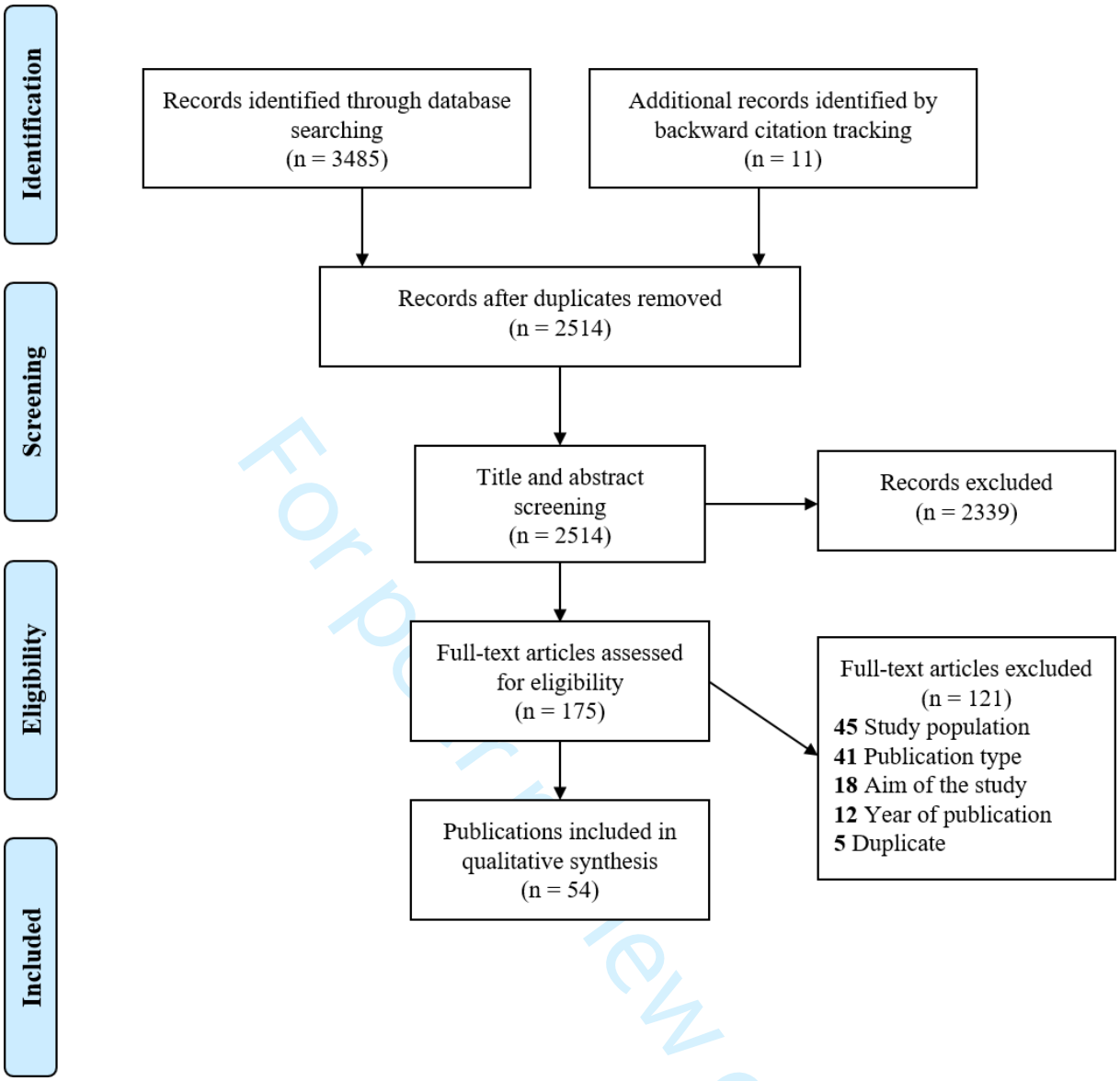
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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	2
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	2
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	2
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	2
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	2
Synthesis of results	6	Specify the methods used to present and synthesise results.	2
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	2
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	2
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	3
Interpretation	10	Provide a general interpretation of the results and important implications.	3
OTHER			
Funding	11	Specify the primary source of funding for the review.	NA
Registration	12	Provide the register name and registration number.	3

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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PRISMA 2019 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2, 3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	2, 5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5, 6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5, 6
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Additional file 3
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	NA
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	NA
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2019 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	NA
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8-33, Table 1-3
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	33, Additional file 4
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	NA
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	NA
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	NA
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	34-36
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	37
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	37
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	39

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Assessment of patient-reported outcomes after polytrauma: protocol for a systematic review

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ABSTRACT

Introduction Survivors of polytrauma experience long-term and short-term burden that influences their lives. The patients' view of relevant short-term and long-term outcomes should be captured in instruments that measure quality of life and other patient-reported outcomes (PROs) after a polytrauma. The aim of this systematic review is to (1) collect instruments that assess PROs (quality of life, social participation and activities of daily living) during follow-up after polytrauma, (2) describe the instruments' application (eg, duration of period of follow-up) and (3) investigate other relevant PROs that are also assessed in the included studies (pain, depression, anxiety and cognitive function).

Methods and analysis The systematic review protocol is developed in line with the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols statement. MEDLINE, EMBASE, Cumulative Index to Nursing and Allied Health Literature, PsycINFO, Cochrane Central Register of Controlled Trials and the trials registers ClinicalTrials.gov and WHO International Clinical Trials Registry Platform will be searched. Keywords, for example, 'polytrauma', 'multiple trauma', 'quality of life', 'activities of daily living' or 'pain' will be used. Publications published between January 2005 and the most recent date (currently: August 2016) will be included. In order to present the latest possible results, an update of the search is conducted before publication. The data extraction and a content analysis will be carried out systematically. A critical appraisal will be performed.

Ethics and dissemination Formal ethical approval is not required as primary data will not be collected. The results will be published in a peer-reviewed publication.

PROSPERO registration number CRD42017060825.

Strengths and limitations of this study

- ► To the best of our knowledge, there are no published systematic reviews providing an overview of assessed patient-reported outcomes after polytrauma, different instruments used to measure these as well as the application of these instruments.
- ► This systematic review will report the identified instruments used to assess quality of life, social participation and activities of daily living, and describe their application.
- ► We will show additional measures for further relevant patient-reported outcomes following polytrauma, for example, pain, depression, anxiety and cognitive function.
- ► A limitation of the review might be that publications will be included which were published since 2005. It cannot be ruled out in that case that relevant literature published before 2005 is missing.

INTRODUCTION

Severe injuries represent a leading cause of death and permanent disability.[1] Especially in the central European region, such severely injured patients are termed as 'polytrauma patients' or as being 'polytraumatised'. 'Polytrauma' is defined as having at least two severe injuries in different body regions that are potentially life threatening. In the Anglo-American literature, these patients are mostly referred to as 'multiple-injury', 'multiple-trauma' or 'severely injured'. All these descriptions have in common that a certain degree of injury is required. The severity of trauma is usually indicated by the Injury Severity Score (ISS).[2] In general, an ISS ≥16 falls within the definition of polytrauma, multiple-trauma or severely injured patients.[1 3] In the following, we use the term polytrauma. Comparing the incidence of trauma is challenging, considering the different definitions or conditions of trauma, and due to inconsistencies in the available data.[4] In 2015, according to the TraumaRegister DGU®, a German registry that cover patients with severe injuries, the number of severely multiply injured patients was 17 630 (ISS ≥16).[5] In 2012, the number of severely multiply injured patients (ISS ≥16) reached approximately 18 400 per year. This corresponds to a cumulative incidence of 0.02% per year for Germany.[6] Persons affected are mostly men (72%) with a mean age of 46.5 years.[7]

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Due to strategies for early advanced life support, high quality in healthcare services, progress in treatment options and more traffic safety, survival rates after polytrauma are increasing. However, survivors are faced with long-term and short-term burden after polytrauma: 1 year after polytrauma, patients continue to suffer from persisting problems with mobility (34%), self-care (15%) and activities of daily living (ADL) (51%). Furthermore, they experience pain and/or discomfort (58%) as well as anxiety and/or depression (37%).[8] Relevant disabilities, such as respiration limitations, paraplegia or tetraplegia, are major causes for 40% of those affected not returning to their former workplace[6 9] and impact on socioeconomic and quality-of-life aspects.[9–11] For the necessary acute and rehabilitative care, the social economic impact is estimated to be €106 000 per patient. In Germany, additional costs of €935 000 accrue if patients do not return to their former workplace.[9]

Consequently, the aim to reduce short-term and long-term burden after polytrauma is critically important.[9 11] Thus, clinical parameters are necessary to support treatment processes and services in acute care facilities and to display short-term outcomes. Additionally, evaluations capturing a patient's view of short-term and long-term outcomes in terms of, for example, psychological and physical factors, functional status and social interaction, are becoming increasingly important for doctors and nurses as well as for patients and their family members.[11 12] The measurement of these patient-reported outcomes (PROs), which can also be reported, for example, by relatives in case of mental disability, is important for completing the assessment of relevant clinical outcomes after injury. To assess the impact of a polytrauma on quality of life (QoL) and other relevant PROs, it is necessary to question the patients.[12 13] The preferred gold standard is a measurement at multiple times during the follow-up period to track trends and focus on long-term conditions and outcomes associated with polytrauma.[13] Therefore, it is necessary to know which instruments for assessing PROs are applied and how. So far, no systematic review has been identified providing an overview of assessed PROs after polytrauma, different instruments used to measure these as well as the application of these instruments. Thus, this systematic review will provide an overview of instruments used to measure PROs, including QoL and other currently reported outcome measures for patients with polytrauma, and describe their application in detail.

Currently the TraumaRegister DGU® collects data on emergency care, treatment in shock rooms, intensive care unit and discharge.[3 12] Regarding the comprehensive evaluation of the short-term and long-term burden, the register plans to expand their measurement battery by assessing PROs, for example, QoL.

AIM OF THE STUDY

The aim of this systematic review is to:

1. collect instruments that assess PROs (QoL, social participation and ADL) during the follow-up after polytrauma;
2. describe the application of these instruments in detail (eg, duration of period of follow-up, frequency of application and time of measurements during follow-up);
3. investigate other relevant PROs that are additionally assessed in the included studies (eg, pain, depression, anxiety and cognitive function) without reporting the application of instruments in detail.

METHOD

The systematic review protocol is developed in line with the quality requirements of the Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) statement,[14] and the PRISMA statement will be considered during the review procedure.[15] The protocol was registered on 8 April 2017 at PROSPERO (registration number CRD42017060825). Prior to publication an update of the search will be performed to be able to present the latest results.

Databases

The literature search will be conducted in the following databases: MEDLINE, EMBASE, Cumulative Index to Nursing and Allied Health Literature, PsycINFO, Cochrane Central Register of Controlled Trials and the trials registers ClinicalTrials.gov and WHO International Clinical Trials Registry Platform. To ensure literature saturation, reference lists of eligible studies will be examined for further relevant publications.

Search strategy

The search strategy will be developed in cooperation with a specialist for systematic reviews using database-specific-controlled vocabulary and additional free-text terms. Online supplementary appendix A provides the MEDLINE search strategy that will be adapted to the other databases accordingly.

Inclusion and exclusion criteria

Studies will be included if they match the following inclusion criteria: (1) the study assesses PROs, such as QoL/health-related QoL, social participation, ADL, pain, depression, anxiety or cognitive function in people aged 18–75 years during a temporally clearly defined follow-up period after polytrauma (injuries involving at least two different areas of the body or organ systems, ISS >15, Abbreviated Injury Scale (AIS) ≥3); (2) all kinds of original empirical research that were published between 1 January 2005 and currently 3 August 2016; (3) English or German language publications will be considered and (4) the full text of the study is available (ie, for contacting the authors). The interest in QoL in research and its implication in practice has been growing since 2005.[16] As one of our main aims is to collect instruments measuring QoL, we decided to include publications as of 2005. Our

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procedure is supported by the burgeoning perspective in clinical research on PROs and its relevance for the future in addition to clinical data.[17] A further reason was to capture the most recent developments in research on PROs. Therefore, we looked specifically for publications from the last 11 years.

Studies will be excluded from this review if they mention one of the following criteria: (1) an ISS ≤15 or an AIS <3 or no reporting of ISS or AIS; (2) low-energy injuries, single or mono injuries or geriatric injuries, burn injuries, war injuries (group of veterans or military staff are excluded), cancer and other chronic diseases as secondary diagnosis; (3) publications in which the primary aim does not focus on PROs and (4) grey literature, books, letters/short reports, abstracts, editorials, comments or discussion papers as well as case studies, systematic reviews and meta-analysis. However, systematic reviews and meta-analysis will be screened to identify further appropriate studies.

Study selection process

On the basis of predefined inclusion and exclusion criteria, suitable publications will be selected by title and abstract, and independently screened by two reviewers. A third reviewer will solve differences concerning the inclusion or exclusion of studies. In order to foster the process of suitability decisions, inclusion and exclusion of the first 50 publications by title and abstract screening will be discussed between the two reviewers. Subsequently, two reviewers assessing eligibility for final inclusion in this review will screen the full texts of the remaining publications again independently. A third reviewer will solve conflicts in the final inclusion or exclusion of studies. Inter-rater reliability will be determined after the title abstract as well as after the full-text screening.

Data extraction and synthesis

The data extraction will be performed according to the requirements of Cochrane reviews.[18] For data extraction, two experienced researchers will use a piloted data extraction sheet independently. Extracted data from the included studies will provide an overview of: first author and publication year, study design, country, study population (number of subjects, proportion of men, mean age with SD and range, kind of injury, ISS, AIS and other characteristics), treatment, aim of the study, findings, and, furthermore, for the reported PROs according to aim 1, applied instruments to assess these PROs, description of the instrument, data collection (method of assessment, time of measurements, length of follow-up period, quality criteria of instruments (eg, validity and reliability) and modifications of the instruments. Furthermore, the result of the critical appraisal of the study quality will be added.

Critical appraisal

In the final selection of eligible studies, two reviewers will independently perform a quality appraisal to assess the methodological quality and the risk of bias in each study type using standardised checklists of the UK National Institute for Health and Care Excellence[19] or Scottish Intercollegiate Guidelines Network[20] or the Mixed Method Appraisal Tool.[21] Reviewers will resolve disagreements through discussion.

DISCUSSION

We will perform the proposed systematic review to generate an overview of the instruments used to assess PROs in the field of polytrauma. The results of this systematic review will serve as a basis to expand the TraumaRegister DGU® with a focus on QoL measures. Likewise, the additional knowledge on further PROs, for example, pain, depression, anxiety and cognitive function, might emphasise the patients' perspectives on relevant outcomes after polytrauma and lead to its consideration in the provision of health services. Subsequently, healthcare providers and policy-makers may draw their attention on this topic and will implement the assessment of PROs in decision-making regarding the treatment process.

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Competing interests JS is executive secretary of the Academy for Trauma Surgery GmbH (AUC).

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Additional file 3: Literature search strategy

PubMed (last searched 5 April 2018)

#1
"Multiple Trauma"[mh] OR multiple trauma*[tw] OR polytrauma*[tw]
#2
"Quality of Life"[mh] or "quality of life"[tw] or "life quality"[tw] OR QoL[tw] OR HRQoL[tw] OR satisf*[tw] OR "self reported health"[tw] OR "Activities of Daily Living"[mh] OR "activities of daily living"[tw] OR "daily living activity"[tw] OR "daily living activities"[tw] OR "activity of daily living"[tw] OR "limitation of activity"[tw] OR "self care"[tw] OR "physical functioning"[tw] OR functional abilit*[tw] OR "functional assessment"[tw] OR "Social Participation"[mh] OR "social participation"[tw] OR social activit*[tw] OR "social functioning"[tw] OR "social interaction"[tw] OR "social isolation"[tw] OR "social integration"[tw] OR "Pain"[mh] OR pain[tw] OR "Depression"[mh] OR "Depressive Disorder"[mh] OR depress*[tw] OR "emotional distress"[tw] OR "Stress, Psychological"[mh] OR stress[tw] OR "Anxiety"[mh] OR anxiet*[tw] OR "Independent Living"[mh] OR "independent living"[tw] OR "community dwelling"[tw] OR "wellbeing"[tw] OR "well being"[tw] OR "Cognition Disorders"[mh] OR cognitive function*[tw] OR "mental health"[tw] OR "cognitive impairment"[tw] OR "Patient Outcome Assessment"[mh] OR "patient reported outcome"[tw] OR "patient reported outcomes"[tw] OR "patient related outcome"[tw] OR "patient related outcomes"[tw] OR "patient centered outcomes"[tw] OR "patient centered outcome"[tw] OR rehabilitation outcome*[tw] OR "Work Capacity Evaluation"[mh] OR "work capacity"[tw] OR "occupational function"[tw] OR "Stress Disorders, Post-Traumatic"[mh] OR PTSD[tw] OR "traumatic stress disorder"[tw] OR "posttraumatic stress disorder"[tw]
#3
#1 AND #2
#4
#3 AND Filters: **Publication date from 2005/01/01 to 2016/12/31; English; German**

UPDATE
#3 AND Filters: **Publication date from 2016/08/01 to 2018/12/31; English; German**

Embase Ovid (<1974 to 2016 August 02>, last searched 3 August 2016)

1. multiple trauma/
2. (multiple trauma* or polytrauma*).tw.
3. or/1-2
4. exp quality of life/
5. (quality of life or life quality or QoL or HRQoL or satisf* or self reported health).tw.
6. daily life activity/
7. ("activities of daily living" or "activity of daily living" or daily living activit* or "limitation of activity" or self care or physical functioning or functional abilit*or functional assessment).tw.
8. social participation/
9. (social participation or social activit* or social functioning or social interaction or social isolation or social integration).tw.
10. pain/
11. pain.tw.
12. exp depression/

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13. (depress* or emotional distress).tw.
14. mental stress/
15. stress.tw.
16. anxiety/
17. anxiet*.tw.
18. independent living/
19. (independent living or community dwelling).tw.
20. wellbeing/
21. (wellbeing or well being).tw.
22. cognitive defect/
23. postoperative cognitive dysfunction/
24. (cognitive adj (disorders or function* or impairment)).tw.
25. (mental health).tw.
26. patient satisfaction/
27. (patient adj (outcome assessment or reported outcome* or related outcome* or centered outcome*)).tw.
28. (rehabiliation outcome*).tw.
29. work capacity/
30. (work capacity or occupational function).tw.
31. posttraumatic stress disorder/
32. (PTSD or posttraumatic stress disorder or traumatic stress disorder).tw.
33. or/4-32
34. 3 and 33
35. limit 34 to ((english or german) and yr="2005 - 2016")

CINAHL EBSCO (last searched 5 April 2018)

1. MH "Multiple Trauma"
2. TI (multiple trauma* or polytrauma*)
3. AB (multiple trauma* or polytrauma*)
4. S1 OR S2 OR S3
5. MH ("Quality of Life+" or "Health and Life Quality (Iowa NOC)+" or "Disability Evaluation")
6. TI ("quality of life" or "life quality" or "QoL" or "HRQoL" or satisf* or "self reported health")
7. AB ("quality of life" or "life quality" or "QoL" or "HRQoL" or satisf* or "self reported health")
8. MH "Activities of Daily Living+"
9. MH "Self-Care Deficit (Saba CCC)+"
10. MH "Activities of Daily Living (Saba CCC)"
11. MH "Self-Care (Iowa NOC)+"
12. TI ("activities of daily living" or "daily living activity" or "daily living activities" or "activity of daily living" or "limitation of activity" or "self care" or "physical functioning" or "functional abilit*" or "functional assessment")
13. AB ("activities of daily living" or "daily living activity" or "daily living activities" or "activity of daily living" or "limitation of activity" or "self care" or "physical functioning" or "functional abilit*" or "functional assessment")
14. MH "Social Participation"
15. TI ("social participation" or "social activit*" or "social functioning" or "social interaction" or "social isolation" or "social integration")
16. AB ("social participation" or "social activit*" or "social functioning" or "social interaction" or "social isolation" or "social integration")
17. MH "Pain+"
18. TI pain

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19. AB pain
20. MH "Depression+"
21. TI (depress* or "emotional distress")
22. AB (depress* or "emotional distress")
23. MH "Stress, Psychological+"
24. TI stress
25. AB stress
26. MH "Anxiety+"
27. TI anxiet*
28. AB anxiet*
29. MH "Community Living+"
30. TI ("independent living" or "community living" or "community dwelling"or "wellbeing"or "well being")
31. AB ("independent living" or "community living" or "community dwelling"or "wellbeing"or "well being")
32. MH "Cognition Disorders+"
33. TI ("cognitive function*" or "mental health" or "cognitive impairment")
34. AB ("cognitive function*" or "mental health" or "cognitive impairment")
35. MH "Outcomes (Health Care)+"
36. TI ("patient reported outcome*" or "patient related outcome*"or "patient centered outcome*" or "rehabilitation outcome*")
37. AB ("patient reported outcome*" or "patient related outcome*"or "patient centered outcome*" or "rehabilitation outcome*")
38. MH "Work Capacity Evaluation"
39. TI ("work capacity" or "occupational function")
40. AB ("work capacity" or "occupational function")
41. MH "Stress Disorders, Post-Traumatic+"
42. TI (PTSD or "posttraumatic stress disorder" or "traumatic stress disorder")
43. AB (PTSD or "posttraumatic stress disorder" or "traumatic stress disorder")
44. S5 OR S6 OR S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR S24 OR S25 OR S26 OR S27 OR S28 OR S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36 OR S37 OR S38 OR S39 OR S40 OR S41 OR S42 OR S43
45. S4 AND S44
46. EM 20050101-20161231
47. S45 AND S46

UPDATE
EM 20160801-20181231

PsycINFO Ovid (last searched 5 April 2018)

1. (multiple trauma* or polytrauma*).tw.
2. "Quality of Life"/
3. "Life Satisfaction"/
4. (quality of life or life quality or QoL or HRQoL or satisf* or self reported health).tw.
5. "Activities of Daily Living"/
6. "Self-Care Skills"/
7. ("activities of daily living" or "activity of daily living" or daily living activit* or "limitation of activity" or self care or physical functioning or functional abilit*or functional assessment).tw.
8. (social participation or social activit* or social functioning or social interaction or social isolation or social integration).tw.

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9. Pain/
10. pain.tw.
11. exp Major Depression/
12. (depress* or emotional distress).tw.
13. Psychological Stress/
14. "Stress and Coping Measures"/
15. Coping Behavior/
16. Adjustment Disorders/
17. stress.tw.
18. Anxiety/
19. anxiet*.tw.
20. (independent living or community dwelling).tw.
21. Well Being/
22. (wellbeing or well being).tw.
23. Cognitive Impairment/
24. exp Memory Disorders/
25. Cognitive Ability/
26. (cognitive adj (disorders or function* or impairment)).tw.
27. (mental health).tw.
28. Client Satisfaction/
29. (patient adj (outcome assessment or reported outcome* or related outcome* or centered outcome*)).tw.
30. exp Treatment outcomes/
31. (rehabilitation outcome*).tw.
32. (work capacity or occupational function).tw.
33. Post-Traumatic Stress/
34. Posttraumatic Stress Disorder/
35. (PTSD or posttraumatic stress disorder or traumatic stress disorder).tw.
36. or/2-35
37. 1 and 36
38. limit 37 to yr="2005 -Current"

UPDATE

(201608* or 201609* or 20161* or 2017* or 2018*).up.

Cochrane Central Register of Controlled Trials via Cochrane Register of Studies Online (last searched 5 April 2018)

1. (multiple trauma* or polytrauma*):TI,AB,KY
2. ("quality of life" or "life quality" or QoL or HRQoL or satisf* or "self reported health"):TI,AB,KY
3. ("activities of daily living" or "daily living activity" or "daily living activities" or "activity of daily living" or "limitation of activity" or "self care" or "physical functioning" or "functional abilit*" or "functional assessment"):TI,AB,KY
4. ("social participation" or social activit* or "social functioning" or "social interaction" or "social isolation" or "social integration"):TI,AB,KY
5. (pain):TI,AB,KY
6. (depress* or "emotional distress"):TI,AB,KY
7. (stress):TI,AB,KY
8. (anxiet*):TI,AB,KY
9. ("independent living" or "community dwelling" or "wellbeing" or "well being"):TI,AB,KY

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10. ("cognition disorder*" or "cognitive function*" or "mental health" or "cognitive impairment"):TI,AB,KY
11. ("patient reported outcome" or "patient reported outcomes" or "patient related outcome" or "patient related outcomes" or "patient centered outcomes" or "patient centered outcome" or "rehabilitation outcome*"):TI,AB,KY
12. ("work capacity" or "occupational function"):TI,AB,KY
13. (PTSD or "traumatic stress disorder" or "posttraumatic stress disorder"):TI,AB,KY
14. #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13
15. #1 AND #14

ClinicalTrials.gov (Expert search, last searched 5 April 2018)

Condition: "multiple trauma" OR "multiple traumata" OR polytrauma OR polytraumata

Search terms: "quality of life" OR "life quality" OR QoL OR HRQoL OR satisfaction OR "self reported health" OR "activities of daily living" OR "daily living activity" OR "daily living activities" OR "activity of daily living" OR "limitation of activity" OR "self care" OR "physical functioning" OR "functional ability" OR "functional abilities" OR "functional assessment" OR social OR pain OR depression OR "depressive disorder" OR distress OR stress OR anxiety OR "independent living" OR "community dwelling" OR "wellbeing" OR "well being" OR "cognition disorder" OR "memory disorders" OR "cognition disorders" OR "cognitive function" OR "cognitive functions" OR "cognitive abilities" OR "cognitive ability" OR "mental health" OR "cognitive impairment" OR "patient outcome assessment" OR "patient reported outcome" OR "patient reported outcomes" OR "patient related outcome" OR "patient related outcomes" OR "patient centered outcomes" OR "patient centered outcome" OR rehabilitation OR coping OR adjustment OR "work capacity" OR "occupational function" OR PTSD OR "traumatic stress disorder" OR "posttraumatic stress disorder"

UPDATE

("quality of life" OR "life quality" OR QoL OR HRQoL OR satisfaction OR "self reported health" OR "activities of daily living" OR "daily living activity" OR "daily living activities" OR "activity of daily living" OR "limitation of activity" OR "self care" OR "physical functioning" OR "functional ability" OR "functional abilities" OR "functional assessment" OR social OR pain OR depression OR "depressive disorder" OR distress OR stress OR anxiety OR "independent living" OR "community dwelling" OR "wellbeing" OR "well being" OR "cognition disorder" OR "memory disorders" OR "cognition disorders" OR "cognitive function" OR "cognitive functions" OR "cognitive abilities" OR "cognitive ability" OR "mental health" OR "cognitive impairment" OR "patient outcome assessment" OR "patient reported outcome" OR "patient reported outcomes" OR "patient related outcome" OR "patient related outcomes" OR "patient centered outcomes" OR "patient centered outcome" OR rehabilitation OR coping OR adjustment OR "work capacity" OR "occupational function" OR PTSD OR "traumatic stress disorder" OR "posttraumatic stress disorder") | "multiple trauma" OR "multiple traumata" OR polytrauma OR polytraumata | First posted from 08/03/2016 to 12/31/2018

WHO ICTRP search (Standard search, last searched 5 April 2018)

multiple trauma* OR polytrauma*

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Additional file 4: Table - Results of the critical appraisal

Author, year	**Critical appraisal**
Abraham et al., 2014 [60]	NICE[31]: pp: 3, p: 6, m: /, NA: 4, NR[a]: 4, p[#], m[?]
Ahrberg et al., 2014 [61]	NICE[31]: pp: 1, p: 7, m: 2, NA: 4, NR[a]: 3, m[#], m[?]
Andruszkow et al., 2013 [42]	NICE[31]: pp: 2, p: 7, m: 1, NA: 4, NR[a]: 3, m[#], m[?]
Archer et al., 2012 [62]	NICE[31]: pp: 3, p: 6, m: 1, NA: 5, NR[a]: 2, p[#], m[?]
Attenberger et al., 2012 [47]	NICE[31]: pp: 3, p: 6, m: 1, NA: 5, NR[a]: 2, p[#], m[?]
Baranyi et al., 2010 [63]	NICE[31]: pp: 7, p: 3, m:/ , NA: 4, NR[a]: 3, pp[#], p[?]
Bouman et al., 2017 [38]	NICE[30]: pp: 9, p: 7, m: 3, NA: /, NR[a]: 6, m[#], m[?]
Burghofer et al., 2005 [64]	NICE[31]: pp: 8, p: 2, m: 1, NA: 6, NR[a]: /, p[#], p[?]
Christensen et al., 2011 [65]	NICE[30]: pp: 5, p: 5, m: 2, NA: 10, NR[a]: 3, p[#], p[?]
Dienstknecht et al., 2013 [43]	NICE[31]: pp: 3, p: 6, m: 1, NA: 4, NR[a]: 3, m[#], m[?]
Frénisy et al., 2006 [66]	NICE[31]: pp: 3, p: 8, m: /, NA: 4, NR[a]: 2, p[#], m[?]
Gribnau et al., 2008 [34]	NICE[31]: pp: 2, p: 1, m: 2, NA: 7, NR[a]: 5, m[#], m[?]
Gross et al., 2010 [48]	NICE[31]: pp: 6, p: 4, m: 2, NA: 3, NR[a]: 2, p[#], p[?]
Gross and Amsler, 2011 [49]	NICE[31]: pp: 4, p: 6, m: 1, NA: 4, NR[a]: 2, p[#], p[?]
Gross et al., 2012 [50]	NICE[31]: pp: 6, p: 5, m: /, NA: 4, NR[a]: 2, p[#], m[?]
Hldaki et al., 2009 [67]	NICE[31]: pp: 2, p: 2, m: 4, NA: 6, NR[a]: 3, m[#], m[?]
Holtslag et al., 2006 [53]	NICE[31]: pp: 3, p: 7, m: 1, NA: 4, NR[a]: 2, p[#], p[?]
Holtslag et al., 2007a [51]	NICE[31]: pp: 8, p: 2, m: /, NA: 5, NR[a]: 2, p[#], p[?]
Holtslag et al., 2007b [52]	NICE[31]: pp: 8, p: 1, m: /, NA: 5, NRa: 3, p#, p?

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Author, year	Critical appraisal
Holtslag et al., 2008 [54]	NICE[31]: pp: 5, p: 8, m: /, NA: 4, NRa: /, pp#, pp?
Jackson et al. 2007 [68]	NICE[31]: pp: 5, p: 7, m: /, NA: 4, NR[a]: 1, p[#], p[?]
Kaske et al., 2014 [69]	NICE[31]: pp: 2, p: 5, m: 2, NA: 6, NR[a]: 2, m[#], m[?]
Lefering et al., 2012 [39]	SIGN[32]: y: 8, n: 3, cs: 2, p*: 1, da: 1
Lippert-Grüner et al., 2007 [70]	NICE[31]: pp: 3, p: 4, m: 1, NA: 4, NR[a]: 5, p[#], m[?]
Livingston et al., 2009 [71]	NICE[31]: pp: 1, p: 8, m: 1, NA: 4, NR[a]: 3, m[#], m[?]
Marasco et al., 2015 [37]	NICE[31]: pp: 6, p: 4, m: /, NA: 4, NR[a]: 3, p[#], p[?]
Mc Carthy et al., 2006 [35]	NICE[31]: pp: 9, p: 3, m: /, NA: 5, NR[a]: /, pp[#], p[?]
Ouellet et al., 2009 [55]	NICE[31]: pp: 6, p: 6, m: 1, NA: 3, NR[a]: 1, p[#], p[?]
Overgaard et al., 2011 [72]	NICE[31]: pp: 5, p: 8, m: 1, NA: 3, NR[a]: /, m[#], m[?]
Post et al., 2006 [11]	NICE[31]: pp: 3, p: 8, m: /, NA: 4, NR[a]: 2, p[#], p[?]
Probst et al., 2010 [44]	NICE[31]: pp: 5, p: 5, m: 2, NA: 4, NR[a]: 1, p[#], p[?]
Quale et al., 2009 [73]	NICE[31]: pp: 4, p: 4, m: 1, NA: 4, NR[a]: 4, pp[#], m[?]
Renovell-Ferrer et al., 2017 [74]	NICE[31]: pp: 1, p: 9, m: 1, NA: 2, NR[a]: 5, m[#], m[?]
Richter et al., 2006 [75]	NICE[31]: pp: 5, p: 4 m: 2, NA: 4, NR[a]: 2, m[#], m[?]
Ringburg et al., 2011 [1]	NICE[31]: pp: 6, p: 6, m: /, NA: 4, NR[a]: 1, p[#], p[?]
Sampalis et al. 2006 [76]	NICE[31]: pp: 6, p: 7, m: 1, NA: 3, NR[a]: /, p[#], p[?]
Schmidt-Rohlfing et al., 2011 [45]	NICE[31]: pp: 1, p: 4, m: 3, NA: 5, NR[a]: 4, m[#], m[?]
Simmel et al., 2013 [57]	NICE[31]: pp: 8, p: 3, m: /, NA: 4, NR[a]: 2, p[#], p[?]
Sirois et al., 2009 [56]	NICE[31]: pp: 5, p: 5, m: /, NA: 4, NR[a]: 3, p[#], p[?]

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Author, year	Critical appraisal
Sluys et al., 2005 [77]	NICE[31]: pp: 5, p: 6, m: 3, NA: 3, NR[a]: /, m[#], m[?]
Soberg et al., 2007 [59]	NICE[31]: pp: 8, p: 2, m: /, NA: 5, NR[a]: 2, p[#], p[?]
Soberg et al., 2011 [58]	NICE[31]: pp: 7, p: 6, m: /, NA: 3, NR[a]: 1, p[#], p[?]
Stulemeijer et al., 2006 [78]	NICE[31]: pp: 3, p: 7, m: /, NA: 5, NR[a]: 2, p[#], m[?]
Suzuki et al., 2007 [79]	NICE[31]: pp: 4, p: 9, m: 1, NA: 3, NR[a]: /, p[#], m[?]
Tee et al., 2014 [80]	NICE[31]: pp: 7, p: 3, m: /, NA: 4, NR[a]: 3, p[#], m[?]
Van Delft-Schreurs et al., 2016 [81]	SIGN[32]: y: 7, n: 1, cs: 5, p*: 1, si: 1
Victorson et al., 2008 [82]	MMAT[33]: y: 8, n: 2, ct: 2
Vles et al., 2005 [7]	NICE[31]: pp: 5, p: 6, m: 3, NA: 3, NR[a]: /, p[#], m[?]
von Rüden et al., 2013 [40]	NICE[31]: pp: 3, p: 5, m: 2, NA: 4, NR[a]: 3, m[#], m[?]
Williamson et al., 2009 [83]	NICE[31]: pp: 7, p: 3, m: /, NA: 4, NR[a]: 3, p[#], p[?]
Wurm et al., 2012 [41]	NICE[31]: pp: 3, p: 5, m: 1, NA: 5, NR[a]: 3, m[#], m[?]
Zelle et al., 2005a [46]	NICE[31]: pp: 4, p: 6, m: 1, NA: 4, NR[a]: 2, m[#], m[?]
Zelle et al., 2005b [36]	NICE[31]: pp: 8, p: 3, m: /, NA: 4, NR[a]: 2, p[#], p[?]
Zwingmann et al., 2016 [84]	NICE[31]: pp: 3, p: 3, m: 4, NA: 5, NR[a]: 2, m[#], m[?]

Abbreviations:

n: no; y: yes

MMAT: ct: can´t tell – statement not possible;

NICE: m: there are significant sources of bias.; m[#]: Evaluation of internal validity of entire study – few or no checklist criteria were fulfilled; m[?]: Evaluation of the study's overall external validity – few or no checklist criteria were fulfilled; NA: not applicable – aspects of the study design that are not applicable; NR[a]: not reported – the study does not report how certain aspects have been or should have been taken into account; p: either the checklist question cannot be clearly answered by the way the study is reported, or the study may not have considered all potential sources of bias; p[#:] Evaluation of internal validity of entire study – some checklist criteria were fulfilled; p[?:] Evaluation of the study's overall external validity – some checklist criteria were fulfilled; pp: study was designed or conducted to minimize risk of bias; pp[#:] Evaluation of the study's overall internal validity – all or most checklist criteria were fulfilled; pp[?:] Evaluation of the study's overall external validity – all or most checklist criteria were fulfilled.

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SIGN: cs: can´t say– statement not possible; da: study directly applicable to target population; p*: Most criteria met. Some flaws in study with an associated risk of bias; si: some indirectness in applicability to target population.

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